CS2 AI COACHING SYSTEM

Book-Coach-3-en

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Ultimate CS2 Coach — Part 3: Program, UI, Tools, and Build

Topics: Complete database schema (three-tier storage, SQLite WAL, SQLModel ORM), Training regime and maturity gates (CALIBRATING/LEARNING/MATURE), Loss function catalog, Complete Program Logic (from launch to advice): Session Engine, Digester, Teacher, Hunter, Pulse; Qt/PySide6 desktop interface (13 screens, ViewModels, Qt Signals); Ingestion pipeline (demo and pro); Unified control console; New user onboarding; Storage architecture; Playback engine and tactical viewer; Spatial data and maps; Observability and logging; Reporting; Quotas and limits; Fault tolerance; Complete user journey (4 flows); Tool suite (validation and diagnostics); Test suite; Pre-commit hooks; Build, packaging, deployment; Alembic migrations; HLTV sync service; RASP Guard; MatchVisualizer; Runtime configuration files; Root-level entry points; Glossary.

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This document is the continuation of **Part 2** — *Services, Analysis, and Database*, which covers coaching services, analysis engines, knowledge, and the Database. Part 3 drops down to the program level: how the entire system boots, how the daemons orchestrate, how the desktop interface makes coaching visible, how the code is tested, validated, and deployed.

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9. Database schema and data lifecycle

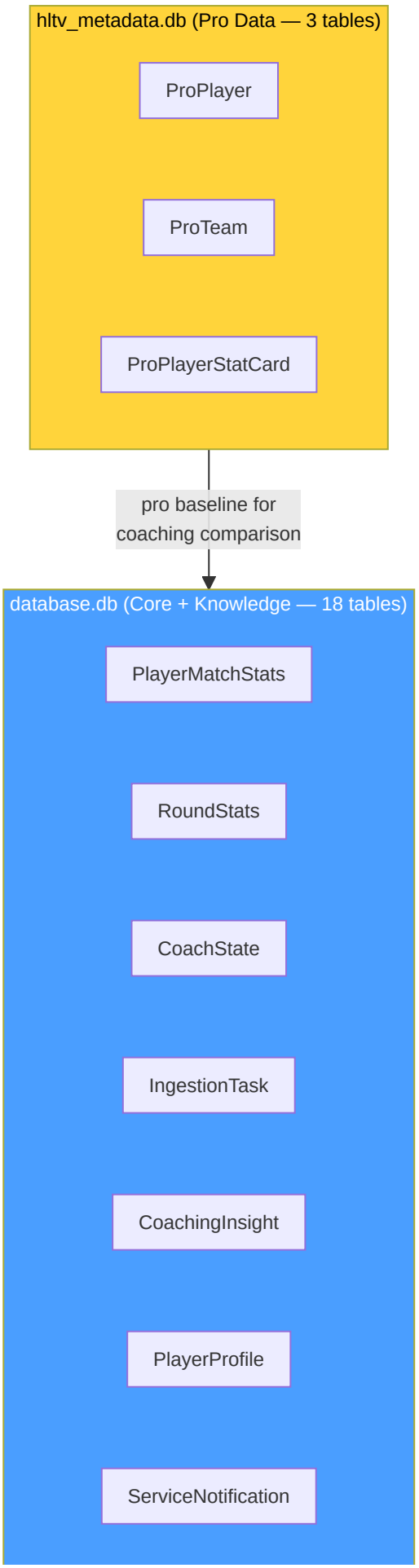
The project uses **SQLModel** (Pydantic + SQLAlchemy) with SQLite (WAL mode) and a specialized **three-tier storage architecture**. In total **24 SQLModel tables** (`backend/storage/db_models.py` : 21 monolith+HLTV tables; `backend/storage/match_data_manager.py` : 3 per-match tables) distributed across 3 storage tiers:

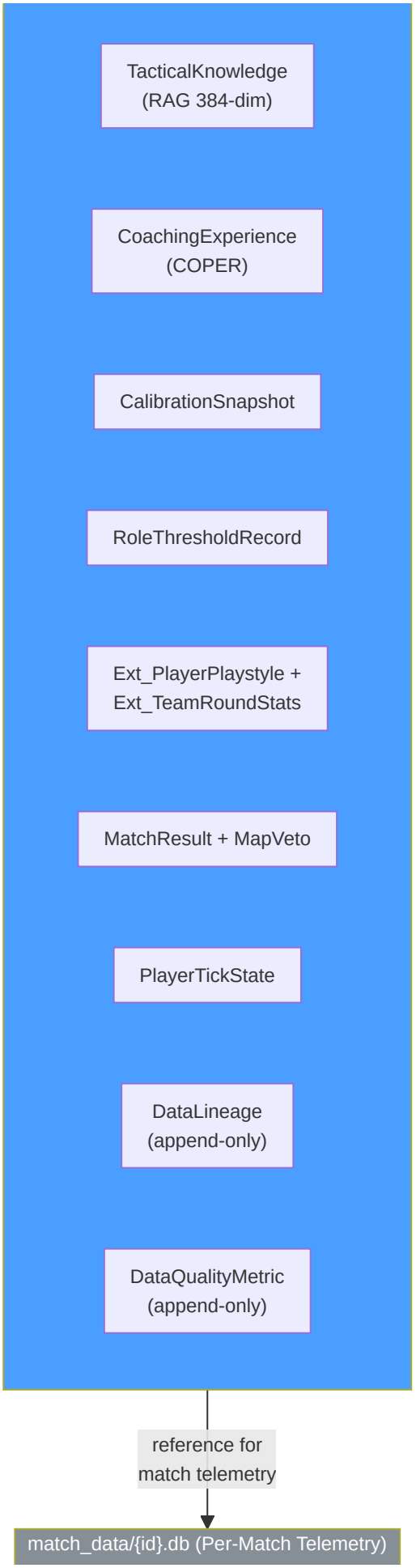
1. **`database.db`** — Main monolithic application database (**18 tables**, explicitly listed in `database.py:_MONOLITH_TABLES` lines 54-73). Contains all core tables: player statistics (`PlayerMatchStats`), coach state (`CoachState`), ingestion tasks (`IngestionTask`), coaching insights (`CoachingInsight`), user profiles (`PlayerProfile`), system notifications (`ServiceNotification`), RAG base (`TacticalKnowledge`), COPER experience bank

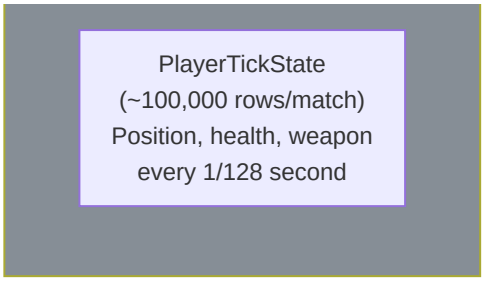
(`CoachingExperience`), match results (`MatchResult` , `MapVeto`), calibrations (`CalibrationSnapshot`), role thresholds (`RoleThresholdRecord`), provenance tracking (`DataLineage` , `DataQualityMetric`), and extended tables for team round (`Ext_TeamRoundStats`) and playstyle (`Ext_PlayerPlaystyle`), plus archival `RoundStats` and `PlayerTickState` .

2. **`hltv_metadata.db`** — Professional metadata database (**3 tables**, `database.py:_HLTV_TABLES` lines 78-82): pro player profiles (`ProPlayer` , `ProTeam`) and stat cards (`ProPlayerStatCard`). Kept separate from the monolith because it is written by a separate process (HLTV sync service) to eliminate WAL contention with the session engine daemons.
3. **`match_data/{id}.db`** — Per-match telemetry database (**3 tables**, defined in `match_data_manager.py` : `MatchTickState` :110, `MatchEventState` :190, `MatchMetadata` :242). Each match has its own dedicated SQLite file containing tick-by-tick data (~100,000 rows per match). Managed by `MatchDataManager` . This separation solves the "Telemetry Cliff" problem — it prevents the monolith database from growing indefinitely with high-frequency data.

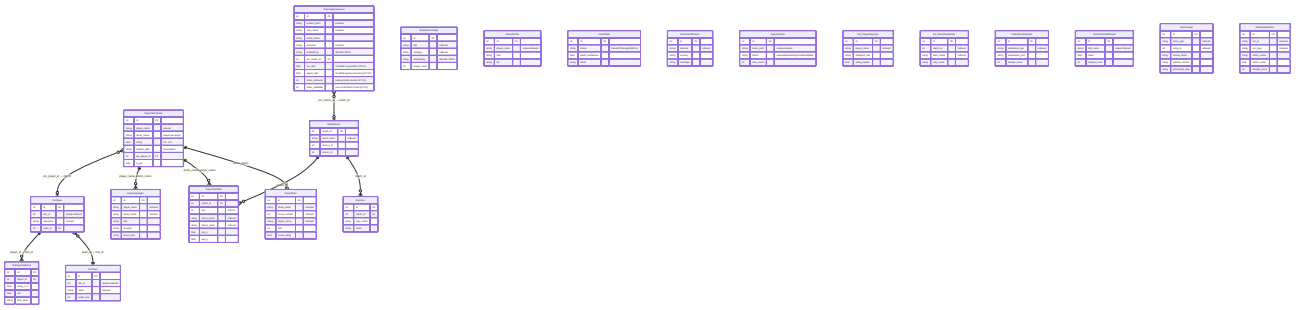
This three-tier separation ensures that the session engine's intensive write operations (demo ingestion, ML training → `database.db`) do not contend for WAL locks with HLTV scraping in a separate process (`hltv_metadata.db`), and that high-frequency per-match telemetry does not bloat the monolith (`match_data/{id}.db`).


hltv_metadata.db (Pro Data — 3 tables)
ProPlayer
ProTeam
ProPlayerStatCard
pro baseline for coaching comparison
database.db (Core + Knowledge — 18 tables)
PlayerMatchStats
RoundStats
CoachState
IngestionTask
CoachingInsight
PlayerProfile
ServiceNotification


TacticalKnowledge
(RAG 384-dim)
CoachingExperience
(COPER)
CalibrationSnapshot
RoleThresholdRecord
Ext_PlayerPlaystyle +
Ext_TeamRoundStats
MatchResult + MapVeto
PlayerTickState
DataLineage
(append-only)
DataQualityMetric
(append-only)
reference for
match telemetry
match_data/{id}.db (Per-Match Telemetry)



Analogy: The database is the system's **archive**: every piece of information has a specific drawer and folder. The three-tier storage architecture is like having **three specialized archives**: the main archive (game data, tactical knowledge, coaching experiences — all in one large WAL filing cabinet), the professionals' filing cabinet (HLTV data, updated by a separate process to avoid contention), and the **per-match folders** (tick-by-tick telemetry, each in a separate file to prevent the main archive from becoming too heavy). By separating them, the main process can write and read from the general archive while the HLTV service updates the pro filing cabinet, and telemetry data stays isolated per match, without blocking each other. SQLite in WAL mode lets multiple programs read each archive simultaneously. SQLModel combines Pydantic (for data validation: "make sure the age field is actually a number") with SQLAlchemy (for database operations: "save this to the right table"). The 21 tables are organized like a school archive: student profiles, test scores, class notes, teacher evaluations, and library books.



> **ER diagram explanation:** Each box represents a **record type** in the database. `PlayerMatchStats` is like a **report card** for each player in each match (how many kills, deaths, their rating, etc.). `PlayerTickState` is like a **frame-by-frame diary**: 128 entries per second recording exactly where the player was, how healthy they were, what direction they were looking. `RoundStats` is the **per-question breakdown**: individual evaluations for each round (kills, deaths, damage, noscope kills, flash assists, round rating), enabling detailed analyses. `CoachingExperience` is the coach's **diary**: every coaching moment, whether the advice worked, and how effective it was. `CoachingInsight` is the **actual advice** given to the player. `TacticalKnowledge` is the **textbook**: tips and strategies the coach can consult. `RoleThresholdRecord` is the **grading rubric**, the learned thresholds for classifying player roles. `CalibrationSnapshot` is the **instrument control log**, recording when the belief model was recalibrated and with how many samples. `Ext_PlayerPlaystyle` is the **external scouting report**, the playstyle metrics derived from CSV data used to train NeuralRoleHead. `ServiceNotification` is the **intercom system**, the error and event messages from background daemons shown in the UI. The lines between tables show relationships: every match record is linked to a player's profile, coaching experiences are linked to specific matches, and RoundStats is linked to PlayerMatchStats via demo_name. `DataLineage` is the **provenance log**: it traces which demo originated each entity and through which pipeline step it was processed (append-only, for full audit trail). `DataQualityMetric` is the **quality metrics panel**: it records numeric quality values for each pipeline run (e.g., percentage of samples discarded, zero-tensor fallback rate), enabling system health monitoring over time.

Data lifecycle:

Phase	Tables written	Volume
Demo ingestion	`PlayerMatchStats`, `PlayerTickState`, `MatchMetadata`	~100,000 ticks/match
Round enrichment	`RoundStats` (per-round, per-player isolation)	~30 rows/match (rounds × players)
HLTV scan	`ProPlayer`, `ProTeam`, `ProPlayerStatCard`	~500 players
CSV import	External tables via `csv_migrator.py`	~10,000 rows
Playstyle data	`Ext_PlayerPlaystyle` (from CSV for NeuralRoleHead)	~300+ players
Feature engineering	`PlayerMatchStats.dataset_split` updated	In place
RAG population	`TacticalKnowledge`	~200 articles
Experience extraction	`CoachingExperience`	~1,000 per match
Coaching output	`CoachingInsight`	~5-20 per match
Role threshold learning	`RoleThresholdRecord`	9 thresholds
Belief calibration	`CalibrationSnapshot` (after retraining)	1 per calibration run
System telemetry	`CoachState`, `ServiceNotification`, `IngestionTask`	Continuous
Provenance tracking	`DataLineage` (append-only per processed entity)	~N rows per ingested demo
Pipeline quality metrics	`DataQualityMetric` (append-only per pipeline run)	~5-10 metrics per run
Backup	Automated via `BackupManager` (7 daily rotations + 4 weekly)	Full database copy

Indexes and query optimization:

The most queried tables have strategic indexes to guarantee fast queries:

Table	Index	Columns	Query type optimized
`PlayerMatchStats`	`idx_pms_player`	`player_name`	Lookup by player
`PlayerMatchStats`	`idx_pms_demo`	`demo_name`	Lookup by match
`PlayerMatchStats`	`idx_pms_processed`	`processed_at`	Chronological sort
`RoundStats`	`idx_rs_demo_round`	`demo_name, round_number`	Lookup for specific round
`IngestionTask`	`idx_it_status`	`status`	Work queue (status=queued)
`CoachingInsight`	`idx_ci_player`	`player_name`	Insights per player
`TacticalKnowledge`	`idx_tk_category`	`category`	RAG lookup by category
`ProPlayer`	`idx_pp_name`	`nickname`	Pro lookup by name
`DataLineage`	`idx_dl_entity`	`entity_type, entity_id`	Provenance tracking per entity
`DataQualityMetric`	`idx_dqm_run`	`run_id, run_type`	Quality metrics per run

Integrity constraints:

Constraint	Tables	Enforcement
`player_name` NOT NULL	PlayerMatchStats, RoundStats	Schema-level
`demo_name` UNIQUE per player	PlayerMatchStats	Prevents duplicates
`status` CHECK IN ('queued', 'processing', 'completed', 'failed')	IngestionTask	Enum enforcement
`dataset_split` CHECK IN ('train', 'val', 'test')	PlayerMatchStats	Split validity
Foreign key demo_name	RoundStats → PlayerMatchStats	Round-match relationship

Key table details:

`PlayerMatchStats` (32 fields) — the most queried table in the system:

The PlayerMatchStats table contains all aggregated per-player, per-match statistics. It is the "report card" for every analyzed match:

Field	Type	Description
`id`	Integer PK	Unique identifier
`player_name`	String	Player name (indexed)
`demo_name`	String	Demo file name (unique per player)
`kills`, `deaths`, `assists`	Integer	Base KDA
`adr`	Float	Average Damage per Round
`kast`	Float	Kill/Assist/Survive/Trade %
`headshot_percentage`	Float	HS%
`hltv_rating`	Float	Computed HLTV 2.0 Rating
`dataset_split`	String	"train" / "val" / "test"
`is_pro`	Boolean	Pro player flag
`processed_at`	DateTime	Processing timestamp
`map_name`	String	Map played
...	...	20+ additional fields for advanced features

`TacticalKnowledge` — the RAG base:

Field	Type	Description
`id`	Integer PK	Identifier
`title`	String	Document title (indexed)
`description`	String	Description of tactical content
`category`	String	"positioning" / "economy" / "utility" / "aim" (indexed)
`map_name`	String	Specific map (indexed, optional)
`situation`	String	Situational context: "T-side pistol round", "CT retake A site"
`pro_example`	String	Reference to pro demo (optional)
`embedding`	String	384-dim JSON vector (sentence-transformers)
`created_at`	DateTime	Creation timestamp
`usage_count`	Integer	Usage counter

RAG semantic search works by computing the **cosine similarity** between the user query embedding and the precomputed embeddings of each document. The top-3 results with similarity > 0.5 are used to enrich the coaching context. The `situation` field enables contextual filtering (e.g., "T-side pistol round") before semantic search.

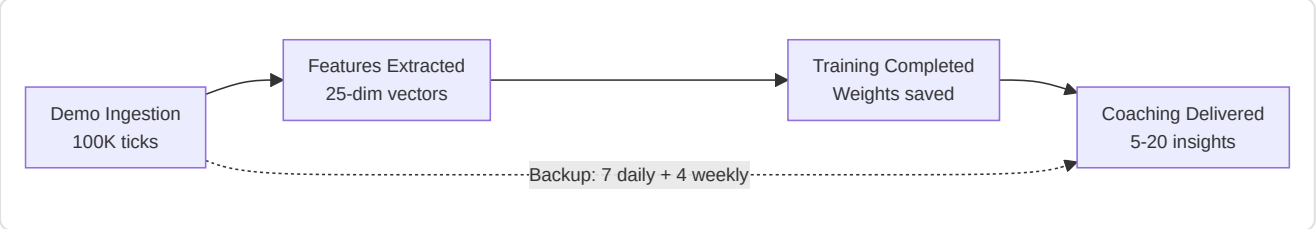
`CoachingExperience` — the COPER bank (22+ fields):

Field	Type	Description
`id`	Integer PK	Identifier
`context_hash`	String	Hash of game state for fast lookup (indexed)
`map_name`	String	Map (indexed)
`round_phase`	String	"pistol" / "eco" / "full_buy" / "force"
`side`	String	"T" / "CT"
`position_area`	String	"A-site" / "Mid" / etc. (indexed, optional)
`game_state_json`	String	Full tick snapshot (max 16KB, validated with `field_validator`)
`action_taken`	String	"pushed" / "held_angle" / "rotated" / "used_utility" / etc.
`outcome`	String	"kill" / "death" / "trade" / "objective" / "survived" (indexed)
`delta_win_prob`	Float	Win probability change from this action
`confidence`	Float	Reliability/generalizability 0.0-1.0
`usage_count`	Integer	How many times retrieved for coaching
`pro_match_id`	Integer FK	Reference to MatchResult (ON DELETE SET NULL)
`pro_player_name`	String	Reference pro player name (indexed, optional)
`embedding`	String	384-dim JSON vector for semantic search (optional)
`source_demo`	String	Source demo (optional)
`created_at`	DateTime	Creation timestamp
`outcome_validated`	Boolean	Whether the outcome was validated
`effectiveness_score`	Float	Score -1.0 to 1.0
`follow_up_match_id`	Integer	Follow-up match for tracking
`times_advice_given`	Integer	How many times the advice was given
`times_advice_followed`	Integer	How many times the advice was followed
`last_feedback_at`	DateTime	Last feedback received
`mu_skill`	Float	TrueSkill μ posterior — experience quality estimate (KT-01)
`sigma_skill`	Float	TrueSkill σ uncertainty — quality uncertainty (KT-01)
`times_retrieved`	Integer	Replay priority counter — how many times retrieved (KT-01)
`times_validated`	Integer	User confirmation counter — how many times validated (KT-01)

The COPER system uses these experiences to **learn from its own advice**: the `context_hash` field enables fast lookup of similar situations, `outcome` and `delta_win_prob` measure action effectiveness, and the feedback loop (`outcome_validated`, `times_advice_given/followed`, `effectiveness_score`) lets the system prioritize advice that has historically produced positive outcomes. The `game_state_json`

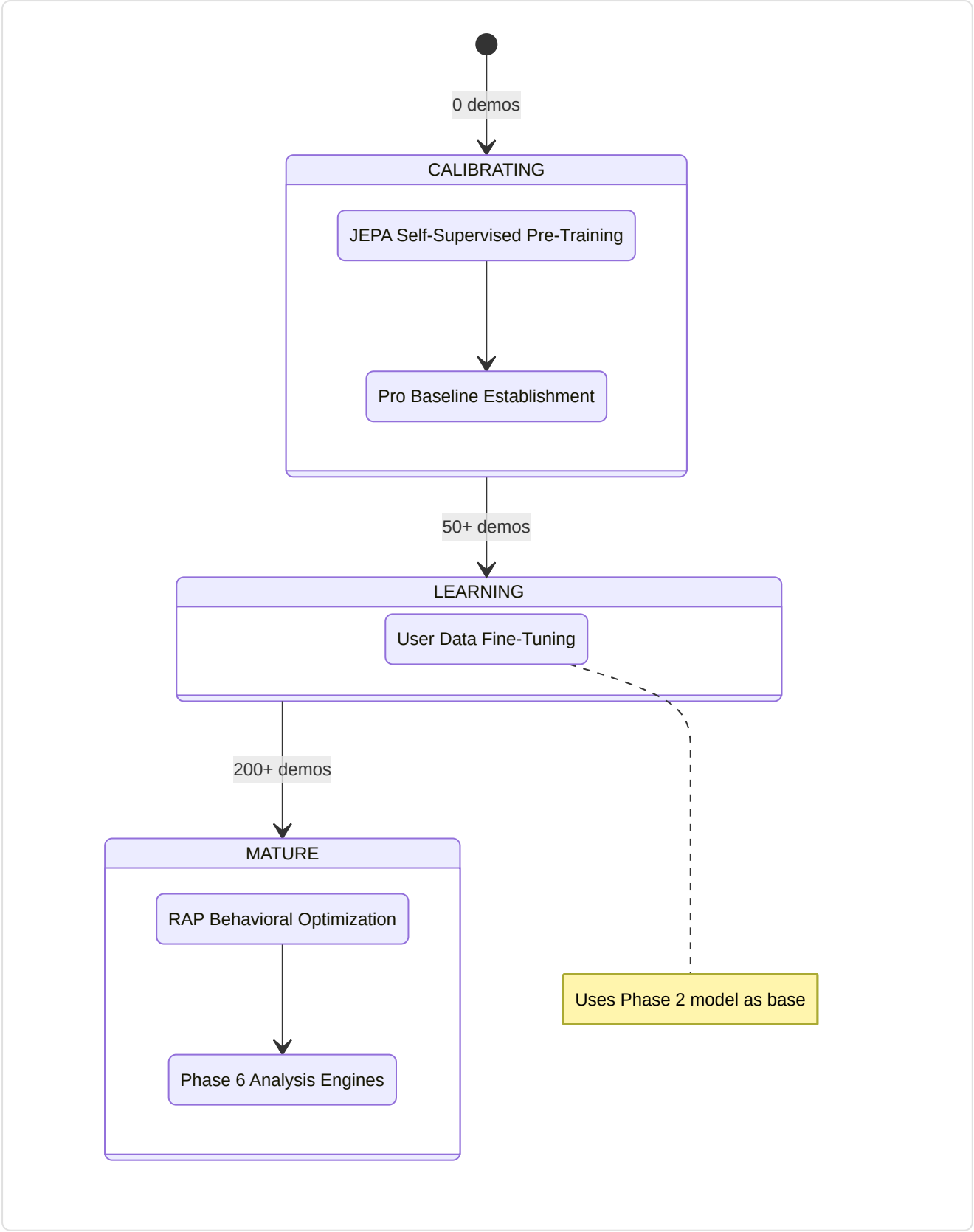
field is capped at 16KB to prevent uncontrolled database growth. The TrueSkill `mu_skill` / `sigma_skill` fields (KT-01) provide a Bayesian estimate of experience quality, used for replay prioritization: `confidence_score = mu_skill - κ × sigma_skill`. The `times_retrieved` counter implements replay priority, while `times_validated` tracks user confirmations for semantic CRUD.

> **Analogy:** The data lifecycle shows **how information flows through the system over time**, like tracking a package from factory to delivery. First come the match recordings (100,000 data points per match!). Then pro player stats are scraped from HLTV (like downloading a sports almanac). External CSV files are imported (like obtaining historical data). The AI processes everything, generates coaching advice, learns role thresholds, and constantly records system health. Backups run automatically: 7 daily copies and 4 weekly copies, like saving your homework both on the computer and on a USB drive, just in case.



10. Training regime and maturity gates

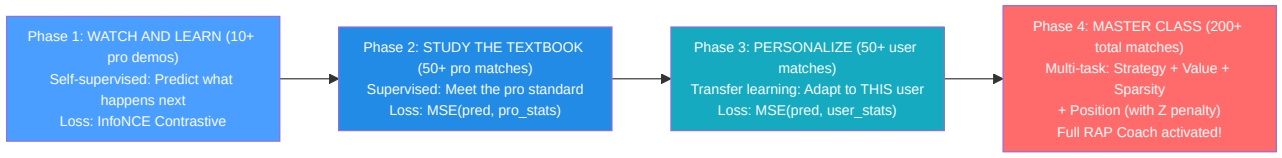
> **Analogy:** The training regime is the **complete school curriculum**, from kindergarten to graduation. A student (the AI model) starts with zero knowledge and gradually learns through 4 phases, unlocking more advanced courses as they prove themselves. Maturity gates are like **grading requirements**: you can't take the AP Physics exam (RAP Optimization) until you've passed Basic Math (JEPA Pre-Training), Algebra (Pro Baseline), and Pre-Calculus (User Fine-Tuning). Each gate checks: "Have you studied enough demos to be ready for the next level?"



Data requirements per phase:

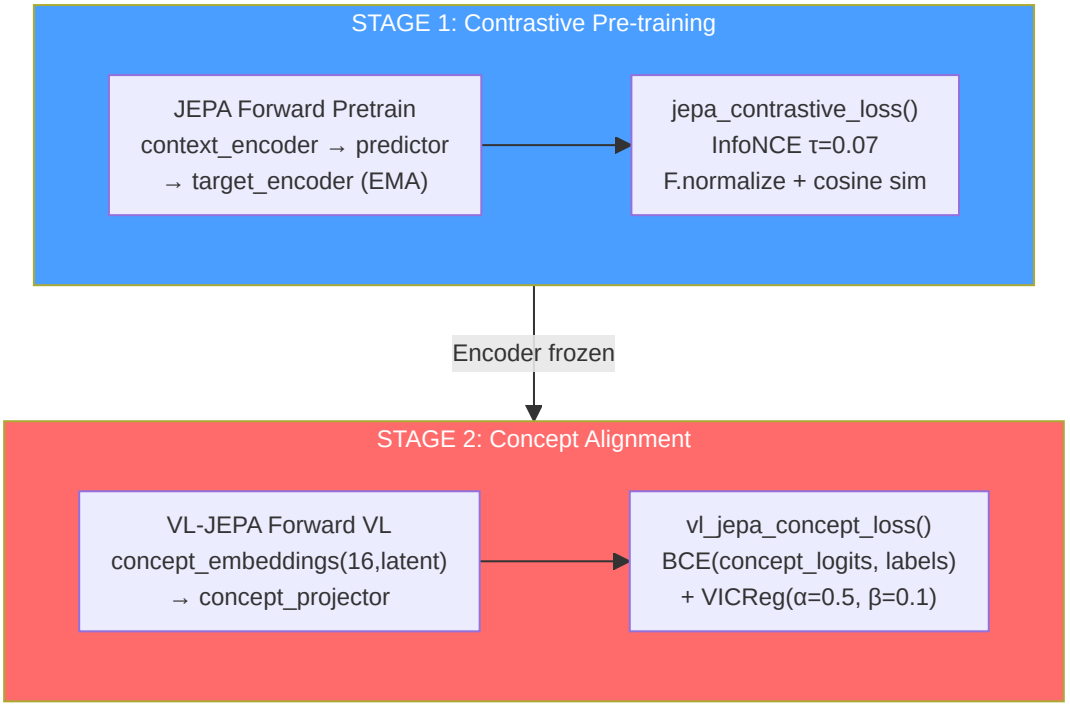
Phase	Minimum data	Training type	Primary loss
1. JEPA Pre-training	10 pro demos	Self-supervised (InfoNCE)	Contrastive with in-batch negatives
2. Pro baseline	50 pro matches	Supervised	MSE(pred, pro_stats)
3. User fine-tuning	50 user matches	Supervised (transfer)	MSE(pred, user_stats)
4. RAP optimization	200 total matches	Multi-task	Strategy + Value + Sparsity + Position

> **Analogy:** Phase 1 is like **watching cooking shows**: the model learns patterns simply by observing (self-supervised, no labels needed). Phase 2 is like **a cooking school with a textbook**: "Here's how a professional makes pasta" (supervised with professional data). Phase 3 is like **cooking for your family**: "Your family prefers it spicy, so let's adapt the recipe" (fine-tuning on user data). Phase 4 is **master chef training**: learning to balance flavor, presentation, timing, and nutrition simultaneously (multi-task: strategy + value + sparsity + position). You need at least 10 cooking shows to start, 50 recipes to learn from, and 200 total dishes prepared before you graduate.



VL-JEPA Two-Stage Protocol (Concept Alignment):

When VL-JEPA is active, Phases 1-2 are extended with a **two-stage protocol** that aligns latent representations with the 16 coaching concepts:



Stage	What is trained	What is frozen	Loss	Purpose
1	Context encoder, predictor	Target encoder (EMA)	InfoNCE (τ=0.07)	General latent representations
2	Concept embeddings, concept projector, concept temperature	Encoder (optional fine-tuning)	BCE + VICReg diversity	Alignment with 16 coaching concepts

The 16 Coaching Concepts (taxonomy):

Index	Concept	Category	Description
0	Positioning Quality	Positioning	Position quality relative to context
1	Trade Readiness	Positioning	Readiness for trade kill
2	Rotation Speed	Positioning	Rotation speed between sites
3	Utility Usage	Utility	Grenade use frequency and quality
4	Utility Effectiveness	Utility	Effectiveness of utilities used
5	Decision Quality	Decision	In-game decision quality
6	Risk Assessment	Decision	Pre-action risk evaluation
7	Engagement Timing	Engagement	Timing of engagements
8	Engagement Distance	Engagement	Optimal engagement distance
9	Crosshair Placement	Engagement	Pre-peek crosshair placement
10	Recoil Control	Engagement	Recoil control
11	Economy Management	Decision	Team economy management
12	Information Gathering	Decision	Information gathering (peek, info utility)
13	Composure Under Pressure	Psychology	Composure under pressure
14	Aggression Control	Psychology	Aggression control
15	Adaptation Speed	Psychology	Speed of adaptation to opponent meta

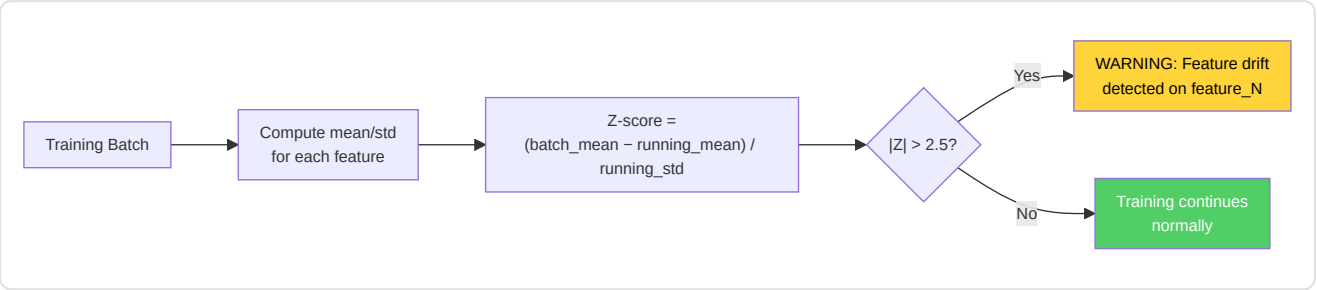
> **Analogy for the 16 Concepts:** The 16 coaching concepts are like the **16 subjects of a complete school curriculum**. Three subjects cover **geography** (positioning — where you are). Two cover **chemistry** (utility — how you use your tools). Five cover **strategy** (decision and economy — the choices you make). Four cover **gymnastics** (engagement — physical skills: aim, recoil, timing). Three cover **psychology** (composure, aggression, adaptation). Stage 2 of VL-JEPA teaches the model to "understand" these 16 subjects and give a grade for each in every player action.

AdamW + CosineAnnealing (JEPA Trainer):

Hyperparameter	Value	Purpose
Optimizer	AdamW	Weight decay separated from gradients
Learning rate	1e-4 (default)	Initial learning rate
Weight decay	0.01	L2 regularization
Scheduler	CosineAnnealingLR	Cosine LR decay down to 0
EMA decay	0.996	Target encoder momentum update
Gradient clip	1.0	Gradient explosion prevention

DriftMonitor (z_threshold=2.5):

The `DriftMonitor` integrated in the JEPA Trainer monitors **feature drift** during training. If a feature's Z-score exceeds 2.5, it emits a warning indicating possible distribution shift:



Retraining trigger: The Teacher daemon monitors growth in the pro demo count; it triggers retraining when `count ≥ last_count × 1.10`.

> **Analogy:** The retraining trigger is like a **school that updates its curriculum when enough new textbooks arrive**. The Teacher daemon (a background process) constantly checks: "How many pro demos do we have now?". When the count grows 10% or more since the last training, it says: "We have enough new material: it's time to retrain the model so it stays current with the evolving pro meta."

JEPAPretrainDataset (`jepa_train.py`):

The JEPA pre-training dataset uses **temporal windows** to create context-target pairs:

Parameter	Value	Purpose
`context_len`	10 ticks	Context window length (input)
`target_len`	10 ticks	Target window length (to predict)
Gap	0-5 ticks (random)	Variable distance between context and target
Batch size	32	Number of pairs per batch

> **Dataset Analogy:** The pre-training dataset is like a **speed reading exercise**. The model is shown one page of the book (10 ticks of context) and then asked: "What does the next page say?" (10 target ticks). The variable gap makes the exercise harder — sometimes the next page is immediately after, sometimes they are separated by a few blank pages. This forces the model to learn general patterns, not specific sequences.

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11. Loss function catalog

Analogy: Loss functions are the **test scores** the AI tries to minimize. Every model has its own type of test. A lower score means better performance, the opposite of school grades! Think of each loss function as a specific test question: "How close was your prediction to the correct answer?" (MSE), "Did you pick the correct answer among multiple options?" (InfoNCE/BCE), "Did you use too many resources?" (Sparsity). The table below is like the **complete exam calendar**: every test, for every model, with the exact grading formula.

Model	Loss name	Formula	Purpose
JEPA	InfoNCE Contrastive	`−log(exp(sim(pred, target)/τ) / Σ exp(sim(pred, neg_i)/τ))`, τ=0.07, `F.normalize` before cosine similarity	Aligning context predictions with target embeddings
JEPA	Fine-tuning	`MSE(coaching_head(Z_ctx), y_true)`	Supervised coaching score
AdvancedCoachNN	Supervised	`MSELoss(MoE_output, y_true)`	Match-level training
RAP	Strategy	`MSELoss(advice_probs, target_strat)`	Correct tactical recommendation
RAP	Value	`0.5 × MSE(V(s), true_advantage)`	Accurate advantage estimation
RAP	Sparsity	`L1(gate_weights)`	Expert specialization
RAP	Position	`MSE(xy) + 2× MSE(z)`	Optimal positioning with Z-axis penalty
WinProb	Prediction	`BCEWithLogitsLoss(pred, outcome)`	Round outcome prediction
NeuralRoleHead	KL-Divergence	`KLDivLoss(log_softmax(pred), target)` with label smoothing ε=0.02	Role probability distribution match
VL-JEPA	Concept Alignment	`BCE(concept_logits, concept_labels)` + `VICReg(concept_diversity)`	Visual-language concept grounding

> **Analogy for key loss functions: InfoNCE** is like a multiple-choice test: "Here are 32 possible answers: which is the correct one?" The model gets a higher score for picking the right one AND for being confident about it. **MSE** (Mean Squared Error) is like measuring how far your dart is from the bullseye: closer = lower loss. **BCE** (Binary Cross-Entropy) is like a true/false quiz: "Did your team win? Yes or no?". **Sparsity Loss** is like a teacher saying "Use fewer words in your essay": it encourages the model to activate fewer experts, making it more efficient and interpretable. **Position Loss with 2x Z penalty** is like saying "missing left or right is bad, but falling off a cliff (wrong floor) is twice as bad". **KL-Divergence** is like comparing two rankings: "Does your role ranking match the real one?" — it measures how much the predicted distribution deviates from the target.

Detail: InfoNCE Contrastive Loss (JEPA)

InfoNCE is the main loss for JEPA pre-training. Its purpose is to align context predictions with target embeddings while simultaneously pushing away negatives (other samples in the batch):

```
L_InfoNCE = -log( exp(sim(pred, target⁺) / τ) / Σᵢ exp(sim(pred, targetᵢ) / τ) )
```

Component	Value	Role
`sim()`	Cosine similarity after `F.normalize`	Similarity measure [-1, +1]
`τ` (temperature)	0.07	Distribution sharpness — low values = more selective
`target⁺`	The correct target embedding for this context	The "positive" — the correct answer
`targetᵢ`	All embeddings in the batch	In-batch negatives — the wrong answers
Batch size	32	Number of negatives = batch_size - 1 = 31

> **InfoNCE Analogy:** Imagine you're in a room with 32 people. You're shown a photo (context) and you have to find the right person (positive target) among the 32. Temperature τ=0.07 is like the **sharpness of the glasses**: low values mean very precise glasses — you must be very sure of your choice to get a good score. High values mean blurry glasses — even an approximate choice is fine. JEPA's low τ forces the model to learn very precise representations.

Detail: VL-JEPA Concept Loss (VICReg Components)

The VL-JEPA concept alignment loss combines two components:

```
L_concept = BCE(concept_logits, concept_labels) + α·VICReg_diversity
```

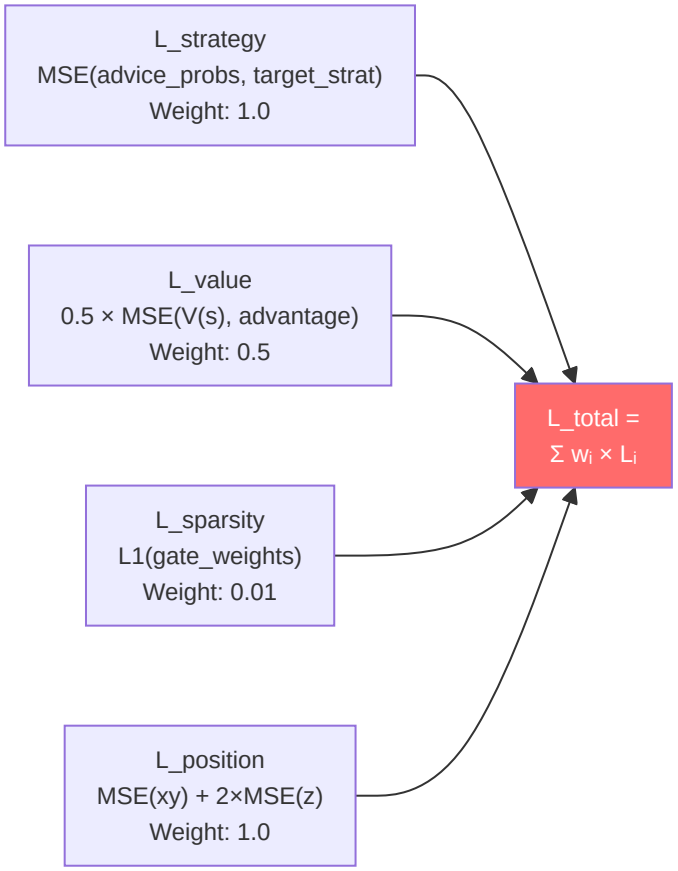
Where `VICReg_diversity` is composed of:

VICReg Term	Formula	Weight	Purpose
Variance	`max(0, γ - std(z))` for each dimension	α=0.5	Prevents representation collapse — each dimension must vary
Covariance	$\Sigma_{i \neq j} \, cov(z_i, z_j)^2$	β=0.1	Decorrelation — each dimension must capture different information

> **VICReg Analogy: Variance** is like a teacher saying "Each of you must have a different opinion — don't all copy the same answer!" (prevents collapse where all concept embeddings become identical). **Covariance** is like saying "Each student must specialize in a different subject — we don't need 16 history experts and zero math experts!" (forces diversity across dimensions). Together, they ensure the 16 concept embeddings are both **distinct from each other** and **individually meaningful**.

Detail: RAP Multi-Task Loss

The RAP Coach combines 4 losses into a total weighted loss:

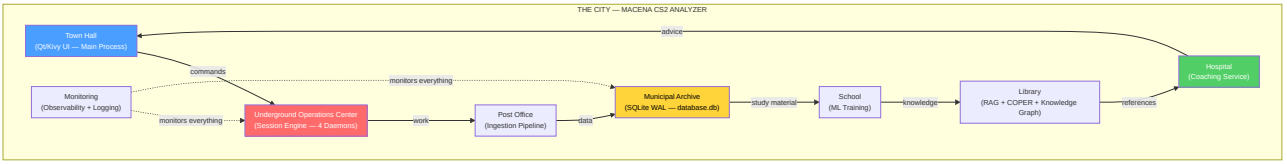


The 2× Z penalty in position loss reflects the fact that in CS2 getting the **floor** wrong (upper vs lower in Nuke/Vertigo) is much worse than getting horizontal position wrong. A 100-unit error on the Z axis (wrong floor) is strategically catastrophic, while the same error on X/Y could be irrelevant.

12. Complete Program Logic — From Launch to Advice

This chapter documents the **complete logic** of Macena CS2 Analyzer, from the moment the user launches the application to when they receive coaching advice. Unlike the previous chapters that focus on AI subsystems, here it is explained how **every component of the program** works together: the desktop interface, the quad-daemon architecture, the ingestion pipeline, the storage system, tactical playback, observability, and the application lifecycle.

> **Analogy:** If chapters 1-11 describe the **individual organs** of a human body (brain, heart, lungs, liver), this chapter describes the **entire body in action**: how it wakes up in the morning, how it breathes, walks, eats, thinks, and speaks. Understanding the organs is essential, but understanding how they work together is what gives the system life. Imagine Macena CS2 Analyzer as a **small city**: it has a town hall (the main Kivy process), an underground operations center (the Session Engine with its 4 daemons), a municipal archive (the SQLite database), a post office (the ingestion pipeline), a school (the ML training system), a library (the RAG/COPER knowledge system), a hospital (the coaching service), and a public health monitoring system (observability). This chapter walks you through each building and shows how the citizens (the data) move from one place to another.



12.1 Entry Point and Startup Sequence

The system has **two main entry points** (Qt primary, Kivy legacy) and three utility entry points:

#	Entry Point	Command	Role
1	**Qt (primary)**	`python -m Programma_CS2_RENAN.apps.qt_app.app`	PySide6 desktop UI
2	Kivy (legacy)	`python -m Programma_CS2_RENAN.main`	Kivy/KivyMD desktop UI
3	Session Engine	`python -m Programma_CS2_RENAN.backend.console`	Headless backend
4	Headless Validator	`python -m Programma_CS2_RENAN.tools.headless_validator`	CI/CD validation
5	HLTV Sync	`python -m Programma_CS2_RENAN.tools.hltv_sync`	Pro data scraping

12.1.1 Qt Entry Point (Primary) — `apps/qt_app/app.py`

File: `Programma_CS2_RENAN/apps/qt_app/app.py`

The Qt startup sequence follows a more modern approach based on **QApplication** and **signal/slot** instead of the Kivy event loop:

1. **High-DPI setup** — Enables automatic scaling for high-density displays
2. **QApplication** — Creates the Qt application instance with args handling
3. **Theme and fonts** — Registers the 3 themes (CS2, CSGO, CS1.6) via QSS and the custom fonts
4. **MainWindow** — Builds `QMainWindow` with sidebar + `QStackedWidget` (14 screens)
5. **Signal wiring** — Connects Qt signals between sidebar, screens, and backend
6. **First-run gate** — If `SETUP_COMPLETED=False`, shows the wizard; otherwise the home
7. **Backend console boot** — Launches the Session Engine as a subprocess
8. **Window show** — Shows the window and starts the Qt event loop
9. **CoachState polling** — Qt timer for periodic state updates

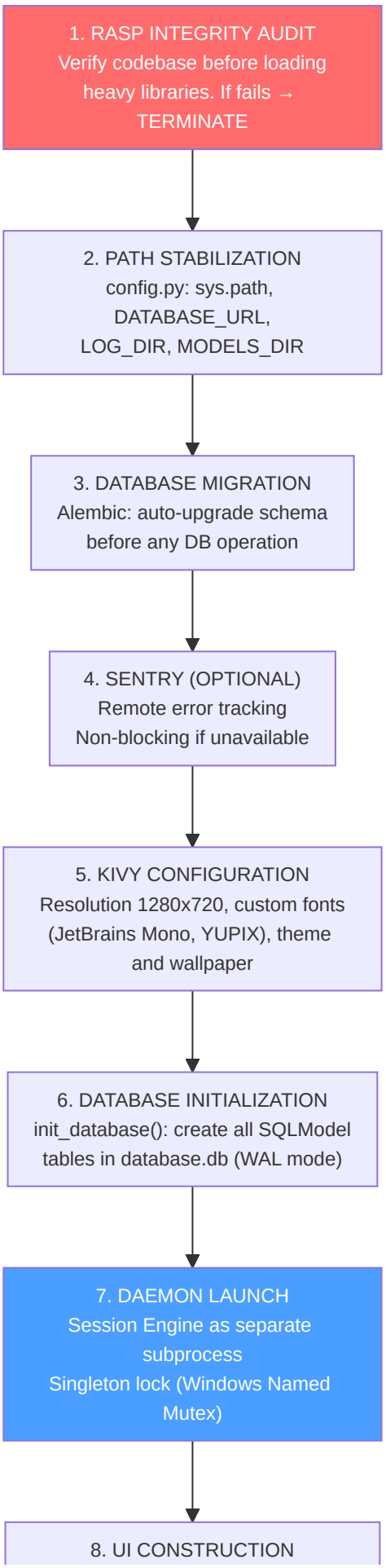
> **Analogy:** The Qt startup is like **turning on a modern car with an electronic start/stop button**. A single button (QApplication) starts the sequence: the onboard computer configures the display (High-DPI), loads the dashboard theme (QSS), assembles all the instruments (14 screens in the QStackedWidget), connects the sensors (signal wiring), checks whether it's the first boot (first-run gate), starts the engine (backend console), and finally lights up the dashboard (window show).

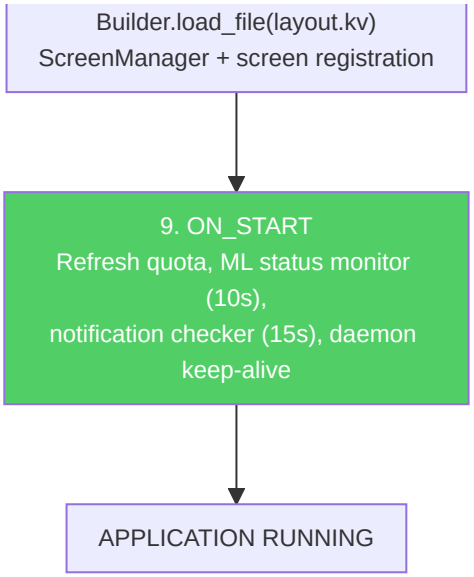
12.1.2 Kivy Entry Point (Legacy) — `main.py`

File: `Programma_CS2_RENAN/main.py`

When the user launches the application via the legacy entry point, `main.py` orchestrates a strictly ordered **9-phase startup sequence**. Each phase must complete successfully before the next can begin. If a critical phase fails, the application terminates with an explicit message — never silently.

> **Analogy:** The program startup is like the **pre-flight checklist of an airplane**. Before the plane can take off, the pilot (main.py) must complete a series of checks in order: verify the fuselage integrity (RASP audit), set up the instruments (path configuration), check the fuel (database migration), start the engines (Kivy initialization), load the passengers (screen registration), engage the autopilot (daemon launch), and finally take off (show the interface). If one check fails — for example, insufficient fuel (corrupt database) — the flight is canceled, they don't try to take off hoping it goes well.


1. RASP INTEGRITY AUDIT
Verify codebase before loading heavy libraries. If fails → TERMINATE
2. PATH STABILIZATION
config.py: sys.path, DATABASE_URL, LOG_DIR, MODELS_DIR
3. DATABASE MIGRATION
Alembic: auto-upgrade schema before any DB operation
4. SENTRY (OPTIONAL)
Remote error tracking
Non-blocking if unavailable
5. KIVY CONFIGURATION
Resolution 1280x720, custom fonts (JetBrains Mono, YUPIX), theme and wallpaper
6. DATABASE INITIALIZATION
init_database(): create all SQLModel tables in database.db (WAL mode)
7. DAEMON LAUNCH
Session Engine as separate subprocess
Singleton lock (Windows Named Mutex)
8. UI CONSTRUCTION



Critical phase details:

Phase	Component	What it does	Failure consequence
1	`integrity.py` (RASP)	Verifies source file hashes against manifest	Immediate termination — possible tampering
2	`config.py`	Stabilizes `sys.path`, defines all path constants	Cascading import errors
3	`db_migrate.py`	Runs pending Alembic migrations	Incompatible schema — crash on DB ops
5	Kivy `Config`	Sets `KIVY_NO_CONSOLELOG`, registers fonts, loads `.kv`	UI not renderable
6	`database.py`	`create_all()` on SQLite engine with `check_same_thread=False`	No persistence possible
7	`lifecycle.py`	Launches subprocess with correct PYTHONPATH, checks mutex	No background automation

> **Analogy:** Failure consequences are arranged like **domino tiles**: if phase 2 (paths) fails, phases 3-9 will all fall because none of them know where to find the database, models, or logs. If phase 6 (database) fails, phases 7-9 will apparently work but won't be able to save or retrieve anything — like a restaurant that opened its doors but forgot to turn on the stoves.

12.2 Lifecycle Management (`lifecycle.py`)

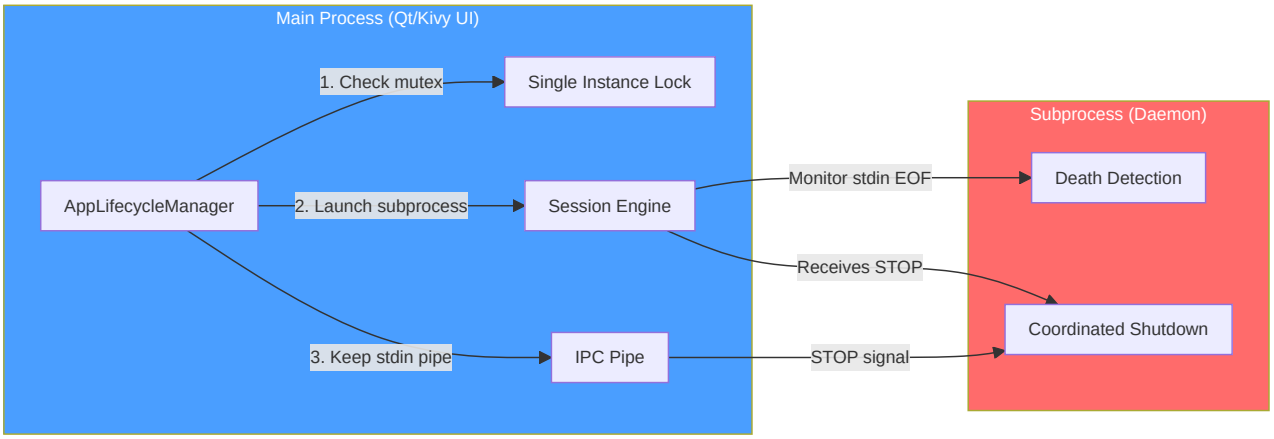
File: `Programma_CS2_RENAN/core/lifecycle.py`

The `AppLifecycleManager` is a **Singleton** that manages the lifecycle of the entire application: from guaranteeing that only one instance is active, to launching the daemon subprocess, to coordinated shutdown.

> **Analogy:** The Lifecycle Manager is like the **director of a theater**. Before the show, he verifies that no other shows are running in the same room (Single Instance Lock). Then he hires the stage director (Session Engine daemon) who will work behind the scenes. During the show, the director is always present in case of emergency. At the end, the director makes sure everyone leaves the theater in order: first the stage director finishes his work, then the lights go out, then the doors close.

Main mechanisms:

Mechanism	Implementation	Purpose
Single Instance Lock	Windows Named Mutex / file lock on Linux	Prevents multiple instances (DB corruption)
Daemon Launch	`subprocess.Popen(session_engine.py)` with PYTHONPATH	Separate process for heavy work
Parent Death Detection	The daemon monitors EOF on `stdin`	If the main process dies, the daemon shuts down
Graceful Shutdown	Sends "STOP" via stdin → daemon terminates within 5s	No data loss or zombie tasks
Status Polling	UI queries `CoachState` every 10s	Daemon state update without direct IPC
Error Recovery	If daemon dies, error logged + ServiceNotification	User informed, no silent crash
Keep-Alive	UI checks heartbeat every 15s	If heartbeat > 30s stale → warning "Daemon not responding"



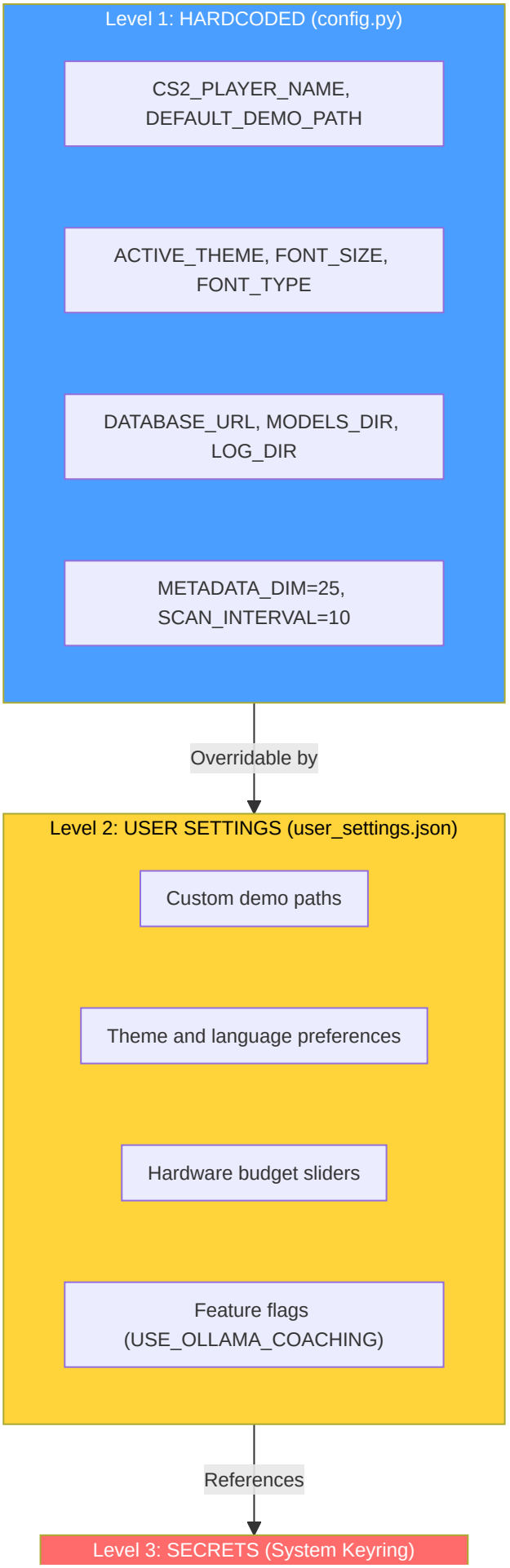
> **Analogy:** The `stdin` pipe is like a **telephone line** between the director and the stage director. As long as the line is connected, the stage director knows the director is still present. If the line suddenly breaks (EOF — the main process crashes), the stage director realizes the show is over and closes everything in an orderly fashion. If the director wants to end normally, they send the "STOP" message through the line and wait for the stage director to confirm they have finished.

12.3 Configuration System (`config.py`)

File: `Programma_CS2_RENAN/core/config.py`

The system uses **three configuration levels**, each with a different level of persistence and security:

> **Analogy:** The three configuration levels are like the **three layers of medieval armor**. The inner layer (hardcoded) is the base armor that never changes — the foundations of the system. The middle layer (JSON) is the customizable chainmail — the user can adjust it as they prefer. The outer layer (Keyring) is the helmet with visor — it protects the most precious secrets (API keys) in an OS vault, inaccessible to prying eyes.


Level 1: HARDCODED (config.py)
CS2_PLAYER_NAME, DEFAULT_DEMO_PATH
ACTIVE_THEME, FONT_SIZE, FONT_TYPE
DATABASE_URL, MODELS_DIR, LOG_DIR
METADATA_DIM=25, SCAN_INTERVAL=10
Overridable by
Level 2: USER SETTINGS (user_settings.json)
Custom demo paths
Theme and language preferences
Hardware budget sliders
Feature flags (USE_OLLAMA_COACHING)
References
Level 3: SECRETS (System Keyring)

Critical system constants:

Constant	Value	File	Purpose
`METADATA_DIM`	25	`config.py`	Feature vector dimension — unified contract
`SCAN_INTERVAL`	10	`config.py`	Scanner scan interval (seconds)
`MAX_DEMOS_PER_MONTH`	10	`config.py`	Monthly demo upload quota
`MAX_TOTAL_DEMOS`	100	`config.py`	Lifetime total demo limit
`MIN_DEMOS_FOR_COACHING`	10	`config.py`	Threshold for full personalized coaching
`TRADE_WINDOW_TICKS`	192	`trade_kill_detector.py`	Trade kill time window (~3 seconds at 64 tick)
`HLTV_BASELINE_KPR`	0.679	`demo_parser.py`	HLTV 2.0 baseline for KPR
`HLTV_BASELINE_SURVIVAL`	0.317	`demo_parser.py`	HLTV 2.0 baseline for survival
`FOV_DEGREES`	90	`player_knowledge.py`	Simulated player field of view
`MEMORY_DECAY_TAU`	160	`player_knowledge.py`	Memory decay constant in ticks
`CONFIDENCE_ROUNDS_CEILING`	300	`correction_engine.py`	Round ceiling for max confidence
`SILENCE_THRESHOLD`	0.2	`explainability.py`	Threshold below which silence is valid action
`MIN_SAMPLES_FOR_VALIDITY`	10	`role_thresholds.py`	Minimum samples for a valid role threshold
`HALF_LIFE_DAYS`	90	`pro_baseline.py`	Temporal decay for pro data
`Z_LEVEL_THRESHOLD`	200	`connect_map_context.py`	Z threshold for floor classification

> **Constants Analogy:** Constants are like the **reference vital signs** in medicine. `METADATA_DIM=25` is like saying "normal systolic pressure is 120" — all doctors (ML models) use the same reference. `TRADE_WINDOW_TICKS=192` is like the "golden hour" in trauma medicine: if a teammate is killed and you kill their killer within 3 seconds, it's a trade kill. `SILENCE_THRESHOLD=0.2` is like "primum non nocere" (first, do no harm): if the coach has nothing significant to say, it's better to be silent than to give useless advice.

Path architecture:

The system handles paths with particular attention to **Windows/Linux portability**. The heart is `BRAIN_DATA_ROOT`: a user-configurable directory that contains models, logs, and derived data. If it doesn't exist, the system falls back to the project folder.

Path	Contents	Configurable
`DATABASE_URL`	Main monolithic database (`database.db`)	No — always in the project folder
`BRAIN_DATA_ROOT`	Root for derived data (models, logs)	Yes — via `user_settings.json`
`MODELS_DIR`	Model checkpoints `.pt`	Derived from `BRAIN_DATA_ROOT`
`LOG_DIR`	Application log files	Derived from `BRAIN_DATA_ROOT`
`MATCH_DATA_PATH`	Per-match database (`match_XXXX.db`)	Derived from `BRAIN_DATA_ROOT`
`DEFAULT_DEMO_PATH`	User demo folder	Yes — via UI Settings
`PRO_DEMO_PATH`	Pro demo folder	Yes — via UI Settings

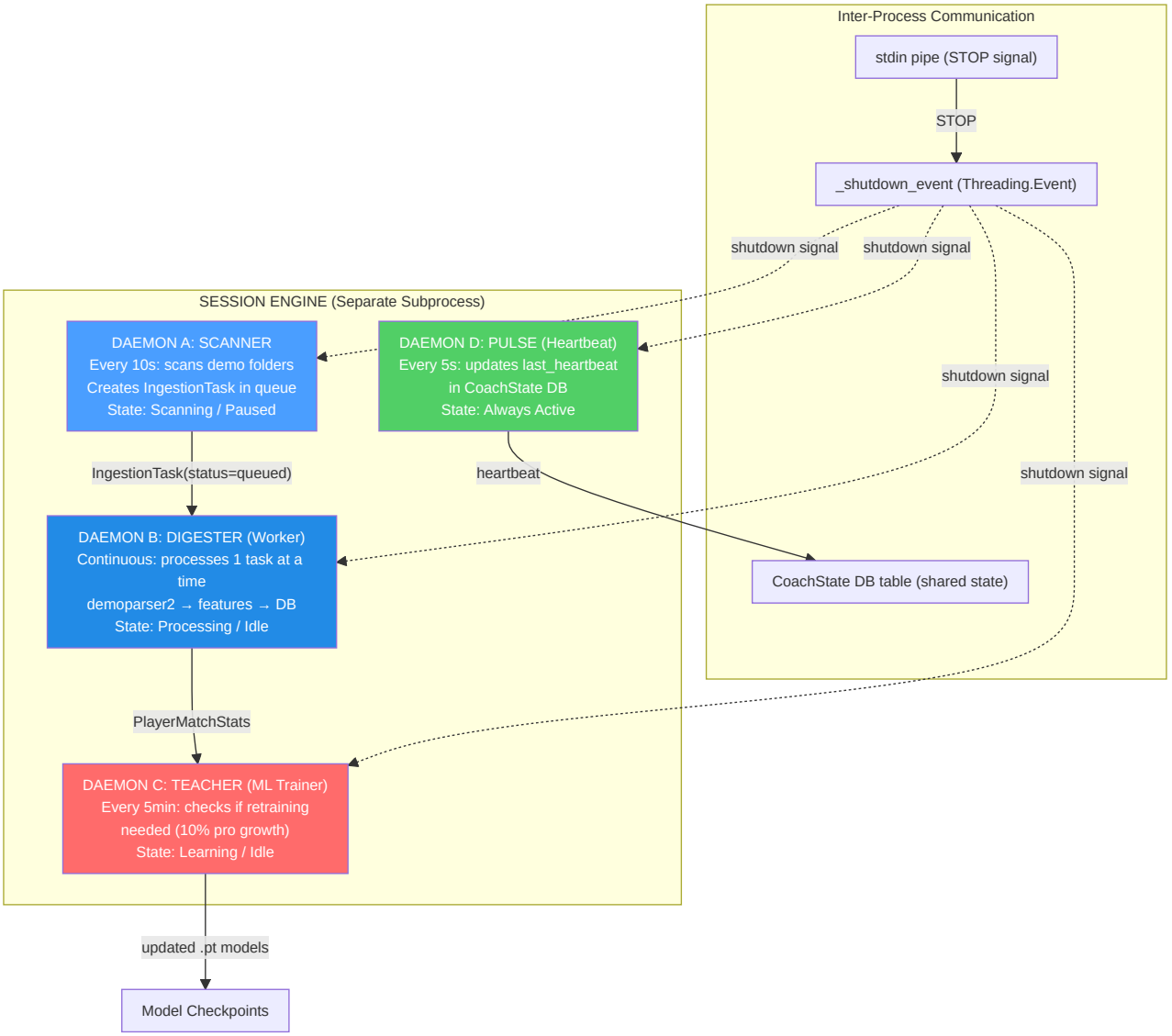
> **Analogy:** `BRAIN_DATA_ROOT` is like the **coach's brain home address**. You can move it to a larger disk (external SSD) simply by changing the address, and everything else — models, logs, match data — will follow automatically. The main database (`database.db`), on the other hand, is like the **municipal registry**: it always stays in the same place for integrity reasons.

12.4 Session Engine — Quad-Daemon Architecture (`session_engine.py`)

File: `Programma_CS2_RENAN/core/session_engine.py`

The Session Engine is the **beating heart** of the system's automation. It lives as a separate subprocess and hosts **4 daemon threads** that work in parallel, each with a well-defined responsibility. This design completely separates the heavy work (demo parsing, ML training) from the user interface, guaranteeing that the Kivy GUI always remains responsive.

> **Analogy:** The Session Engine is like an **underground nuclear power plant** that powers the entire city. It has 4 reactors (daemons), each producing a different type of energy. Reactor 1 (Scanner) is the **radar scanner**: it constantly scans the horizon looking for new demos to process. Reactor 2 (Digester) is the **refinery**: it takes raw demos and transforms them into structured data. Reactor 3 (Teacher) is the **research lab**: it uses refined data to train the coach's brain. Reactor 4 (Pulse) is the **heart monitor**: it emits a beat every 5 seconds to confirm the plant is alive. If the surface city (Kivy GUI) is destroyed by an earthquake (crash), the plant detects the loss of communication and shuts down safely.



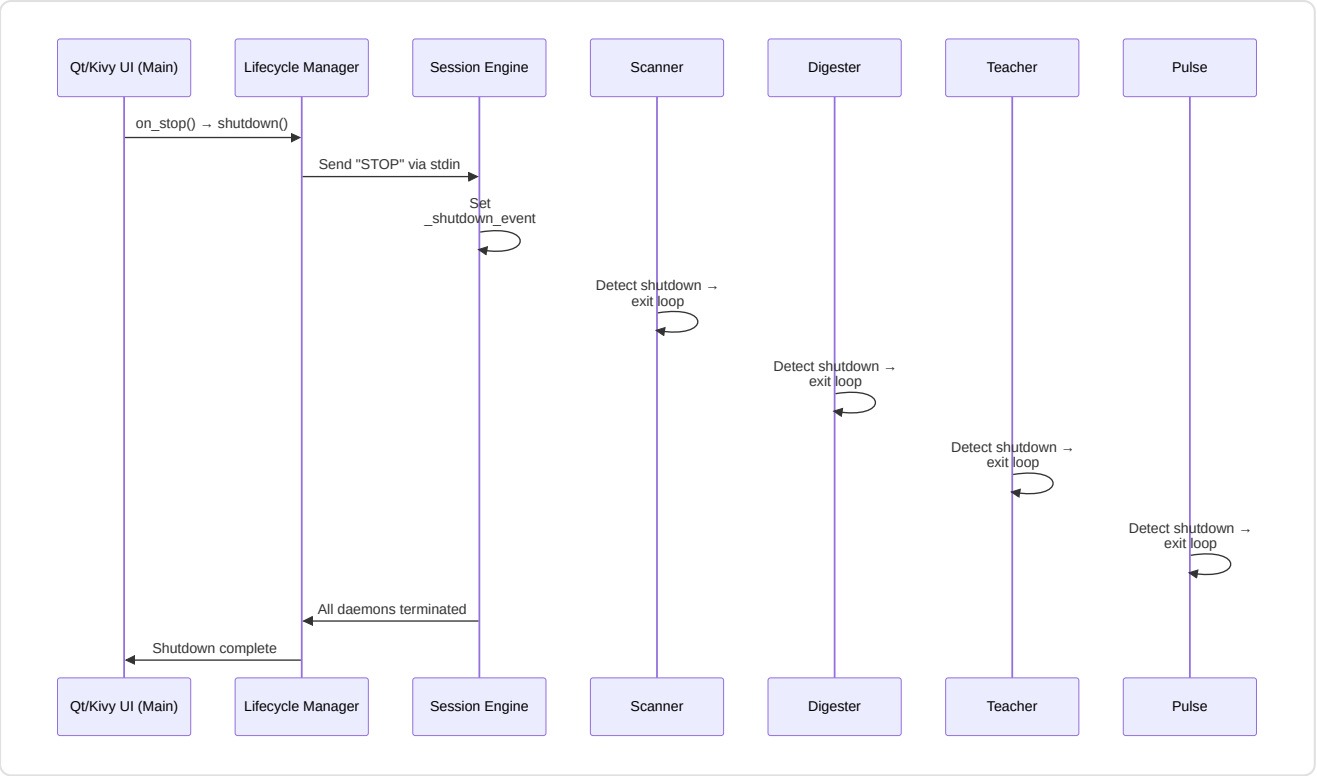
Lifecycle of each daemon:

Daemon	Interval	Work per cycle	Trigger
Scanner	10 seconds	Scans pro and user folders, creates `IngestionTask` for new `.dem`	Always active (if state = Scanning)
Digester	Continuous	Picks 1 task from queue, performs full parsing	`_work_available_event` (signaled by Scanner)
Teacher	300 seconds (5 min)	Checks pro sample growth; if ≥10% → `run_full_cycle()`	`pro_count >= last_count × 1.10`
Pulse	5 seconds	Updates `CoachState.last_heartbeat` in the database	Always active

> **Digester Analogy:** The Digester is like a **tireless dishwasher**: it takes a dirty dish (raw demo) from the pile, washes it carefully (parsing with demoparser2, feature extraction, HLTV 2.0 rating computation, RoundStats enrichment), dries it (normalization), puts it on the right shelf (saves to database), and then takes the next dish. It never takes 2 dishes at a time — one at a time, to avoid mistakes. If the pile is empty, it waits patiently (sleep 2s + `_work_available_event`) until someone brings new dirty dishes.

> **Teacher Analogy:** The Teacher is like a **university professor who updates his courses**. Every 5 minutes he checks: "Have enough new scientific articles (pro demos) arrived?" If the count has grown 10% since the last update, he says: "It's time to rewrite the handouts!" and kicks off a full training cycle. After training, he also runs a meta-shift check: "Has the pro average changed? Has the game meta shifted?" — ensuring the coaching always stays current.

Shutdown sequence:



Daemon implementation details:

Daemon	Main Method	Internal Loop	Exit Condition
Scanner	`_scanner_daemon_loop()`	`while not _shutdown_event.is_set()` → `scan_all_paths()` → `sleep(10)`	`_shutdown_event` set
Digester	`_digester_daemon_loop()`	`while not _shutdown_event.is_set()` → `_work_available_event.wait(2)` → `process_next_task()`	`_shutdown_event` set
Teacher	`_teacher_daemon_loop()`	`while not _shutdown_event.is_set()` → `_check_retrain_needed()` → `sleep(300)`	`_shutdown_event` set
Pulse	`_pulse_daemon_loop()`	`while not _shutdown_event.is_set()` → `state_mgr.heartbeat()` → `sleep(5)`	`_shutdown_event` set

Error handling in daemons:

Each daemon is protected by a global `try/except`. If a daemon crashes:

1. The error is logged with a full traceback
2. The `StateManager` registers the error (`set_error(daemon, message)`)
3. A `ServiceNotification` is created for the user
4. The daemon **is not automatically restarted** (by design: a daemon crash indicates a bug, not a transient error)
5. The other daemons continue to function independently

> **Analogy:** Each daemon is a **worker with their own office and their own door**. If a worker has a medical incident (crash), they close their door and post a "Temporarily unavailable" notice (ServiceNotification). The other workers in the other offices continue working normally. The director (Session Engine) takes note of the incident but does not try to revive the worker — they prefer that a technician (the developer) investigate the cause before putting them back to work.

Zombie Task Cleanup: On startup, the Session Engine looks for tasks with `status="processing"` left over from a previous crash and resets them to `status="queued"`, enabling automatic recovery without data loss.

Automatic Backup: On Session Engine startup, `BackupManager.should_run_auto_backup()` checks whether a backup is needed and, if so, creates a checkpoint labeled `"startup_auto"`. Backups follow a rotation of 7 daily + 4 weekly copies.

12.5 Desktop Interface

The desktop interface has two implementations: **Qt/PySide6** (primary) and **Kivy/KivyMD** (legacy). Both follow the **MVVM** pattern (Model-View-ViewModel).

12.5.1 Qt Interface (Primary) — `apps/qt_app/`

Directory: `Programma_CS2_RENAN/apps/qt_app/` **Key files:** `app.py`, `main_window.py`, `core/i18n_bridge.py`, `core/theme_engine.py`, `screens/`

The Qt interface is built with **PySide6 (Qt 6)** and uses an **MVVM pattern with Qt Signals/Slots**. The `MainWindow` (`QMainWindow`) is composed of a **navigation sidebar** and a `QStackedWidget` that hosts the 14 screens.

> **Analogy:** The Qt interface is like a **digital dashboard of a modern sports car**. The dashboard (QStackedWidget) has several view modes selectable from the sidebar: the "Trip" view (Home), the "Navigation" view (Tactical Viewer with QPainter), the "Diagnostic" view (Coach), the "Settings" view (Settings). The MVVM pattern with Signals/Slots ensures that every user interaction emits a "signal" that is caught by the appropriate "slot" — like the car's sensors communicating with the onboard computer via the CAN bus.

Specification	Detail
Framework	PySide6 (Qt 6 for Python)
Pattern	MVVM with Qt Signals/Slots
Platforms	Windows, macOS, Linux
Resolution	Adaptive, native High-DPI
Themes	3: CS2 (orange), CSGO (blue-gray), CS1.6 (green) — QSS + QPalette
i18n	3 languages: EN, IT, PT — JSON + `QtLocalizationManager`
Charts	QPainter for tactical map, native Qt widgets for charts

i18n system: The `QtLocalizationManager` (`core/i18n_bridge.py`) loads per-language JSON files (`en.json` , `it.json` , `pt.json`) and handles runtime language switching via Qt signals. Every UI string is dynamically resolved via a localization key.

Theme system: The `ThemeEngine` (`core/theme_engine.py`) applies QSS stylesheets and configures Qt's `QPalette` for each theme:

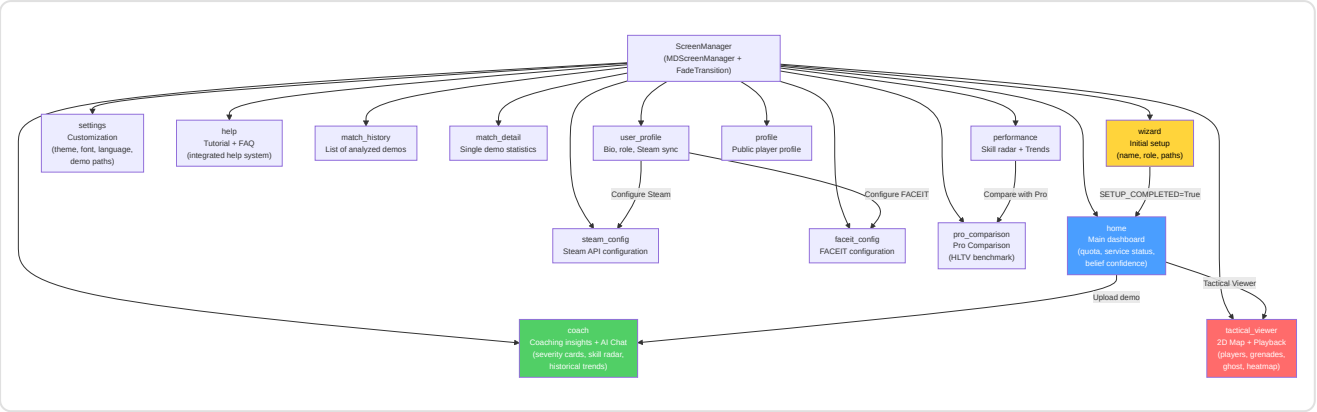
- **CS2** — Orange palette (#FF6600) with dark background, inspired by the CS2 UI
- **CSGO** — Blue-gray palette (#4A90D9) with cool tones, inspired by CS:GO
- **CS1.6** — Green palette (#33CC33) on black background, inspired by the retro look of CS 1.6

12.5.2 Kivy Interface (Legacy) — `apps/desktop_app/`

Directory: `Programma_CS2_RENAN/apps/desktop_app/` **Key files:** `layout.kv` , `wizard_screen.py` , `player_sidebar.py` , `tactical_viewer_screen.py` , `tactical_viewmodels.py` , `tactical_map.py` , `timeline.py` , `widgets.py` , `help_screen.py` , `ghost_pixel.py`

The legacy interface is built with **Kivy + KivyMD** and follows the **MVVM** pattern (Model-View-ViewModel). The `ScreenManager` handles navigation between screens with `FadeTransition` transitions.

> **Analogy:** The Kivy interface is like a **dashboard of a classic sports car**. The dashboard (ScreenManager) has several view modes you can select: the "Trip" view (Home — general dashboard), the "Navigation" view (Tactical Viewer — 2D map), the "Diagnostic" view (Coach — detailed analysis), the "Settings" view (Settings — customization). Each view has its specialized instruments. The MVVM pattern ensures that the "engine" (ViewModel) and the "display" (View) are separate: if you change the dashboard design, the engine continues to work identically, and vice versa.



14 interface screens:

Screen	Role	Key components
Wizard	First-time setup	Player name, role, demo folder paths
Home	Dashboard	Monthly quota (X/10), service status (green/red), belief confidence (0-1), active tasks, processed matches counter
Coach	Coaching insights	Color-coded cards by severity, multi-dimensional skill radar, historical trends, AI chat (Ollama/Claude), active tasks
Tactical Viewer	Tactical playback	2D map with players/grenades/ghost, timeline with event markers, CT/T player sidebar, speed controls (0.25x→8x)
Settings	Customization	Theme (CS2/CSGO/CS1.6), font, text size, language, demo paths, wallpaper
Help	User support	Interactive tutorial, FAQ, troubleshooting
Match History	Match archive	List of analyzed demos with filters and sorting
Match Detail	Match detail	Detailed statistics for a single analyzed demo
Performance	Progress	5-axis skill radar, trend charts, temporal comparisons
User Profile	User profile	Bio, preferred role, Steam/FACEIT sync
Profile	Public profile	Public player profile view
Steam Config	Steam configuration	Entering and validating Steam API key
Pro Comparison	Pro comparison	User statistics comparison with HLTV pro players, performance benchmark
FACEIT Config	FACEIT configuration	Entering and validating FACEIT API key

> **Home Screen Analogy:** The Home is like the **command deck of a spaceship**. The quota indicator ("5/10 demos this month") is the **fuel gauge**. The service status (green/red) is the **vital systems panel**: green = all systems operational, red = alarm. Belief confidence (0.0-1.0) is the **AI stability level**: 0.0 = the AI knows nothing, 1.0 = the AI is confident in its analyses. The processed matches counter is the **odometer**: how far the system has traveled.

Custom widgets:

Widget	File	Function
`PlayerSidebar`	`player_sidebar.py`	CT/T list with role icons, health/armor, current weapon, money, and alive/dead status
`TacticalMap`	`tactical_map.py`	2D canvas with multi-layer rendering: map texture → heatmap → players → grenades → ghost (QPainter in Qt, Canvas Kivy in legacy)
`Timeline`	`timeline.py`	Horizontal scrubber with tick numbers, color-coded event markers, drag-to-seek, double-click jump
`GhostPixel`	`ghost_pixel.py`	Rendering of the semi-transparent ghost circle (optimal position predicted by RAP)

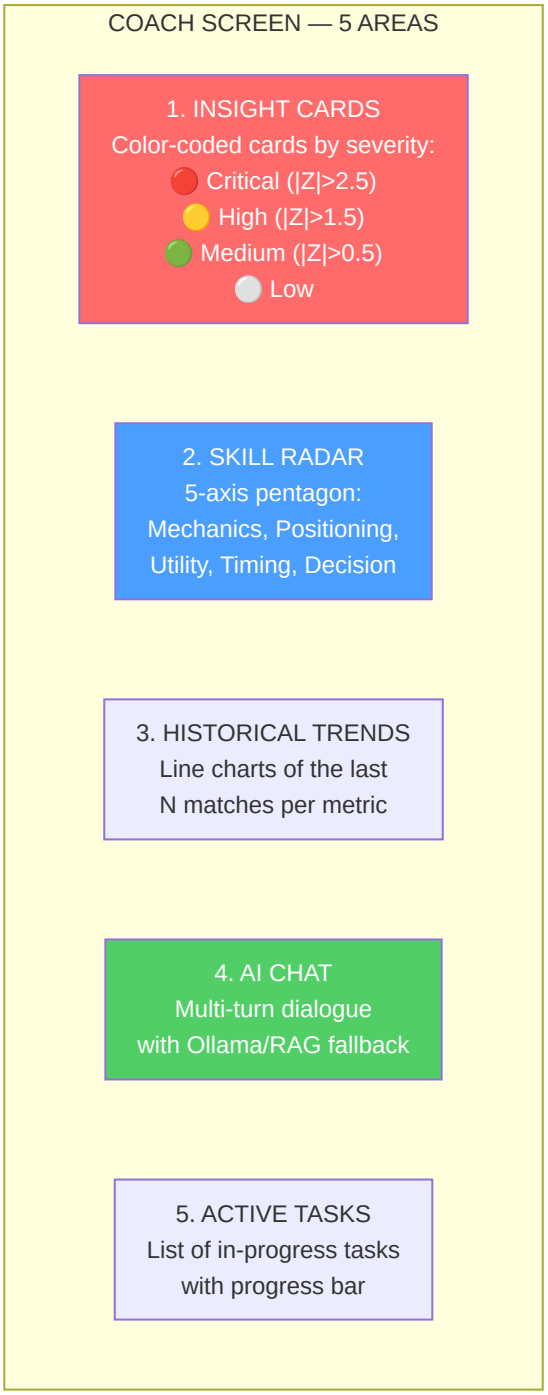
Available themes:

The application supports **3 themes** selectable from the Settings screen, each with a custom color palette and wallpaper:

Theme	Primary palette	Inspiration
CS2 (default)	Steel blue + orange	Modern Counter-Strike 2 UI
CSGO	Military green + yellow	Counter-Strike: Global Offensive
CS 1.6	Dark brown + lime green	Classic Counter-Strike 1.6 (nostalgia)

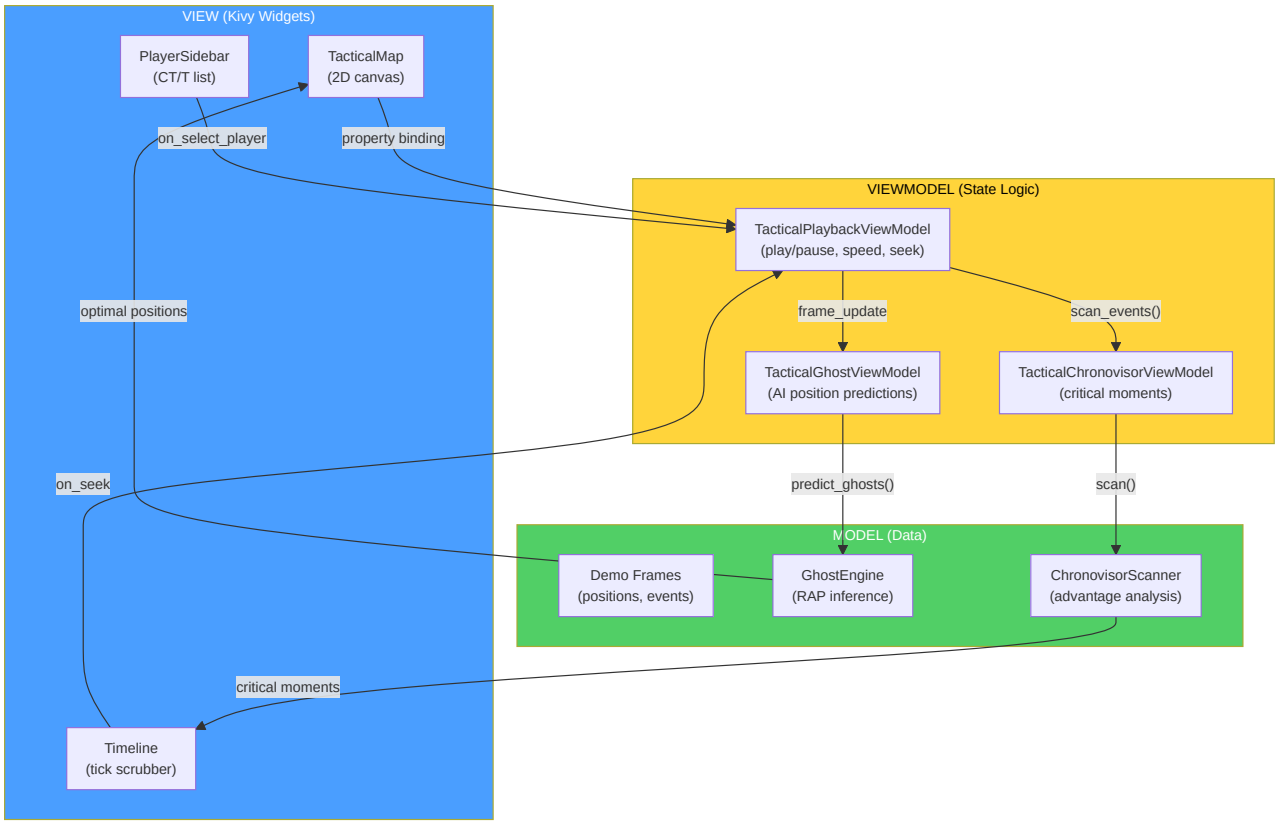
Coach Screen — Detailed layout:

The Coach screen is the most complex in the application, with 5 functional areas:



> **Coach Screen Analogy:** The Coach screen is like the **doctor's office after the visit**. The **Insight Cards** are the immediate report: "High blood pressure!" (red), "Cholesterol to monitor" (yellow), "Good physical shape" (green). The **Radar** is the visualization of physical abilities. The **Trends** are the comparison with previous visits. The **AI Chat** is the opportunity to ask the doctor questions. The **Active Tasks** show the exams still in progress.

MVVM pattern in the Tactical Viewer:



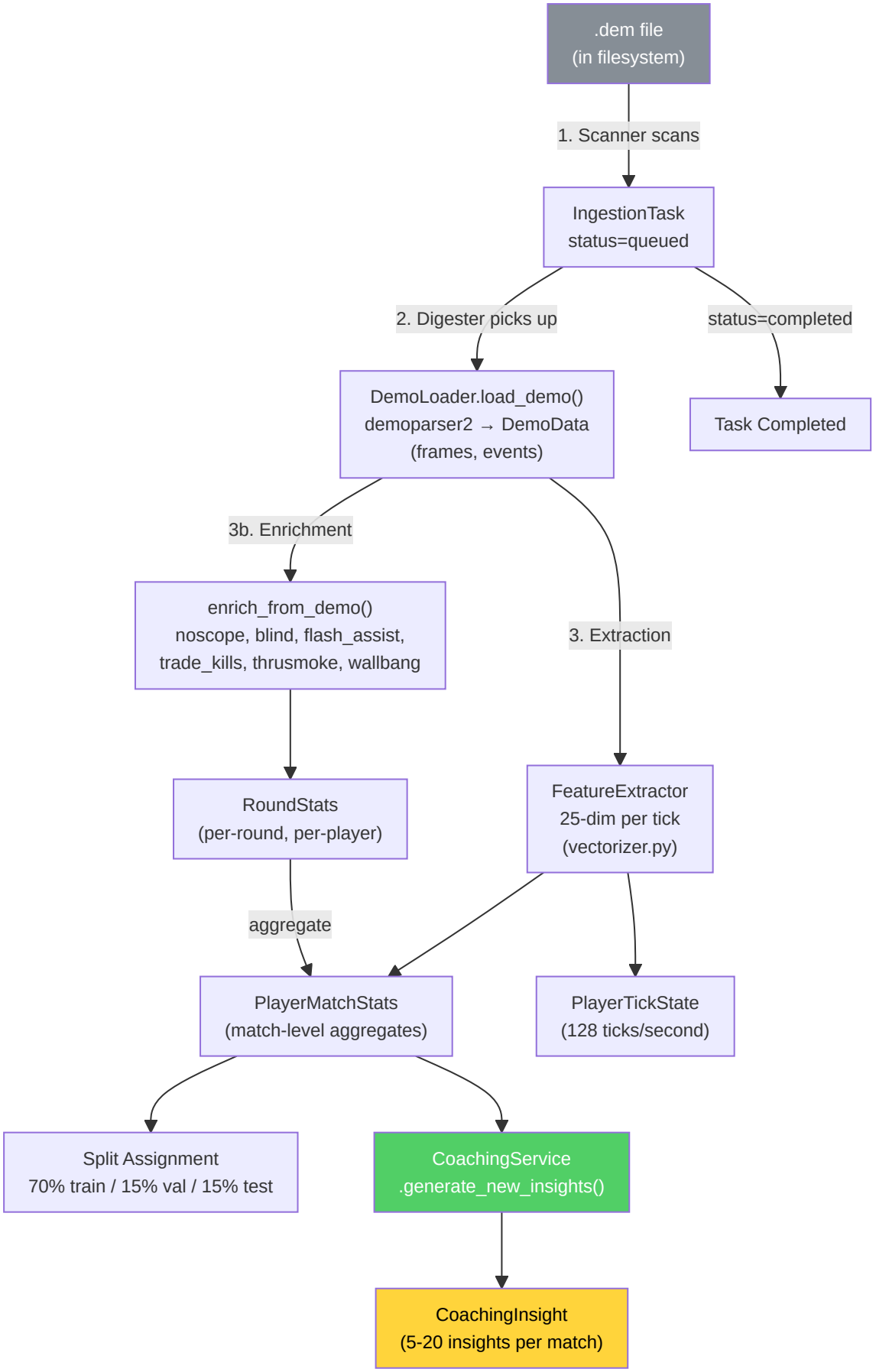
> **MVVM Analogy:** The MVVM pattern is like a **TV newscast**: the **Model** is the reporter in the field who gathers the facts (demo data, AI inference). The **ViewModel** is the editor who organizes the news and decides what's important (playback state, speed, selection). The **View** is the presenter who reads the news to the audience (Kivy on-screen rendering). The reporter doesn't know how the news will be presented. The presenter doesn't know how it was gathered. The editor is the bridge between the two. If you change the presenter (new UI design), the news stays the same.

12.6 Ingestion Pipeline (`ingestion/`)

Directory: `Programma_CS2_RENAN/ingestion/` **Key files:** `demo_loader.py`, `steam_locator.py`, `integrity.py`, `registry/`, `pipelines/user_ingest.py`, `pipelines/json_tournament_ingestor.py`

The ingestion pipeline is the **complete path** a `.dem` file takes from the filesystem to becoming coaching insights in the database. It is orchestrated by the Scanner daemon (discovery) and the Digester daemon (processing).

> **Analogy:** The ingestion pipeline is like the **journey of a letter through the post office**. (1) The postman (Scanner) collects the letter (.dem file) from the mailbox (demo folder). (2) The sorting office (DemoLoader) opens the envelope and extracts its contents (parsing with demoparser2). (3) The archivist (FeatureExtractor) measures and catalogs every detail (25 features per tick). (4) The historian (RoundStatsBuilder) writes a summary per chapter (per-round statistics). (5) The librarian (data_pipeline) classifies and orders the material (normalization, dataset split). (6) The doctor (CoachingService) examines everything and writes a diagnosis (coaching insights). (7) Finally, everything is archived (database persistence) for future reference.


.dem file
(in filesystem)
1. Scanner scans
IngestionTask
status=queued
2. Digester picks up
status=completed
DemoLoader.load_demo()
demoparser2 → DemoData
(frames, events)
Task Completed
3b. Enrichment
enrich_from_demo()
noscope, blind, flash_assist,
trade_kills, thrusmoke, wallbang
3. Extraction
RoundStats
(per-round, per-player)
FeatureExtractor
25-dim per tick
(vectorizer.py)
aggregate
PlayerMatchStats
(match-level aggregates)
PlayerTickState
(128 ticks/second)
Split Assignment
70% train / 15% val / 15% test
CoachingService
.generate_new_insights()
CoachingInsight
(5-20 insights per match)

DemoLoader — The parser at the heart of the pipeline:

The `DemoLoader` is the wrapper around **demoparser2** (high-performance Rust library) that transforms a binary `.dem` file into Python data structures:

Parsing Phase	Output	Typical size
1. Header parsing	Metadata (map, server, duration)	~100 bytes
2. Frame extraction	List of frames (position, health, weapon per tick)	~100,000 frames/match
3. Event extraction	List of events (kill, death, bomb_plant, round_start, etc.)	~500-2,000 events/match
4. Player summary	Aggregated per-player statistics	~10 records

FeatureExtractor (`backend/processing/feature_engineering/vectorizer.py`):

The FeatureExtractor is the component that transforms raw demo data into **25-dimensional numeric vectors** (`METADATA_DIM=25`) usable by the neural networks. **Important:** these are **tick-level** features (128 Hz), not match-level aggregated statistics. Each individual game frame produces a 25-dim vector that captures the instantaneous state of the player:

Dim	Feature	Type	Range	Description
0	health	Float	[0, 1]	Normalized health
1	armor	Float	[0, 1]	Normalized armor
2	has_helmet	Binary	0/1	Helmet equipped
3	has_defuser	Binary	0/1	Defuse kit equipped
4	equipment_value	Float	[0, 1]	Normalized equipment value
5	is_crouching	Binary	0/1	Crouched
6	is_scoped	Binary	0/1	Scope active
7	is_blinded	Binary	0/1	Flashed
8	enemies_visible	Float	[0, 1]	Visible enemies (normalized, clamped)
9	pos_x	Float	[-1, 1]	X position (normalized ±pos_xy_extent)
10	pos_y	Float	[-1, 1]	Y position (normalized ±pos_xy_extent)
11	pos_z	Float	[0, 1]	Z position (normalized, handles Nuke/Vertigo)
12	view_x_sin	Float	[-1, 1]	sin(yaw) — cyclic continuity of horizontal angle
13	view_x_cos	Float	[-1, 1]	cos(yaw) — cyclic continuity of horizontal angle
14	view_y	Float	[-1, 1]	Normalized pitch (vertical angle)
15	z_penalty	Float	[0, 1]	Vertical level distinction (floor penalty)
16	kast_estimate	Float	[0, 1]	KAST estimate (participation ratio)
17	map_id	Float	[0, 1]	Deterministic map hash
18	round_phase	Float	{0, 0.33, 0.66, 1}	Economic phase: pistol/eco/force/full_buy
19	weapon_class	Float	{0–1.0}	Weapon class: 0=knife, 0.2=pistol, 0.4=SMG, 0.6=rifle, 0.8=sniper, 1.0=heavy
20	time_in_round	Float	[0, 1]	Seconds in round / 115 (clamped)
21	bomb_planted	Binary	0/1	Bomb planted
22	teammates_alive	Float	[0, 1]	Teammates alive (count / 4)
23	enemies_alive	Float	[0, 1]	Enemies alive (count / 5)
24	team_economy	Float	[0, 1]	Team money average / 16000 (clamped)

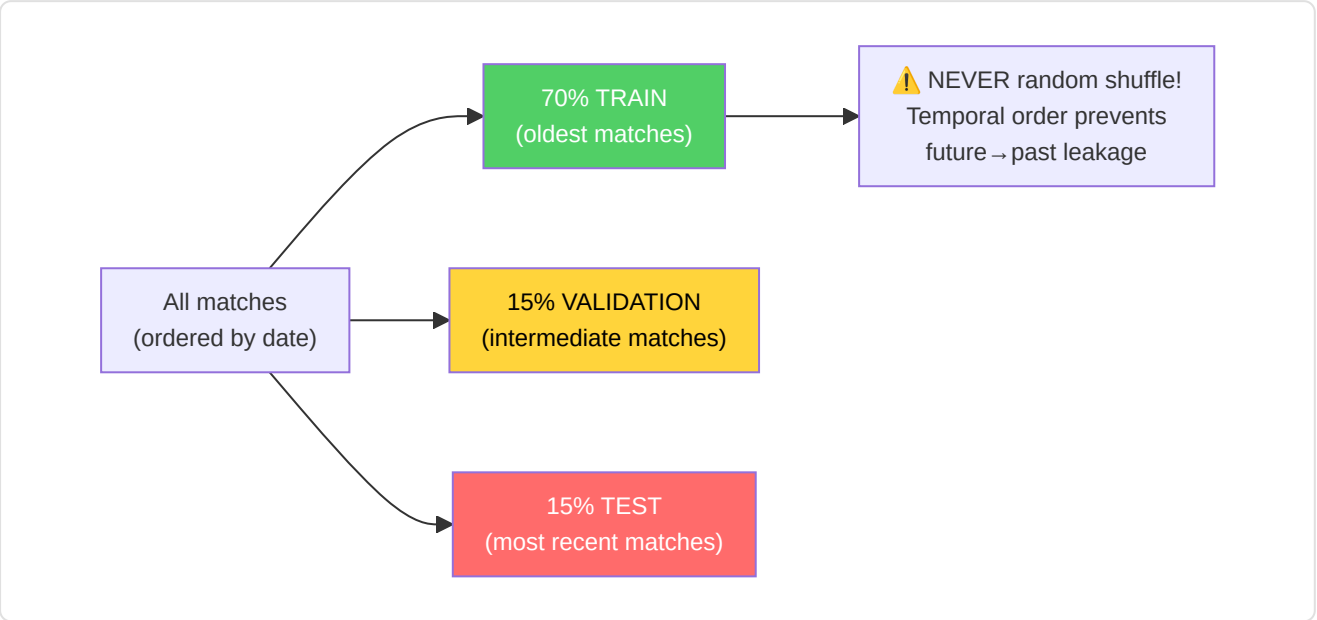
Normalization and bounds:

Normalization is integrated directly into the FeatureExtractor, with bounds configurable via `HeuristicConfig` (externalized to JSON). The cyclic encoding of view angles (sin/cos for yaw) prevents discontinuities at the 0°/360° boundary. The `z_penalty` automatically distinguishes floors on multi-level

maps (Nuke, Vertigo). Weapon class uses an ordinal categorical mapping (6 classes) defined in the `WEAPON_CLASS_MAP` constant (also includes grenades=0.1 and special equipment=0.05).

Temporal Dataset Split:

The dataset split follows a strict **chronological ordering** to prevent temporal data leakage:



> **Analogy:** The temporal split is like preparing a **fair school exam**. Test questions (15% test) must cover topics taught AFTER the exercises (70% train) and homework (15% val). If the test questions covered topics studied before the exercises, the student (the model) would look better than they really are — a trick, not real knowledge.

> **Feature Vector Analogy:** The 25-dimensional vector is like the **instantaneous reading of 25 sensors** attached to the player at every moment: 5 "body" sensors (health, armor, helmet, defuse kit, equipment value), 3 "postural" sensors (crouched, scoped, flashed), 1 "visual" sensor (visible enemies), 6 "spatial" sensors (X/Y/Z position and view angles sin/cos/pitch), 1 "level" sensor (Z penalty for Nuke/Vertigo), 1 "tactical" sensor (KAST), 3 "contextual" sensors (map, economic phase, weapon class), and 5 "situational" sensors (time in round, bomb planted, teammates/enemies alive, team economy). Unlike match-level aggregated statistics, every tick produces a new vector — 128 readings per second.

Enrich From Demo — Post-parsing enrichment:

After base parsing, `enrich_from_demo()` adds advanced metrics computed from events:

Enriched Metric	Computation	Source
Trade kills	`TradeKillDetector.detect()` with TRADE_WINDOW_TICKS=192	Kill/death events
Flash assists	Count of blinds within time window before a kill	Blind + kill events
Noscope kills	Kills with sniper weapon without scope active	Kill event + weapon state
Wallbang kills	Kills through penetrable surfaces	Kill event with penetration flag
Through-smoke kills	Kills with smoke active in line of fire	Kill event + smoke position
Blind kills	Kills while the player is flashed	Kill event + flash state

Specific components:

Component	File	Role
DemoLoader	`demo_loader.py`	Wraps `demoparser2`, extracts frames and events from the `.dem` file
SteamLocator	`steam_locator.py`	Automatically locates the CS2 demo folder via Steam registry / libraryfolders.vdf
IntegrityChecker	`integrity.py`	Verifies that demo files are valid, complete, and not corrupt before parsing
UserIngestPipeline	`pipelines/user_ingest.py`	Complete pipeline for user demos: parse → enrich → stats → coaching
JsonTournamentIngestor	`pipelines/json_tournament_ingestor.py`	Imports tournament data from structured JSON files
Registry	`registry/registry.py`	Tracks all processed demos, prevents duplicates
ResourceManager	`ingestion/resource_manager.py`	Hardware resource management: CPU/RAM throttling, disk space
JsonTournamentIngestor	`pipelines/json_tournament_ingestor.py`	Imports tournament data from structured JSON files
RegistryLifecycle	`registry/lifecycle.py`	Lifecycle management of ingestion records

SteamLocator (`ingestion/steam_locator.py`) — automatic CS2 demo location:

The SteamLocator implements a **cross-platform discovery** algorithm to automatically find the CS2 demo folder:

Platform	Strategy	Typical path
Windows	System registry → `libraryfolders.vdf`	`C:\Program Files (x86)\Steam\steamapps\common\Counter-Strike Global Offensive\game\csgo\replays`
Linux	`~/.steam/steam/` → `libraryfolders.vdf`	`~/.steam/steam/steamapps/common/...`
Fallback	Prompts user via UI Settings	Custom path

IntegrityChecker (`ingestion/integrity.py`):

Preliminary check of each demo file before expensive parsing:

- **Magic bytes**: `PBDEMS2\0` (CS2 Source 2) or `HL2DEMO\0` (legacy Source 1)
- **Size bounds**: minimum 1KB (not empty), maximum 5GB (not corrupt/excessive)
- **Read test**: attempts to read the first N bytes to verify the file is accessible

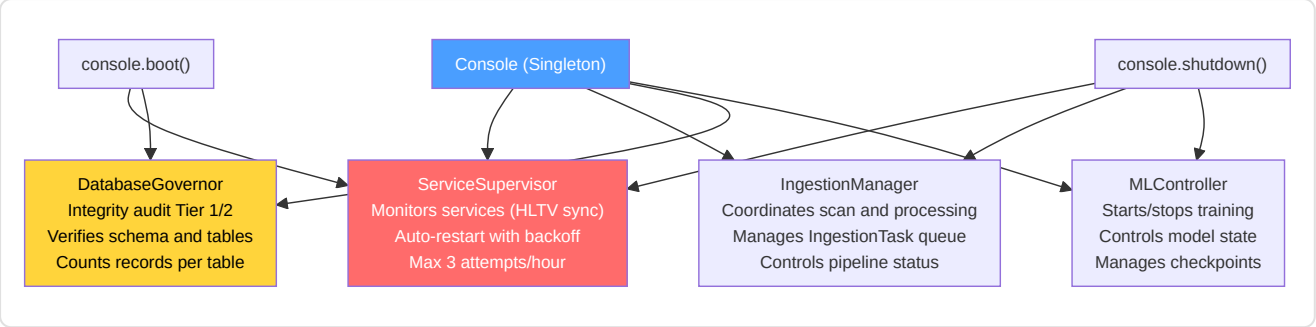
> **SteamLocator Analogy:** The SteamLocator is like a **bloodhound that sniffs out the CS2 folder** on your computer. It knows Steam stores its libraries in specific places (Windows registry, `libraryfolders.vdf` on Linux/Mac), and follows the trail to the `csgo/replays` folder where demos are saved. If it can't find it automatically, it asks the user to indicate the path manually — but in most cases, it finds it on its own.

12.7 Unified Control Console (`backend/control/`)

File: `Programma_CS2_RENAN/backend/control/console.py` , `ingest_manager.py` , `db_governor.py` , `ml_controller.py`

The Console is a **Singleton** that serves as a central coordination point for all backend subsystems. It is the "control panel" through which every part of the system can be controlled.

> **Analogy:** The Console is like the **control tower of an airport**. It has 4 screens: one for the radar (ServiceSupervisor — monitors running services), one for the runways (IngestionManager — coordinates demo arrival), one for maintenance (DatabaseGovernor — verifies storage integrity), and one for pilot training (MLController — manages the machine learning lifecycle). The air traffic controller (Console Singleton) coordinates everything from a single station.



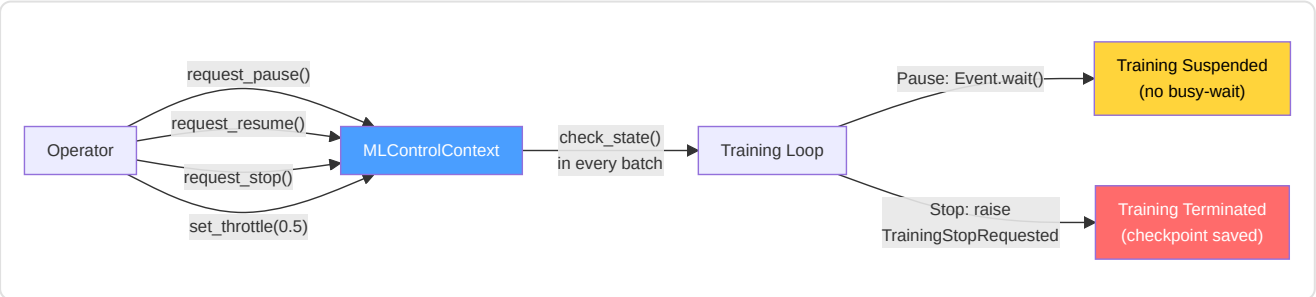
Console boot sequence:

1. `ServiceSupervisor` starts the "hunter" service (HLTV sync) as a monitored process
2. `DatabaseGovernor` performs an integrity audit: verifies all tables, counts records, checks schema
3. `MLController` remains on standby — training is handled by the Teacher daemon
4. `IngestionManager` stays idle — active work is handled by the Scanner/Digester daemons

MLControlContext — Live Training Control:

The `MLController` uses a control token called `MLControlContext` (`backend/control/ml_controller.py`) that is passed to training cycles to enable **real-time intervention** by the operator. This replaces the previous approach based on `StopIteration` with a more robust thread-safe system.

> **Analogy:** MLControlContext is like the **remote control of a video player**. The operator can press **Pause** (training stops immediately, without data loss), **Play** (training resumes from the exact point where it stopped), **Stop** (training terminates with a controlled `TrainingStopRequested` exception), or adjust the **speed** (throttle: 0.0 = max speed, 1.0 = max delay). The mechanism uses `threading.Event` to avoid busy-wait during pause — the training thread simply sleeps until it receives the resume signal.



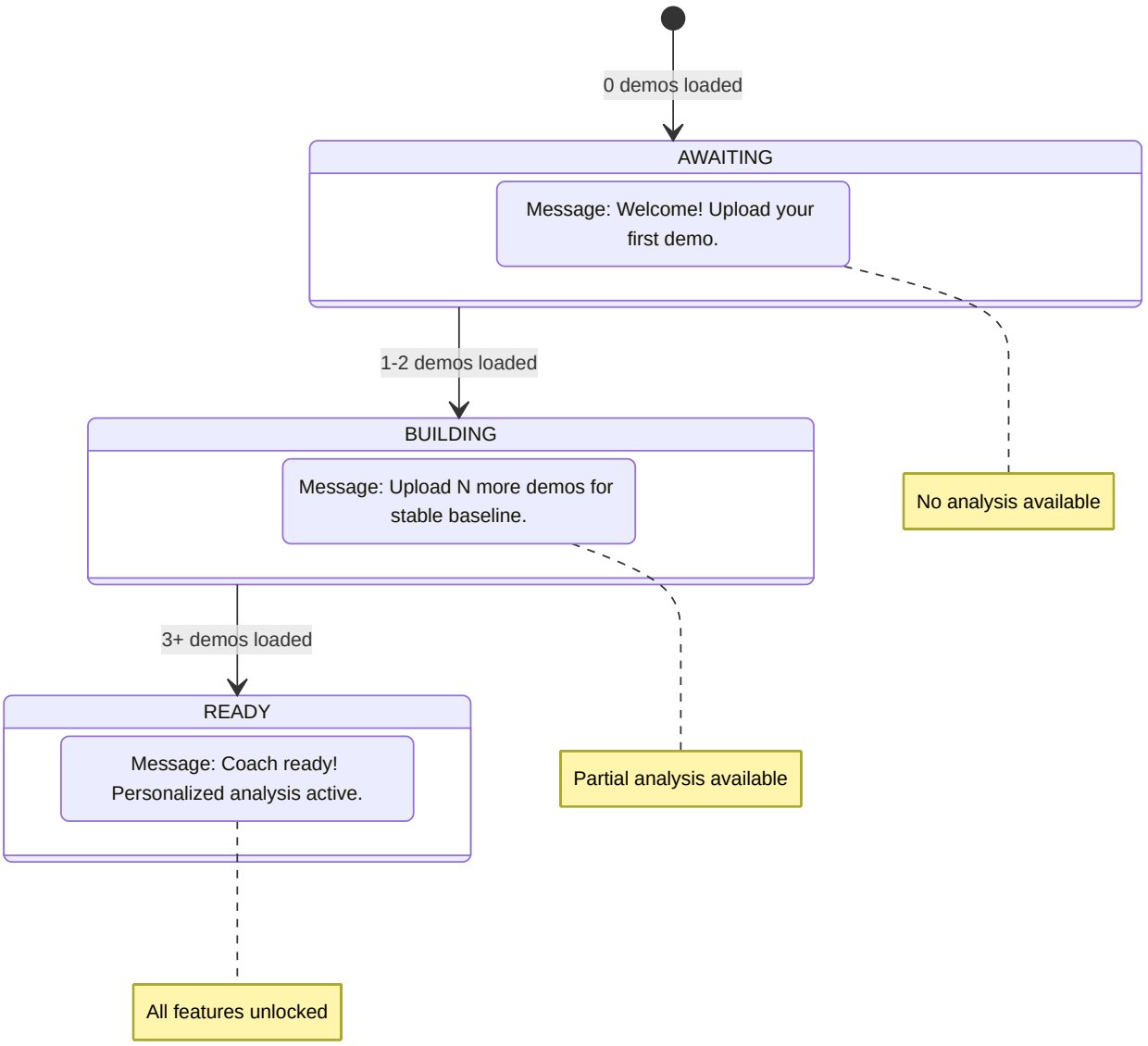
Command	Method	Effect
Pause	`request_pause()`	`_resume_event.clear()` → blocks `check_state()`
Resume	`request_resume()`	`_resume_event.set()` → unblocks training
Stop	`request_stop()`	Raises `TrainingStopRequested` (custom exception)
Throttle	`set_throttle(factor)`	Adds `time.sleep(factor)` after each batch

12.8 Onboarding and New User Flow

File: `Programma_CS2_RENAN/backend/onboarding/new_user_flow.py`

The `OnboardingManager` guides new users through a **3-phase progression** that automatically adapts to the amount of available data.

> **Analogy:** Onboarding is like the **tutorial of an RPG video game**. When you start a new game (no demos loaded), the game guides you step by step: "Welcome, adventurer! Upload your first demo to begin." After 1-2 demos, the system says: "Good start! Upload N more to unlock stable analysis." After 3+ demos, the system announces: "Your coach is ready! Personalized analyses are now active." Each phase gradually unlocks the program's features, preventing the system from displaying unreliable results when it doesn't have enough data.



Wizard Screen (First-time Setup):

On first run (`SETUP_COMPLETED = False`), the user is guided through the Wizard:

1. **Player name** — The name that will appear in analyses
2. **Preferred role** — Entry Fragger, AWPer, Lurker, Support, or IGL
3. **Demo folder paths** — Where the system automatically looks for demos
4. On completion: `SETUP_COMPLETED = True` , redirect to Home

Quota cache (Task 2.16.1): The demo count is cached for 60 seconds to avoid repeated DB queries. `invalidate_cache()` is called after every new upload, ensuring the UI always shows the correct count without overloading the database.

Help System (`backend/knowledge/help_system.py`):

The integrated help system provides contextual support to the user:

Feature	Implementation
Interactive tutorial	Step-by-step guides for the main features
Contextual FAQ	Frequently asked questions filtered by current screen
Troubleshooting	Decision tree for common problems (Steam path not found, demo not parsed, empty coaching)
Tooltips	Inline explanations for complex metrics (KAST, HLTV 2.0 Rating, ADR)

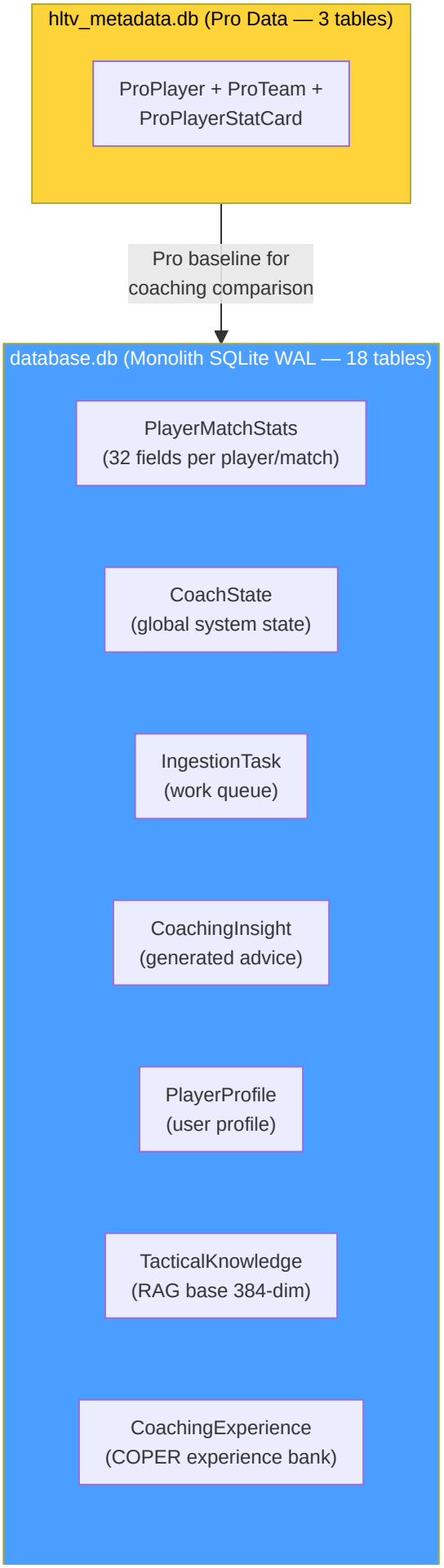
> **Analogy:** The Help System is like having a **librarian assistant** always available. If you're in the "Fiction" section (Tactical Viewer), it helps you with map reading questions. If you're in "Science" (Coach), it explains what the metrics mean. If something doesn't work, it guides you step by step through troubleshooting — like a flow chart "If the faucet doesn't give water → check the valve → check the pipe → call the plumber."

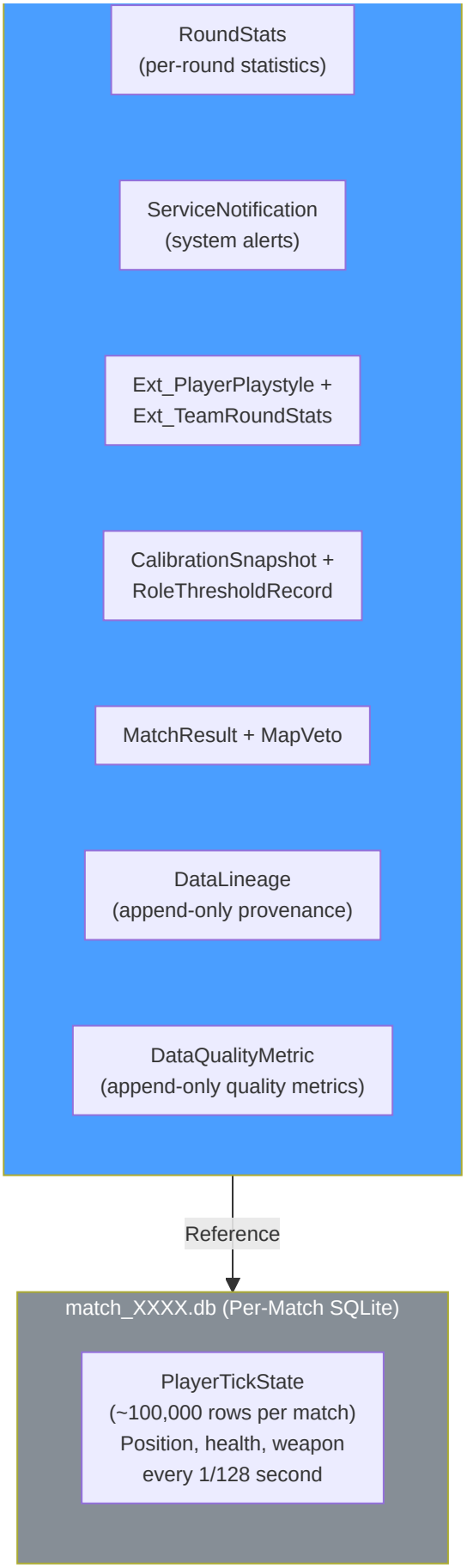
12.9 Storage Architecture (`backend/storage/`)

Directory: `Programma_CS2_RENAN/backend/storage/` **Key files:** `database.py` , `db_models.py` , `match_data_manager.py` , `storage_manager.py` , `maintenance.py` , `state_manager.py` , `stat_aggregator.py` , `backup.py`

The storage system uses a **three-tier storage** architecture based on SQLite in WAL mode (Write-Ahead Logging), which enables concurrent reads and writes without blocking.

> **Analogy:** The storage architecture is like a **3-level library system**. The **ground floor** (`database.db` , 18 tables) holds the general catalog, all readers' (players') cards, the tactical knowledge base (RAG), the COPER experience bank, the critics' reviews (coaching insights), and the loan register (ingestion tasks) — all in one large filing cabinet always available. The **separate filing cabinet** (`hltv_metadata.db` , 3 tables) contains the pro player profiles and their statistics — separated because it is updated by a different process (HLTV sync) to avoid lock contention. Beyond these, the **per-match databases** (`match_XXXX.db`) contain the full original manuscripts (tick-by-tick match data) — each in a separate box to prevent the main filing cabinet from becoming too heavy. WAL mode is like having a **revolving door**: many people can enter to read at the same time, and someone can write without blocking the entrance.


hltv_metadata.db (Pro Data — 3 tables)
ProPlayer + ProTeam + ProPlayerStatCard
Pro baseline for coaching comparison
database.db (Monolith SQLite WAL — 18 tables)
PlayerMatchStats (32 fields per player/match)
CoachState (global system state)
IngestionTask (work queue)
CoachingInsight (generated advice)
PlayerProfile (user profile)
TacticalKnowledge (RAG base 384-dim)
CoachingExperience (COPER experience bank)



The 21 SQLModel tables:

#	Table	Database	Category	Description
1	`PlayerMatchStats`	database.db	Core	Aggregated per-player/match statistics (32 fields)
2	`PlayerTickState`	database.db	Core	Per-tick state (128 Hz), also archived in separate per-match DBs
3	`PlayerProfile`	database.db	User	User profile (name, role, Steam ID, monthly quota)
4	`RoundStats`	database.db	Core	Per-round isolated statistics (kills, rating, enrichment)
5	`CoachingInsight`	database.db	Coaching	Advice generated by the coaching service
6	`CoachingExperience`	database.db	Coaching	COPER experience bank (context, outcome, effectiveness, TrueSkill μ/σ, replay priority — KT-01)
7	`IngestionTask`	database.db	System	Work queue for the Digester daemon
8	`CoachState`	database.db	System	Global state (training metrics, heartbeat, status)
9	`ServiceNotification`	database.db	System	Error/event messages from daemons → UI
10	`TacticalKnowledge`	database.db	Knowledge	RAG base (384-dim embedding in JSON)
11	`ProPlayer`	hltv_metadata.db	Pro	Pro player profiles
12	`ProTeam`	hltv_metadata.db	Pro	Pro team metadata
13	`ProPlayerStatCard`	hltv_metadata.db	Pro	Seasonal statistics per pro player
14	`Ext_PlayerPlaystyle`	database.db	External	Playstyle data from CSV (for NeuralRoleHead)
15	`Ext_TeamRoundStats`	database.db	External	External tournament statistics
16	`MatchResult`	database.db	Matches	Match outcomes
17	`MapVeto`	database.db	Matches	Map selection history
18	`CalibrationSnapshot`	database.db	System	Belief model calibration log (timestamp, samples, outcome)
19	`RoleThresholdRecord`	database.db	System	Learned thresholds for role classification (persistent across restarts)
20	`DataLineage`	database.db	Provenance	Append-only data provenance log: entity_type, entity_id, source_demo, pipeline_version, processing_step
21	`DataQualityMetric`	database.db	Provenance	Append-only quality metrics per run: run_id, run_type, metric_name, metric_value, sample_count

Support enums (not tables):

Enum	Type	Description
`DatasetSplit`	`str, Enum`	Split categories (train/val/test/unassigned) — used as constraint on `PlayerMatchStats.dataset_split`
`CoachStatus`	`str, Enum`	Coach states (Paused/Training/Idle/Error) — used as constraint on `CoachState.status`

Detailed Storage Components:

MatchDataManager (`backend/storage/match_data_manager.py`) — the largest component of the storage layer:

The MatchDataManager is responsible for managing high-density per-match data (PlayerTickState with ~100,000 rows per match). To prevent the main database from growing uncontrollably, each match has its own separate SQLite database (`match_XXXX.db`).

Method	Description
`create_match_db(demo_name)`	Creates a new per-match database with `PlayerTickState` schema
`store_tick_data(demo_name, ticks)`	Bulk insert of tick data into the dedicated DB
`load_match_frames(demo_name)`	Loads all frames for the Tactical Viewer
`get_match_db_path(demo_name)`	Resolves the path of the per-match DB
`list_available_matches()`	Lists all matches with DB available
`delete_match_data(demo_name)`	Removes the per-match DB and updates the registry
`get_match_statistics(demo_name)`	Computes aggregate statistics from tick data

> **Analogy:** The MatchDataManager is like an **archivist who manages the original manuscript boxes**. Each match is a manuscript too bulky to fit in the general filing cabinet (database.db), so it is kept in a separate box with a label (match_XXXX.db). The archivist knows exactly where every box is, can open it on request, and when the box becomes too old, can move it to cold storage.

StorageManager (`backend/storage/storage_manager.py`):

The StorageManager is the **high-level storage coordinator** that manages quotas, uploads, and data lifecycle:

Responsibility	Implementation
Monthly quota	`can_user_upload()` → checks `MAX_DEMOS_PER_MONTH=10` and `MAX_TOTAL_DEMOS=100`
Upload flow	`handle_demo_upload(path)` → validation → copy to working dir → create IngestionTask
Cleanup	`cleanup_old_data(days)` → removes old match DBs and tasks
Disk space	`get_storage_usage()` → reports size for each database and directory

StateManager (`backend/storage/state_manager.py`):

The StateManager is a **Singleton** that maintains the system's runtime state and persists it to the database via the `CoachState` table:

Method	Purpose
`update_status(daemon, text)`	Updates the state of a specific daemon
`heartbeat()`	Updates the `last_heartbeat` timestamp
`get_state()`	Returns the current state as a `CoachState` object
`set_error(daemon, message)`	Registers an error for a daemon with timestamp
`update_training_metrics(epoch, loss, val_loss, eta)`	Updates training metrics in real time
`get_belief_confidence()`	Returns the belief model's confidence level (0.0-1.0)

StatAggregator (`backend/storage/stat_aggregator.py`):

Computes aggregate statistics from raw per-round data:

Aggregation	Formula	Use
`avg_kills`	`mean(RoundStats.kills)`	Dashboard, radar chart
`avg_adr`	`mean(RoundStats.damage_dealt / rounds)`	Pro comparison
`avg_kast`	`mean(rounds_with_kast / total_rounds)`	HLTV metric
`accuracy`	`sum(hits) / sum(shots_fired)`	Mechanical performance
`trade_kill_rate`	`trade_kills / team_deaths`	Team play

BackupManager (`backend/storage/backup.py`):

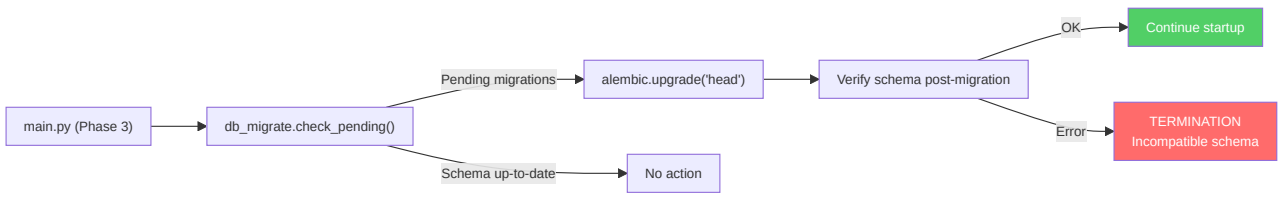
Feature	Detail
Daily rotation	7 copies — the oldest is overwritten
Weekly rotation	4 copies — additional weekly backup
Automatic trigger	On Session Engine startup via `should_run_auto_backup()`
Manual trigger	Via Console or UI Settings
Format	Full copy of the `.db` file (not SQL dump)
Labeling	`startup_auto` , `manual` , `pre_migration`

Maintenance (`backend/storage/maintenance.py`):

Operation	Frequency	Purpose
`vacuum()`	Monthly	Compacts the database, reclaims space from deleted records
`analyze()`	After every bulk insert	Updates SQLite query optimizer statistics
`wal_checkpoint()`	On startup	Forces merge of WAL into the main database
`integrity_check()`	On startup	`PRAGMA integrity_check` — verifies structural consistency

DbMigrate (`backend/storage/db_migrate.py`):

Wrapper around Alembic that automates the execution of migrations:



Connection Pooling and Concurrency:

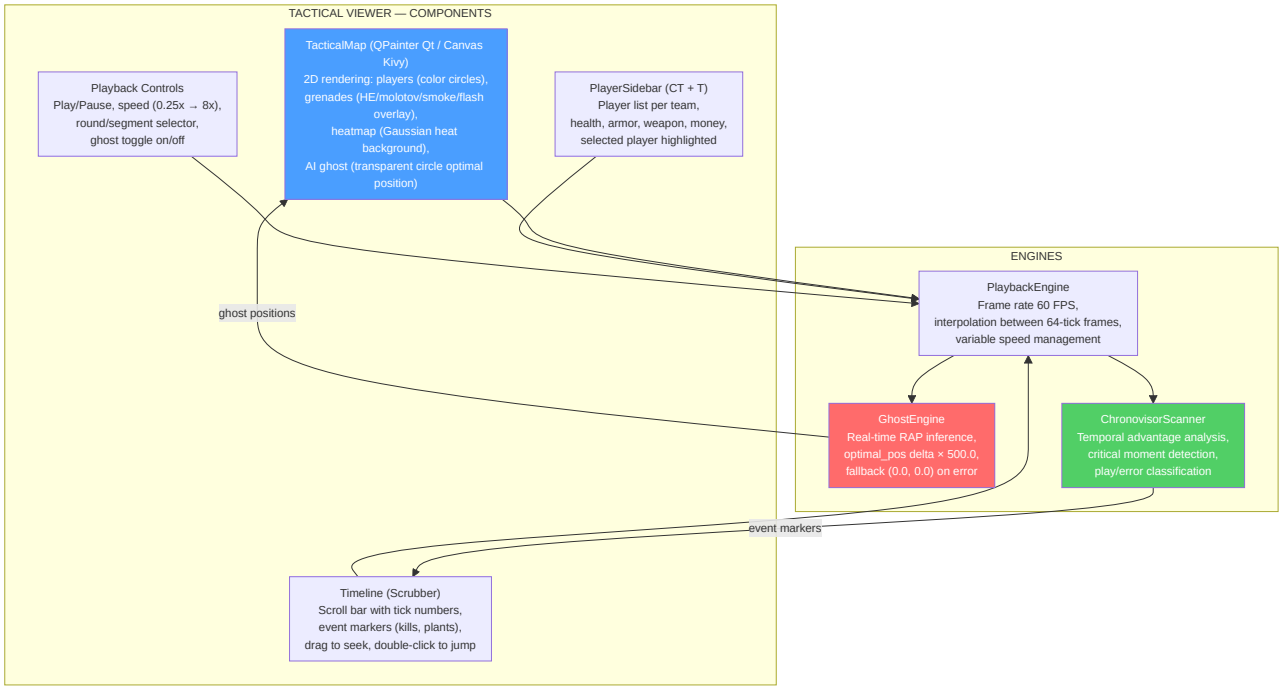
Parameter	Value	Purpose
`check_same_thread`	`False`	Enables multi-thread access
`timeout`	30 seconds	Busy timeout for WAL contention
`pool_size`	1	Single SQLite writer (single-writer safety)
`max_overflow`	4	Overflow connections for load spikes
WAL mode	Enabled	Unlimited concurrent reads

12.10 Playback Engine and Tactical Viewer

File: `Programma_CS2_RENAN/core/playback.py` , `playback_engine.py` , `apps/desktop_app/tactical_viewer_screen.py` , `tactical_map.py` , `timeline.py` , `player_sidebar.py`

The tactical playback system allows the user to **relive their matches** on an interactive 2D map, with AI overlay (optimal ghost position), event markers (kills, bomb plants), and full playback controls.

> **Analogy:** The Tactical Viewer is like a **professional-grade sports replay system**. Imagine being able to watch your soccer matches from the perspective of an aerial drone, with the ability to slow down, speed up, pause, and with an AI assistant showing you "where you should have been" as a transparent shadow on the field. Additionally, a smart timeline automatically highlights the key moments: "Minute 23:15 — you lost the advantage here" (red marker) or "Minute 34:02 — excellent play!" (green marker). You can click on any marker and the replay jumps directly to that moment.



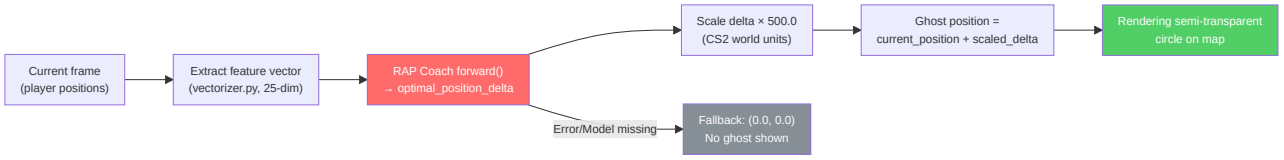
PlaybackEngine — Internal architecture:

The PlaybackEngine handles frame-by-frame playback with temporal interpolation. In the Qt implementation, the update timer uses `QTimer` instead of `Kivy Clock.schedule_interval`, keeping the same interpolation and buffering logic:

Feature	Detail
Frame rate	60 FPS (interpolation from demo's native 64-tick)
Variable speed	0.25x (slow-mo), 0.5x, 1x (normal), 2x, 4x, 8x (fast-forward)
Interpolation	Linear between adjacent frames for smooth movement
Buffering	Pre-loads 120 frames ahead to avoid lag
Seek	Direct access to any tick via index
Round selection	Jumps directly to the start of a specific round

GhostEngine — Real-time inference:

The GhostEngine is the component that transforms the RAP Coach's predictions into **ghost positions visible on the map**:



> **GhostEngine Analogy:** The ghost is like an **invisible coach** running on the field next to you. He watches the same match you watch, but he knows the optimal position where you should be. His semi-transparent shadow on the map says: "You would have been safer and more effective here." The ×500.0 factor converts the model's small deltas (numbers between -1 and +1) into real distances on the map (hundreds of world units). If the model isn't loaded, the ghost simply disappears — it doesn't show wrong positions.

ChronovisorScanner — Critical moment detection:

The ChronovisorScanner analyzes the entire match and identifies the **decisive moments** based on advantage change:

Moment type	Condition	Marker color
Critical error	Death while in numerical advantage (e.g., 4v3 → 3v3)	🔴 Red
Excellent play	Kill while in numerical disadvantage (e.g., 2v3 → 2v2)	🟢 Green
Bomb plant	bomb_planted event with timer	🟡 Yellow
Clutch	1vN victory (N ≥ 2)	⭐ Gold
Eco round win	Victory with equipment < $2,000	🔵 Blue

Each moment is placed on the Timeline as a clickable marker. Clicking jumps playback directly to that tick.

Viewer loading flow (One-Click):

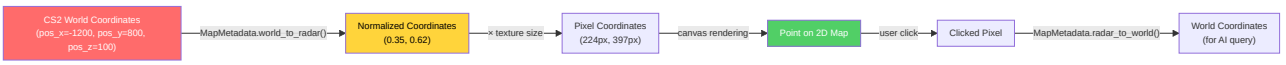
1. The user clicks "Tactical Viewer" from Home
2. If no demo is loaded → `trigger_viewer_picker()` automatically opens the file picker
3. The user selects a `.dem` file
4. A "Dynamic 2D Reconstruction in Progress..." dialog appears
5. A background thread runs `_execute_viewer_parse(path)` → `DemoLoader.load_demo()`
6. On completion: dialog dismissal, loading frames into PlaybackEngine
7. The map is rendered with players at frame 0

12.11 Spatial Data and Map Management

File: `Programma_CS2_RENAN/core/spatial_data.py`, `spatial_engine.py`, `data/map_config.json`

The map management system translates **CS2 world coordinates** (typical values: -2000 to +2000 on X/Y) into **pixel coordinates** on the map texture (0.0 to 1.0 normalized), and vice versa.

> **Analogy:** Map management is like a **GPS system for the CS2 world**. Each map has its "cartographic projection": an origin point (top-left corner), a scale (how many game units per pixel), and, for multi-level maps like Nuke, a **floor separator** (z_cutoff = -495 for Nuke). The GPS knows that if your Z coordinate is above -495, you're on the upper floor, otherwise you're on the lower one. This information is crucial for the GhostEngine and for correct rendering on the tactical map.



MapMetadata (immutable dataclass for each map):

Field	Example (Dust2)	Purpose
`pos_x`	-2476	X of top-left corner in world units
`pos_y`	3239	Y of top-left corner in world units
`scale`	4.4	Game units per texture pixel
`z_cutoff`	`None`	Level separator (multi-level maps only)
`level`	`"single"`	Type: "single", "upper", "lower"
`texture_width`	1024	Radar texture width in pixels
`texture_height`	1024	Radar texture height in pixels

MapManager (`core/map_manager.py`):

The MapManager is the high-level component that coordinates map loading for the interface:

Method	Purpose
`get_map_metadata(map_name)`	Returns MapMetadata for the requested map
`load_radar_texture(map_name)`	Loads PNG texture of the radar map from cache or disk
`get_available_maps()`	List of supported maps with metadata
`world_to_pixel(pos, map_name)`	Shortcut for world → pixel coordinate conversion

Supported multi-level maps:

Map	z_cutoff	Levels	Notes
Nuke	-495	upper / lower	Two plant sites on different floors
Vertigo	11700	upper / lower	Skyscraper with two playable areas
All others	`None`	single	Single-level maps

SpatialEngine (`core/spatial_engine.py`):

The SpatialEngine adds **spatial reasoning** capabilities to the base coordinate system:

Method	Input	Output	Use
`distance_2d(pos_a, pos_b)`	Two world coordinates	Float (world units)	Engagement distance computation
`is_visible(pos_a, pos_b, obstacles)`	Two pos + obstacles	Bool	Line of sight (simplified)
`get_zone(pos, map_name)`	Coordinate + map	String (e.g., "A_site")	Tactical zone classification
`nearest_cover(pos, map_name)`	Coordinate + map	Cover coordinate	Positional suggestion

AssetManager (`core/asset_manager.py`):

Handles loading of map textures and UI assets:

Asset	Format	Typical size	Cache
Map texture (radar)	PNG 1024×1024	~500KB	Yes (in-memory)
Player icons	PNG 32×32	~2KB	Yes
Fonts (JetBrains Mono, YUPIX)	TTF	~200KB	Yes (registered in Kivy)
Theme wallpapers	PNG 1920×1080	~2MB	Lazy load

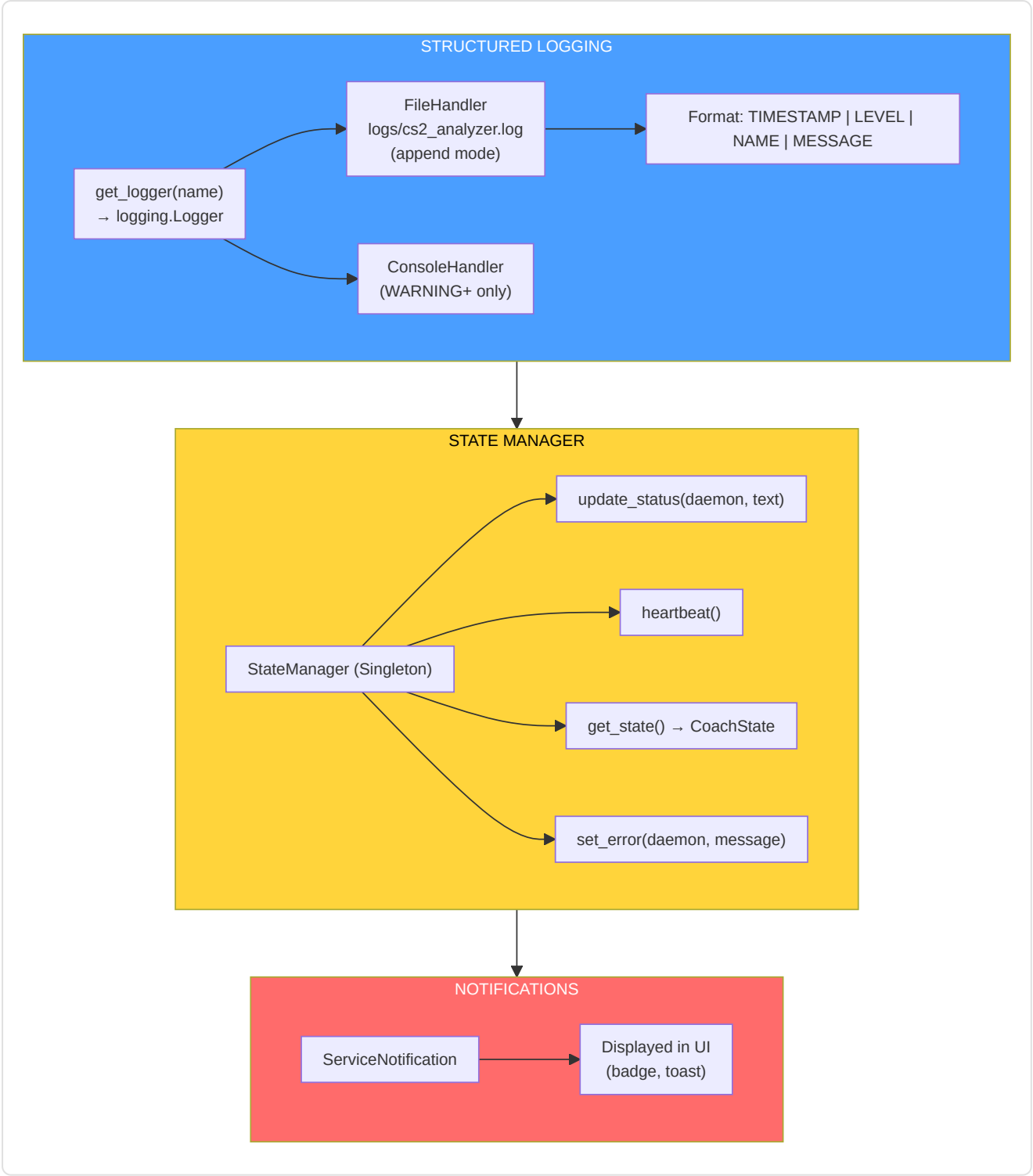
> **AssetManager Analogy:** The AssetManager is like the **theater warehouseman**. He knows exactly where all the props (textures, fonts, icons) are kept, loads them on demand, and keeps the most used in his **pocket** (cache) to avoid having to go back to the warehouse every time. If an asset doesn't exist, it returns a generic placeholder — the show must go on.

12.12 Observability and Logging

File: `Programma_CS2_RENAN/observability/logger_setup.py` **Related files:** `backend/storage/state_manager.py` , `backend/services/telemetry_client.py`

The observability system ensures that every significant event is **traceable, structured, and persistent**. The telemetry client (`telemetry_client.py`) uses `**httpx**` (asynchronous HTTP) for non-blocking sending of metrics and events, avoiding network latency affecting application performance.

> **Analogy:** Observability is like the **security camera system and registers of a building**. Structured logging (`get_logger()`) is the camera that records everything with timestamps and labels ("who did what, where, and when"). The StateManager is the **lobby whiteboard** showing the current status of every floor (daemon): "Floor 1 (Scanner): Scanning. Floor 2 (Digester): Idle. Floor 3 (Teacher): Learning." ServiceNotifications are the **intercom announcements** that inform residents (the user) of important events or errors.



Daemons monitored by StateManager:

Daemon	Field in CoachState	Typical values
Scanner	`hltv_status`	"Scanning", "Paused", "Error"
Digester (worker)	`ingest_status`	"Processing", "Idle", "Error"
Teacher (trainer)	`ml_status`	"Learning", "Idle", "Error"
Global	`status`	"Paused", "Training", "Idle", "Error"

Sentry Integration (`observability/sentry_setup.py`):

The system includes optional integration with **Sentry** for remote error tracking, with a **double opt-in** approach for privacy:

Feature	Implementation
Double opt-in	The user must (1) set `SENTRY_DSN` as an environment variable AND (2) activate the flag in settings
PII Scrubbing	All personal data (player names, Steam IDs, paths) are removed before sending
Breadcrumb sanitization	Navigation breadcrumbs are cleaned of sensitive information
Non-blocking	If Sentry is not configured, the system continues normally (Phase 4 of startup)
Enriched context	Every error includes: app version, daemon state, demo count, operating system

> **Analogy:** Sentry is like an **optional telemedicine service**. If you enable it (double opt-in), when your body (the application) has a serious malfunction, it automatically sends an anonymized report to the remote doctor (Sentry server). The report doesn't contain your name or address (PII scrubbing) — only the symptoms and circumstances of the incident. If you prefer total privacy, you don't enable the service and everything stays local.

Logger Setup (`observability/logger_setup.py`):

The centralized logging system provides structured logs with:

Feature	Detail
Format	`TIMESTAMP
File handler	`logs/cs2_analyzer.log` (append mode, automatic rotation)
Console handler	Only `WARNING+` to avoid polluting output
Naming convention	`get_logger("cs2analyzer.<module>")` — hierarchical namespace
Levels used	`DEBUG` (development), `INFO` (normal operations), `WARNING` (non-critical anomalies), `ERROR` (failures), `CRITICAL` (termination)

Training metrics exposed in CoachState: `current_epoch` , `total_epochs` , `train_loss` , `val_loss` , `eta_seconds` , `belief_confidence` , `system_load_cpu` , `system_load_mem` .

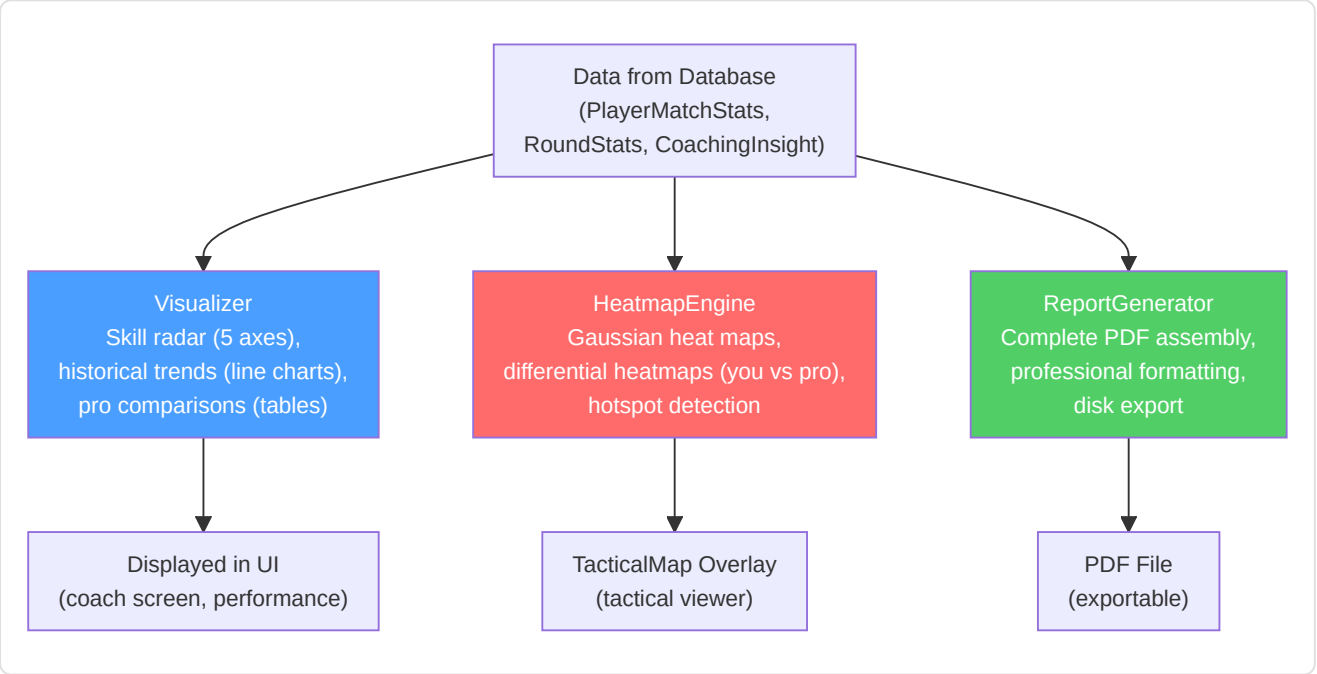
> **Analogy:** Training metrics are like a **car's instrument panel during a race**: the current epoch is the **odometer** (how far you've come), train_loss is the **fuel consumption** (lower = more efficient), val_loss is the **lap timer** (your time on the test track), and ETA is the **GPS** estimating how far to destination. The user can see all this in real time on the Home screen, updated every 10 seconds.

12.13 Reporting and Visualization

File: `Programma_CS2_RENAN/reporting/visualizer.py`, `report_generator.py` **Related files:** `backend/processing/heatmap_engine.py`

The reporting system transforms raw data into **understandable visualizations** for the user.

> **Analogy:** The reporting system is like a **graphic designer** who takes dry numbers and transforms them into colorful posters and infographics. The **Visualizer** creates radar charts (pentagons showing your 5 skills vs pros), trend charts (how you're improving over time), and comparison tables. The **HeatmapEngine** creates heat maps (where you stay too much vs where you should be). The **ReportGenerator** assembles everything into a complete PDF document, like a medical report that the "patient" (player) can study carefully.



HeatmapEngine — Technical details:

Feature	Detail
Thread-safety	`generate_heatmap_data()` runs in a separate thread (does not block UI)
Texture creation	Only in the main thread (Kivy OpenGL requirement)
Type	2D Gaussian occupancy (parametric blur kernel)
Differential	Subtracts pro heatmap from user heatmap → red (too much time), blue (too little)
Hotspot	Identifies position clusters for positional training

MatchVisualizer (`reporting/visualizer.py`) — specialized rendering methods:

The MatchVisualizer extends the reporting capability with 6 high-quality rendering methods (cf. section 12.26 for algorithmic details):

Method	Input	Output	Description
`render_skill_radar(user, pro)`	Metrics dict	PNG radar chart	5-axis pentagon with pro baseline overlay
`render_trend_chart(history, metric)`	Historical list	PNG line chart	Temporal trend with moving average
`render_heatmap(positions, map_name)`	Tick coordinates	PNG heatmap	2D Gaussian on map texture
`render_differential(user_pos, pro_pos)`	Two position sets	PNG diff heatmap	Red (excess) / Blue (deficit) / Green (alignment)
`render_critical_moments(events)`	Round events list	PNG timeline	Color markers for kill/death/bomb
`render_comparison_table(user, pro)`	Two stats profiles	HTML/PNG table	Percentage delta for each metric

ReportGenerator (`reporting/report_generator.py`):

Assembles all visual elements into a **complete PDF document**:

1. **Header**: Player name, date, demo analyzed
2. **Executive Summary**: Key metrics (KPR, ADR, KAST, Rating)
3. **Radar Chart**: 5-axis comparison with pro baseline
4. **Trend Charts**: Trends of the last N matches
5. **Differential Heatmap**: Where to position better
6. **Critical Moments**: The 5 decisive moments of the match
7. **Coach's Advice**: Top-3 prioritized insights
8. **Footer**: Generated by Macena CS2 Analyzer v.X.X

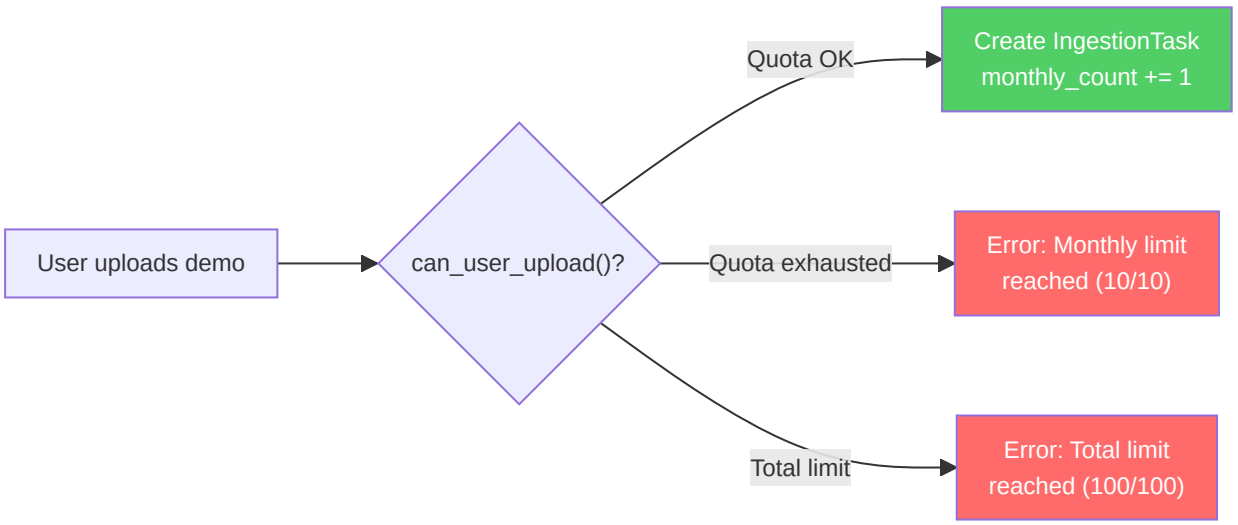
> **PDF Report Analogy:** The report is like a **complete medical report after a check-up**. The first page has the summary ("patient in good health, cholesterol to improve"). Then come the detailed charts (blood analysis = radar chart, CT scan = heatmap, ECG = trend). Finally, the doctor's recommendations (the coach's advice). The patient can study it carefully at home and bring it to the next appointment.

12.14 Quota and Limit Management

The system implements a **monthly quota** mechanism to prevent abuse of processing resources and guarantee fair load distribution.

> **Analogy:** The quota system is like a **gym subscription with limited sessions**. Every month you have 10 sessions (demos) available. Every time you upload a demo, the counter decreases. At the beginning of the next month, the counter resets. Additionally, there is a lifetime total limit of 100 demos: like a diary that only has 100 pages. These limitations ensure that the system isn't overloaded by users uploading hundreds of demos at once, and that the database doesn't grow uncontrollably.

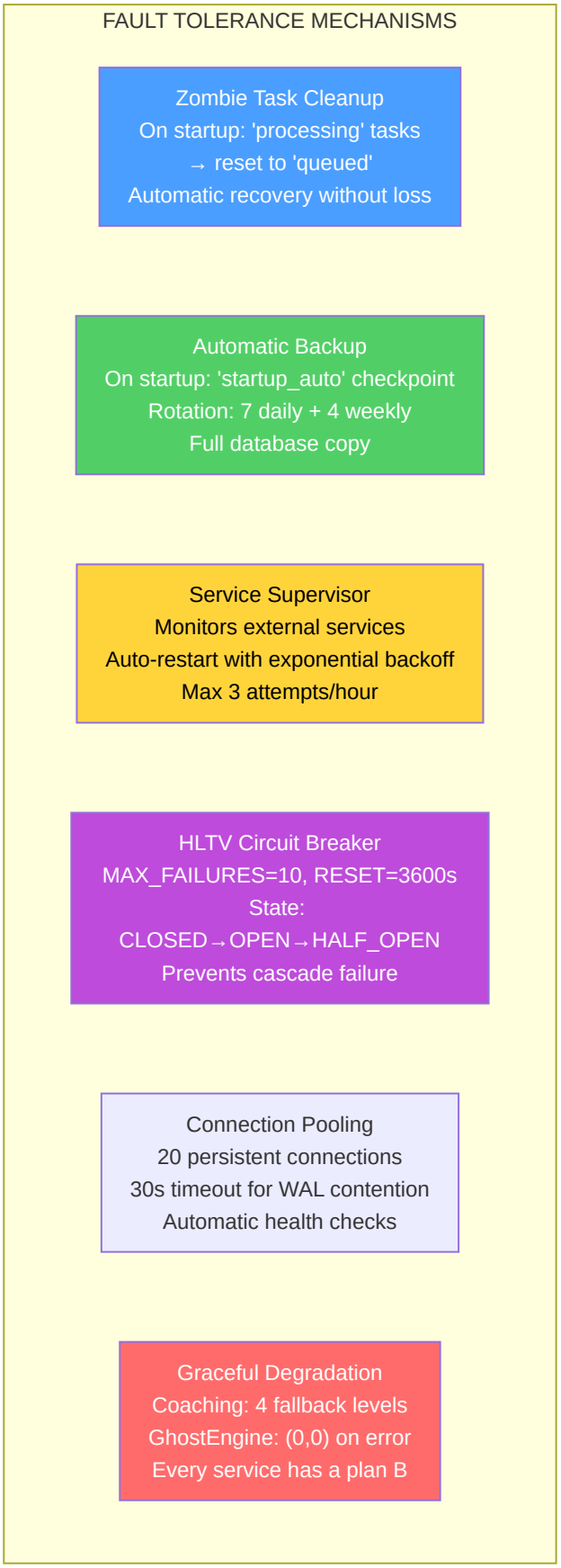
Limit	Value	Enforcement
MAX_DEMOS_PER_MONTH	10	`StorageManager.can_user_upload()`
MAX_TOTAL_DEMOS	100	`StorageManager.can_user_upload()`
MIN_DEMOS_FOR_COACHING	10	Threshold for full personalized coaching
Monthly reset	Automatic	`PlayerProfile.last_upload_month` vs current month



12.15 Fault Tolerance and Recovery

The system is designed to **never lose data** and to **recover automatically** from almost all types of failure.

> **Analogy:** Fault tolerance is like a **safety system in a nuclear power plant**: there are multiple protection levels, each independent of the others. If one level fails, the next one activates. (1) **Zombie Task Cleanup** is like the night shift cleaning up forgotten tools: if the previous shift (previous session) left tasks in "processing" state due to a crash, the next shift resets them and restarts them. (2) **Automatic Backup** is like the fireproof safe: even if the entire building burns down, the documents in the safe survive. (3) **Service Supervisor** is like the guard restarting the emergency generators if they shut down. (4) **Connection Pooling** is like having key reserves: if one lock jams, there are 19 more available.

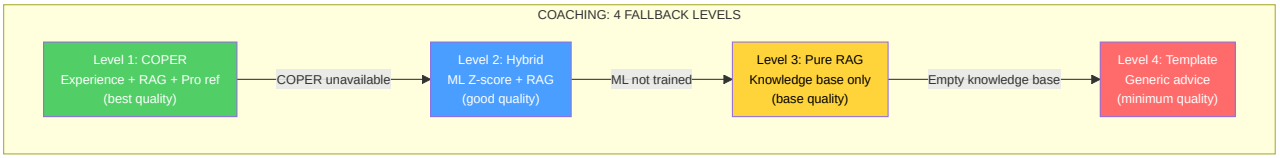


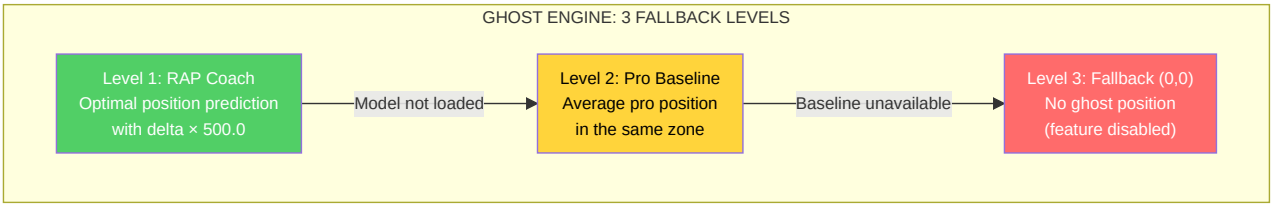
Failure and recovery matrix:

Failure scenario	Recovery mechanism	Data loss
Application crash during parsing	Zombie Task Cleanup on restart	None — task restarted
Database corruption	Restore from most recent automatic backup	Up to 24 hours of data
HLTV service unreachable	Circuit Breaker `_CircuitBreaker` (MAX_FAILURES=10, RESET_WINDOW_S=3600): after 10 consecutive failures the circuit opens and blocks requests for 1 hour, then passes to HALF_OPEN for a test. Prevents cascade failure on external APIs.	None — pro data delayed
ML model not loadable	Fallback to random weights (GhostEngine) or base coaching	None — degraded quality
Insufficient RAM during training	Automatic early stopping, checkpoint saved	None — last valid checkpoint
Disk full	Database Governor detects and notifies via ServiceNotification	Prevention — no data written
Corrupt/truncated demo	IntegrityChecker rejects before parsing	None — demo ignored with warning
Ollama unavailable	LLM fallback to pure RAG, then base coaching	None — degraded narrative quality
Steam/FACEIT API timeout	Retry with exponential backoff (max 3 attempts)	None — profile not updated
Playwright browser crash	BrowserManager: recreates instance, retries operation	None — HLTV scraping delayed
WAL conflict (lock contention)	30s timeout with automatic retry	None — operation delayed
Corrupt ML checkpoint	Fallback to previous checkpoint (versioning)	Partial — loses last training
Zero-norm query vector (M-07)	`VectorIndex.search()` returns `None` with warning — RAG pipeline continues without crash	None — empty RAG result
Corrupt settings.json file (M-08)	`load_user_settings()` validates structure, falls back to default if non-dict	None — default settings
Logger not initialized (C-09)	Early bootstrap uses `print()` before logger init	None — messages on stdout

Failure scenario	Recovery mechanism	Data loss
DB handle leak on integrity check (H-02)	`_verify_integrity()` uses `try/finally` + `PRAGMA quick_check`	None — connection always closed
Masked batch errors in RAP training (H-03)	Removed `except (KeyError, TypeError): continue` — `train_step()` requires dict with `'loss'` key	None — errors now visible

Graceful degradation — The Fallback Chain:





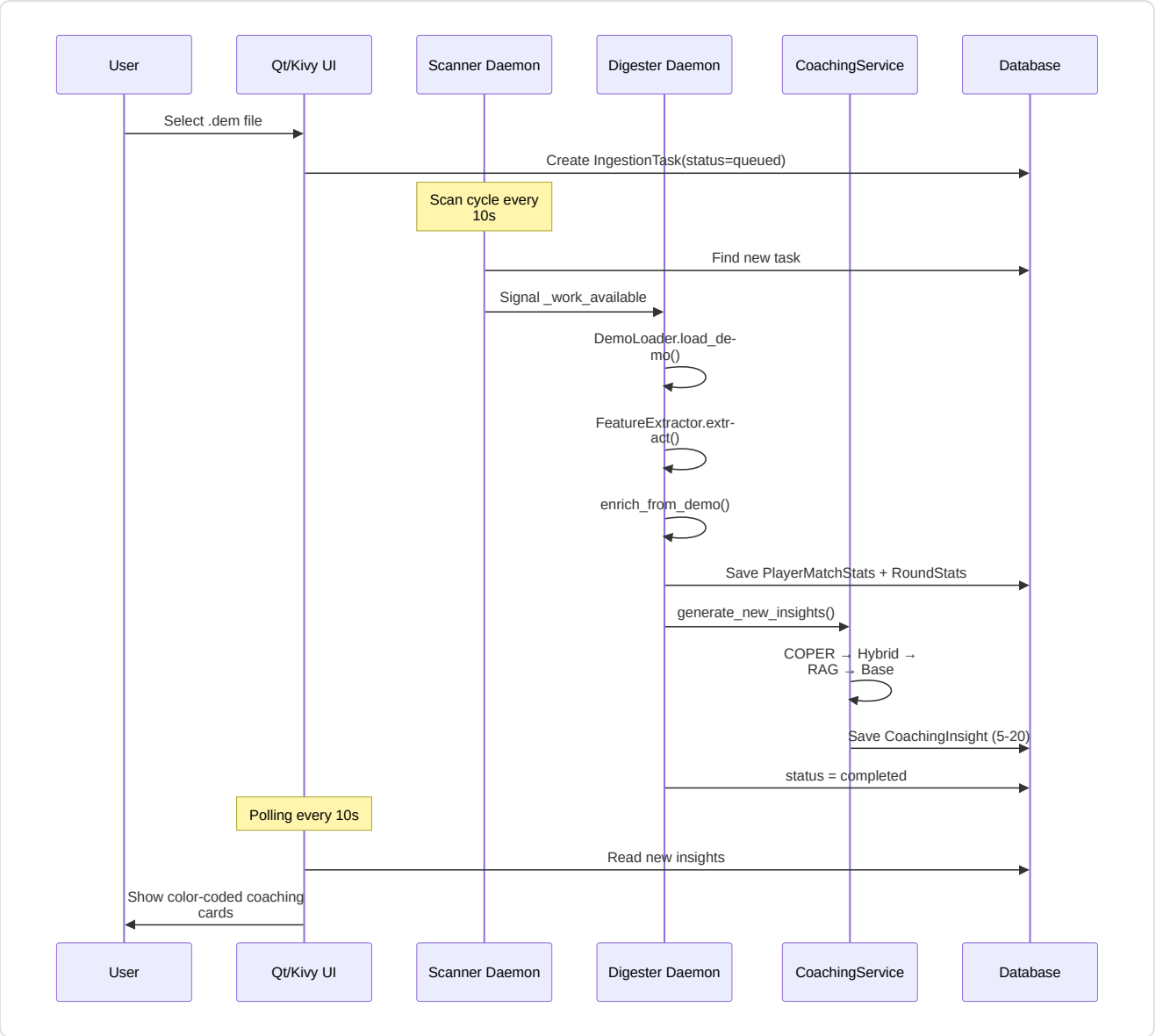
> **Degradation Analogy:** Graceful degradation is like a **4-level lighting system in case of blackout**. Level 1: main power (full COPER — the best experience). Level 2: backup generator (ML+RAG hybrid — good but not perfect). Level 3: emergency batteries (pure RAG — lights are on but dim). Level 4: handheld torches (generic templates — you see just barely, but don't stay in the dark). The system **never crashes** — it always degrades in a controlled and transparent way.

12.16 Complete User Journey — 4 Main Flows

This section describes the **4 main flows** a user goes through while using Macena CS2 Analyzer.

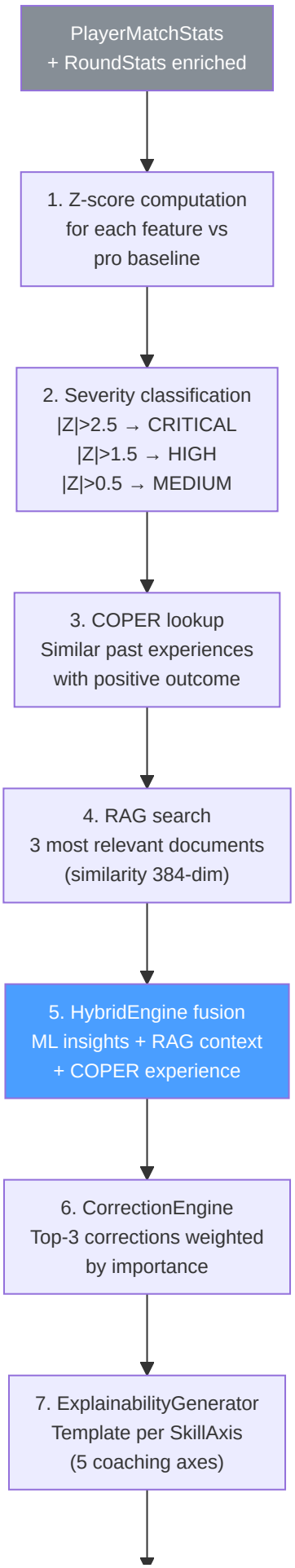
Flow 1: Demo Upload and Analysis

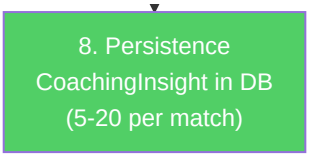
> **Analogy:** This flow is like **bringing a blood sample to the lab**: you hand it in at reception (upload), the technician prepares it (parsing), the chemist analyzes it (feature extraction), the doctor reads the results (coaching service), and you receive the report (insights in the UI). Everything happens automatically once the sample is delivered.



Detail: Internal CoachingService Pipeline

When `generate_new_insights()` is invoked, the CoachingService internally runs:


PlayerMatchStats
+ RoundStats enriched
1. Z-score computation
for each feature vs
pro baseline
2. Severity classification
|Z|>2.5 → CRITICAL
|Z|>1.5 → HIGH
|Z|>0.5 → MEDIUM
3. COPER lookup
Similar past experiences
with positive outcome
4. RAG search
3 most relevant documents
(similarity 384-dim)
5. HybridEngine fusion
ML insights + RAG context
+ COPER experience
6. CorrectionEngine
Top-3 corrections weighted
by importance
7. ExplainabilityGenerator
Template per SkillAxis
(5 coaching axes)



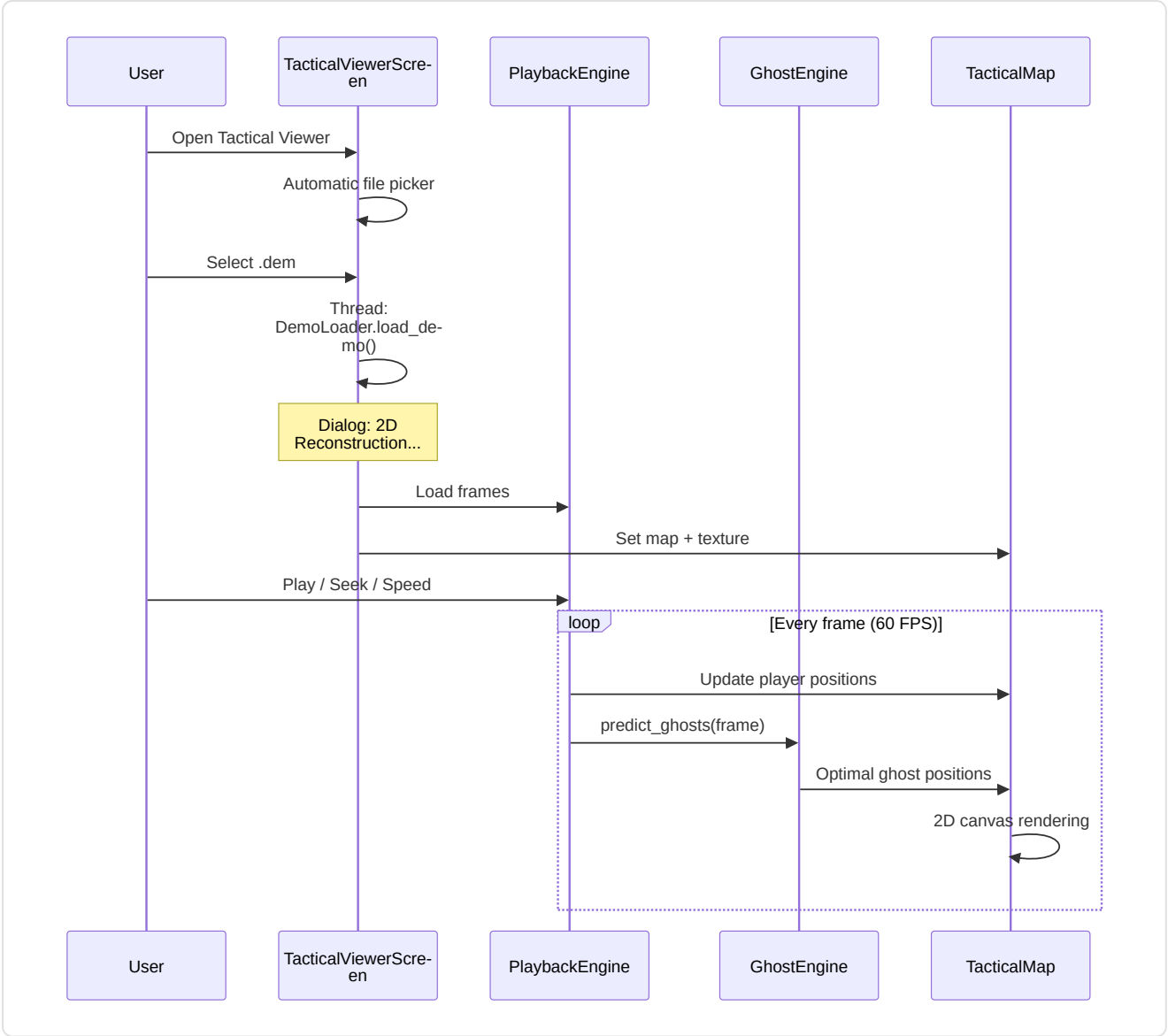
LessonGenerator — generation of structured lessons from demo:

Parameter	Value	Meaning
`ADR_STRONG`	75	ADR ≥ 75 → "Good average damage"
`HS_STRONG`	0.40	HS% ≥ 40% → "Good headshot accuracy"
`KAST_STRONG`	0.70	KAST ≥ 70% → "Good round contribution"
Output format	Section 1 (Overview) + Section 2 (Strengths) + Section 3 (Areas for improvement) + Section 4 (Pro tip)	Standard lesson structure

The LessonGenerator operates in dual mode: if Ollama is available, it generates natural language lessons via `LLMService.generate_lesson()`. If not available, it uses structured templates with numeric data inserted into preformed phrases.

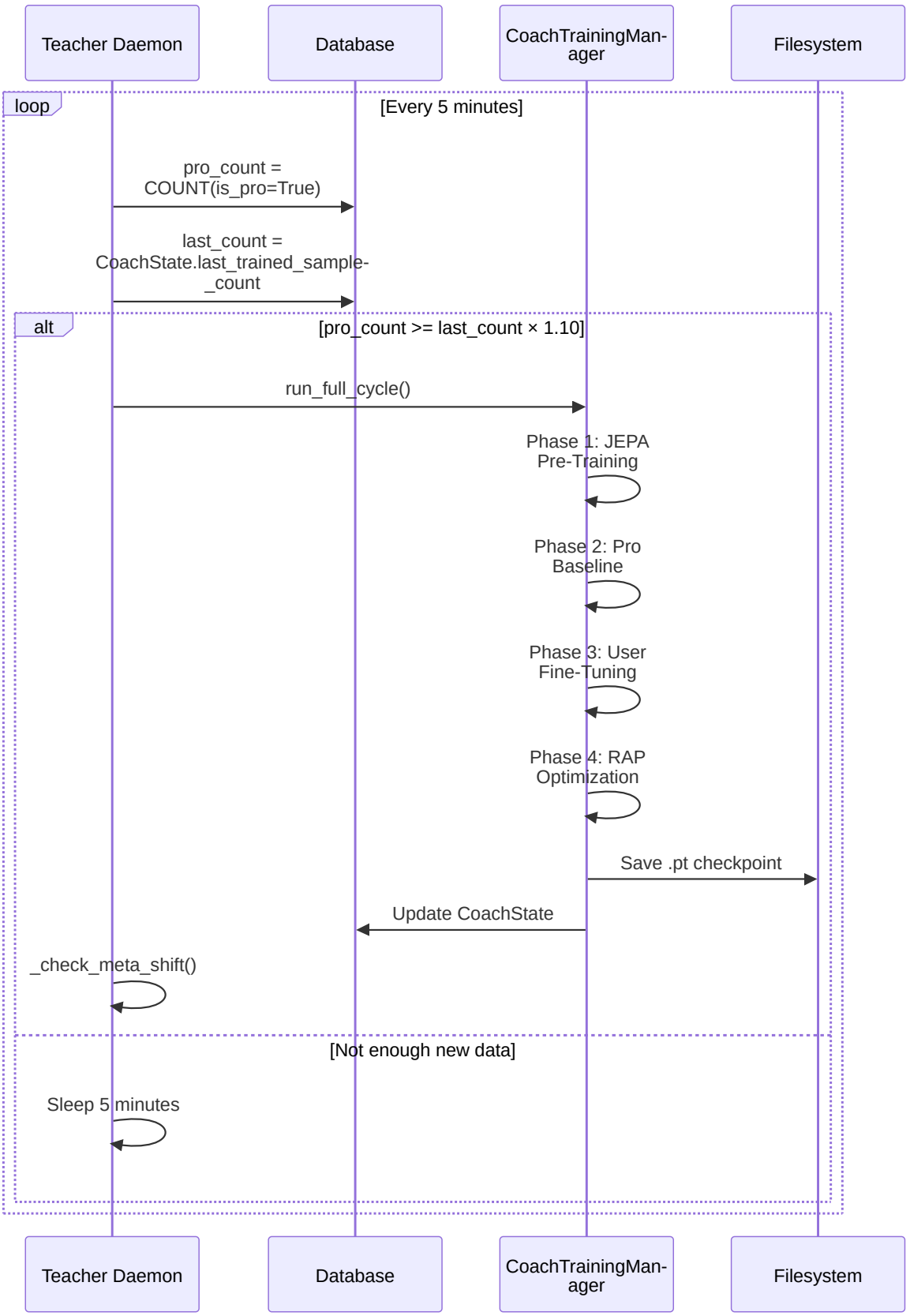
Flow 2: Tactical Visualization

> **Analogy:** This flow is like **watching a recorded match in slow motion** with an AI commentator. You load the video (demo), the system decodes it, and then you can navigate back and forth, slow down, speed up, and see where the commentator thinks you should have positioned.



Flow 3: Automatic ML Retraining

> **Analogy:** This flow is completely **automatic**, like an airplane's autopilot. The user doesn't have to do anything: the system constantly monitors the amount of available data and, when enough new data is available, automatically retrains the model to keep it updated with the current meta.


Teacher Daemon
Database
CoachTrainingManager
Filesystem
loop
[Every 5 minutes]
pro_count = COUNT(is_pro=True)
last_count = CoachState.last_trained_sample_count
alt
[pro_count >= last_count × 1.10]
run_full_cycle()
Phase 1: JEPA Pre-Training
Phase 2: Pro Baseline
Phase 3: User Fine-Tuning
Phase 4: RAP Optimization
Save .pt checkpoint
Update CoachState
_check_meta_shift()
[Not enough new data]
Sleep 5 minutes
Teacher Daemon
Database
CoachTrainingManager
Filesystem

Flow 4: AI Chat with the Coach

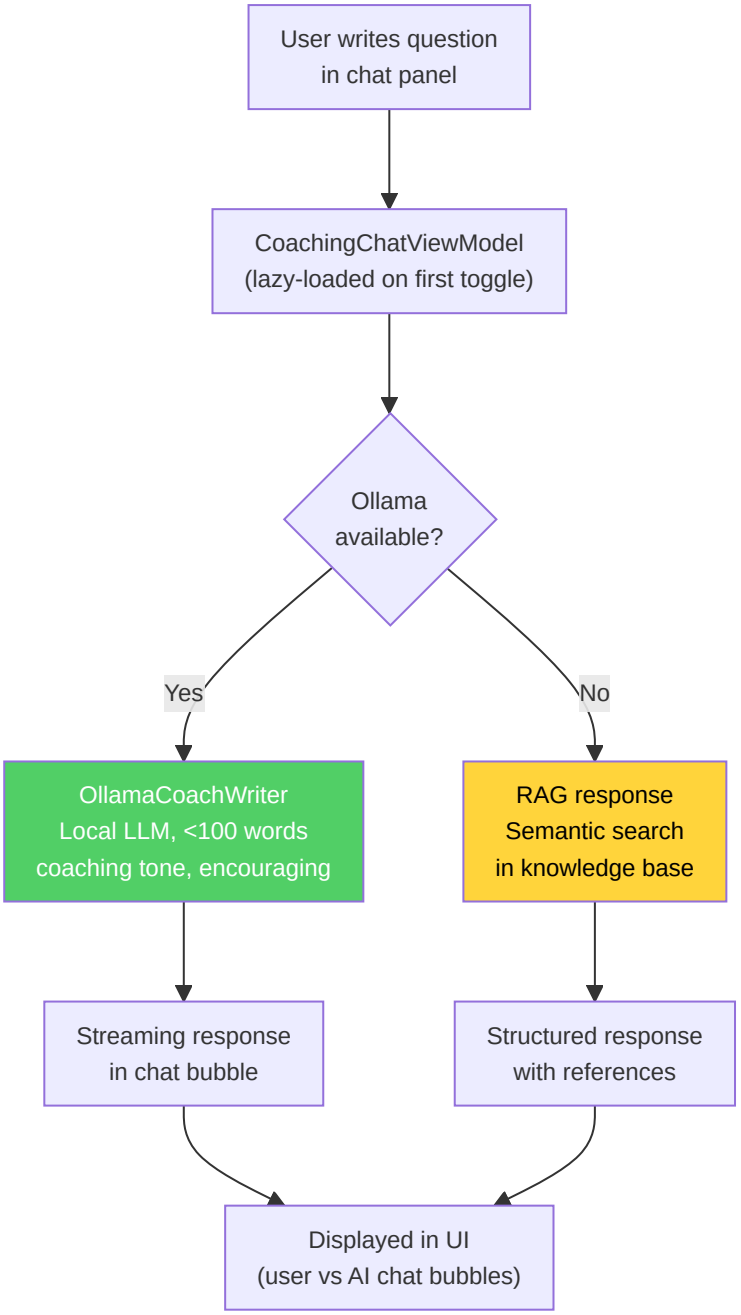
> **Analogy:** The chat is like having a **private conversation with your coach**. You can ask specific questions ("What should I improve on Dust2?"), and they consult their knowledge base, your statistics, and pro references to give you a personalized answer. If Ollama is installed locally, the response sounds natural and motivating. If not available, the system still uses the RAG knowledge base to provide useful answers.

CoachingDialogueEngine — Multi-Turn Pipeline:

The AI dialogue follows a 5-phase pipeline for each user message:

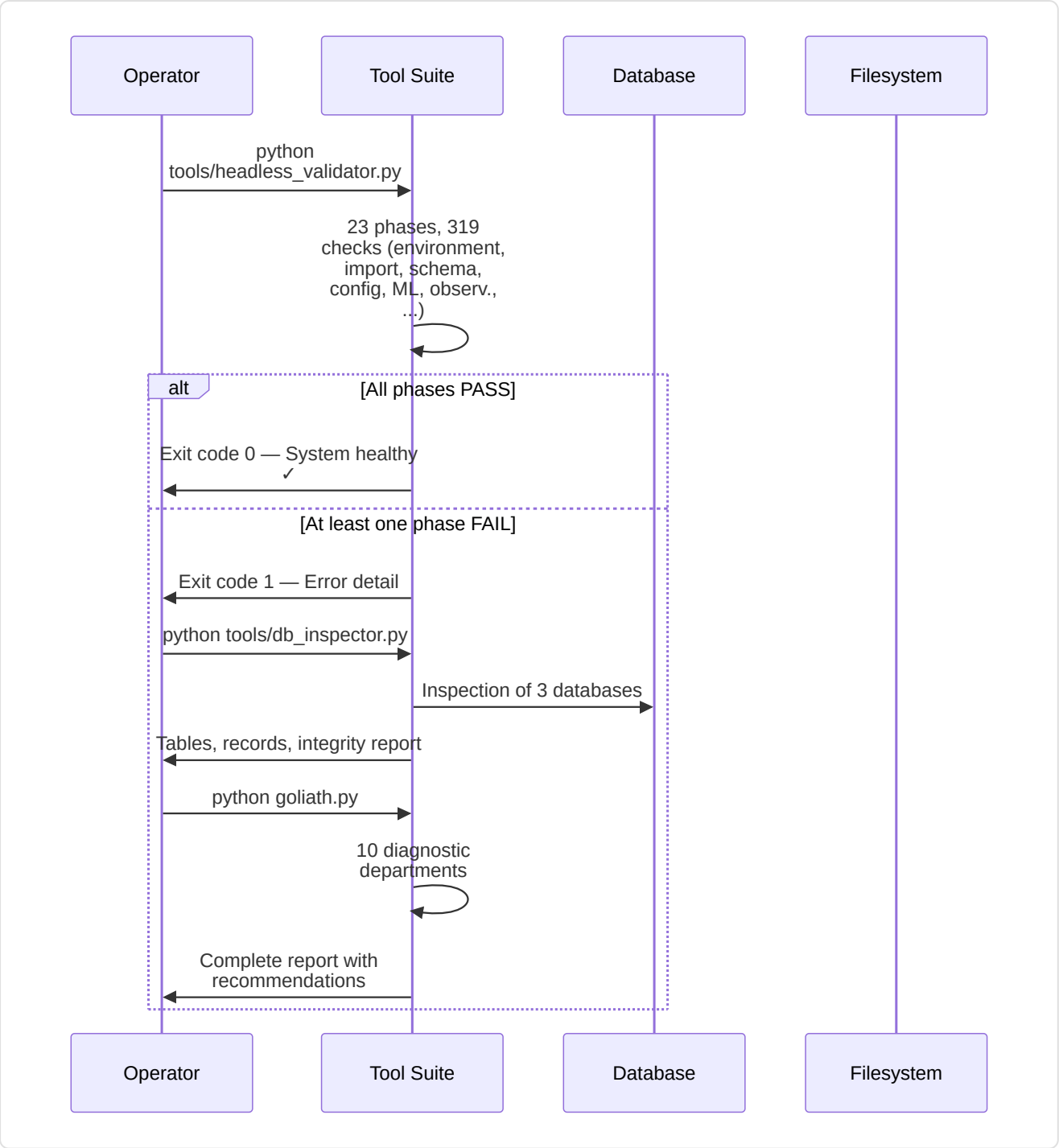
Phase	Component	Description
1	**Intent Classification**	Classifies the question: coaching, stats query, comparison, general
2	**Context Retrieval**	Retrieves the last N matches, recent insights, player profile
3	**RAG Augmentation**	Searches the knowledge base for the 3 most relevant documents (similarity search 384-dim)
4	**COPER Augmentation**	If available, adds past coaching experiences with positive outcome
5	**Response Generation**	Generates response via Ollama (if available) or structured RAG template

Parameter	Value	Purpose
`MAX_CONTEXT_TURNS`	6	Maximum number of turns kept in context
`MAX_RESPONSE_WORDS`	100	Word limit for LLM response
`SIMILARITY_THRESHOLD`	0.5	Minimum threshold for RAG results
Tone	"coaching, encouraging"	System prompt for positive tone



Flow 5: Diagnostics and Maintenance

> **Analogy:** This flow is the **periodic car service**. The operator (developer or advanced user) brings the system to the workshop (console or tools) and has a series of checks performed. It's not necessary to do it every day, but it's recommended after updates, after periods of inactivity, or when something seems to go wrong.



Tool matrix by scenario:

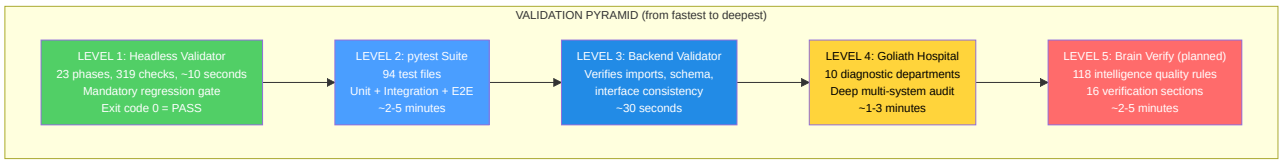
Scenario	Recommended tool	Time
Quick check post-deploy	`headless_validator.py`	~10s
Suspicious database	`db_inspector.py`	~30s
Demo not parsed	`demo_inspector.py`	~5s
Training diverges	`Ultimate_ML_Coach_Debugger.py`	~2min
Complete check-up	`Goliath_Hospital.py`	~3min
Deep ML audit	`brain_verify.py`	~5min

12.17 Tool Suite — Validation and Diagnostics (`tools/`)

Directory: `tools/` (root) + `Programma_CS2_RENAN/tools/` **Main files:** `headless_validator.py`, `db_inspector.py`, `demo_inspector.py`, `brain_verify.py` *(planned, not yet implemented)*, `Goliath_Hospital.py`, `Ultimate_ML_Coach_Debugger.py`, `_infra.py`

The tool suite is a **collection of 41 scripts** distributed across two directories (`tools/` root with 29 scripts and `Programma_CS2_RENAN/tools/` with 12 scripts) that form a multi-level validation pyramid. Each tool has a precise purpose and can be executed independently, but together they form a quality assurance system covering every aspect of the project.

> **Analogy:** The tool suite is like the **quality control department of an automobile factory**. Every vehicle (project build) goes through a series of inspection stations, each specialized in an aspect: the **Headless Validator** is the quick test station that verifies the engine starts and the wheels turn (23-phase, 319-check regression gate). The **DB Inspector** is the mechanic who dismantles and checks the engine piece by piece (database inspection). The **Demo Inspector** is the technician who verifies the bodywork (demo file integrity). **Brain Verify** is the neurologist who tests every cognitive function (118 quality rules for intelligence). The **Goliath Hospital** is the complete hospital with 10 specialized departments. No vehicle leaves the factory without passing all checks.



Headless Validator (`tools/headless_validator.py`) — the mandatory regression gate (Dev Rule 9). Executed after **every** development task with 23 phases and 319 checks:

Phase	Verification	Detail
1	**Environment**	Python ≥ 3.10, critical dependencies present (torch, kivy, sqlmodel, demoparser2)
2	**Core Import**	`config.py`, `spatial_data.py`, `lifecycle.py` — the fundamental modules load without errors
3	**Backend Import**	`nn/`, `processing/`, `storage/`, `services/`, `coaching/` — all backend subsystems importable
4	**DB Schema**	The 21 SQLModel tables are created correctly, the relationships are valid
5	**Configuration**	`METADATA_DIM`, paths, constants — consistent and reachable values
6	**ML Smoke**	Model instantiation (JEPA, RAP, MoE) with random weights — verify dimensions and forward pass
7	**Observability**	`get_logger()` working, `StateManager` initializable, writable log path

> **Analogy:** The Headless Validator is the **thermometer** of the project. It doesn't make an in-depth diagnosis, but if the temperature is high (exit code ≠ 0), you immediately know something is wrong. It's the first check to do, fast and non-invasive. If it passes, you can proceed with confidence. If it fails, stop and investigate before moving on.

Shared Infrastructure (`tools/_infra.py`):

Component	Role
`BaseValidator`	Base class for all validators — unified console output, pass/fail/warning count
`path_stabilize()`	`sys.path` normalization to avoid inconsistent relative imports
`Console` (Rich)	Color output with tables, progress bars, status emoji
`Severity` enum	4 levels: `INFO`, `WARNING`, `ERROR`, `CRITICAL`

DB Inspector (`tools/db_inspector.py`) — deep database inspection:

- Opens `database.db` and `hltv_metadata.db` separately
- For each table: counts records, verifies schema, checks index integrity
- Displays space metrics (file size, WAL pages, fragmentation)
- Detects anomalies: unexpected empty tables, orphan records, out-of-range timestamps
- Output: color Rich table with status for each table

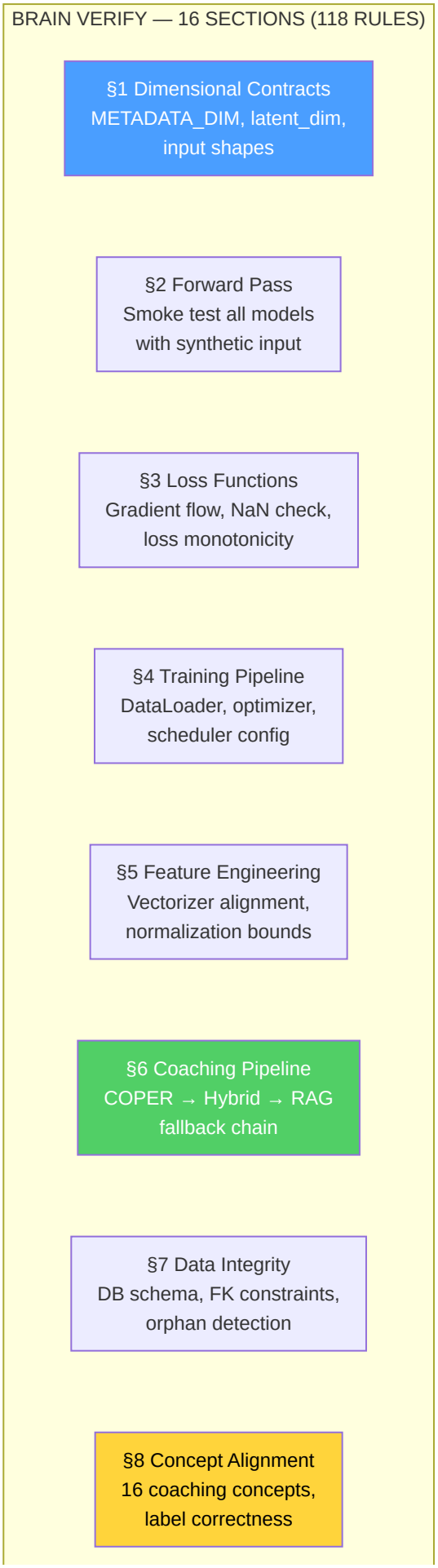
Demo Inspector (`tools/demo_inspector.py`) — demo file inspection:

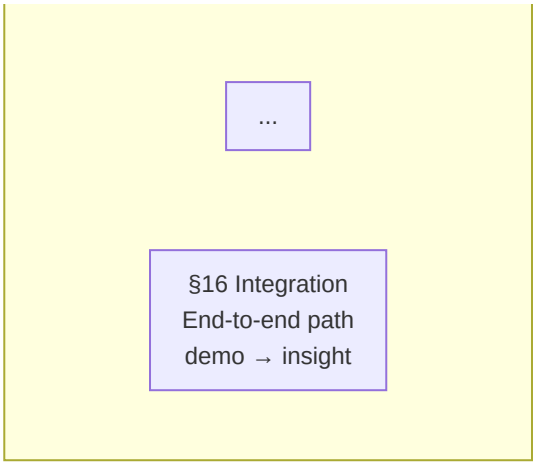
- Verifies magic bytes (`PBDEMS2\0` for CS2, `HL2DEMO\0` for legacy)
- Checks file size (1KB min, 5GB max)
- Extracts header metadata without full parsing
- Detects partial corruption (truncation, damaged header)

- Output: report on demo validity with error details if invalid

Brain Verify (`tools/brain_verify.py` + `brain_verification/`, 16 sections) — *(planned, not yet implemented)*:

The intelligence verification system is designed with **16 thematic sections** that will cover **118 rules** of quality:


BRAIN VERIFY — 16 SECTIONS (118 RULES)
§1 Dimensional Contracts
METADATA_DIM, latent_dim, input shapes
§2 Forward Pass
Smoke test all models with synthetic input
§3 Loss Functions
Gradient flow, NaN check, loss monotonicity
§4 Training Pipeline
DataLoader, optimizer, scheduler config
§5 Feature Engineering
Vectorizer alignment, normalization bounds
§6 Coaching Pipeline
COPER → Hybrid → RAG fallback chain
§7 Data Integrity
DB schema, FK constraints, orphan detection
§8 Concept Alignment
16 coaching concepts, label correctness



Goliath Hospital (`tools/Goliath_Hospital.py`) — multi-department diagnostics:

The "Goliath Hospital" is the **most complete diagnostic system** of the project, organized as a hospital with 10 specialized departments:

Department	Name	Checks
1	**Emergency Room (ER)**	Critical imports, immediate crashes, path resolution
2	**Radiology**	File/directory structure, missing files, permissions
3	**Pathology**	DB schema, data integrity, anomalous records
4	**Cardiology**	Session Engine, daemon heartbeat, IPC
5	**Neurology**	ML models, forward pass, gradient flow
6	**Oncology**	Dead code, unused imports, circular dependencies
7	**Pediatrics**	Onboarding, first-user flows, wizard
8	**Intensive Care (ICU)**	Concurrency, race conditions, WAL contention
9	**Pharmacy**	Dependencies, versions, compatibility
10	**Tool Clinic**	Validation of other tools (meta-test)

> **Analogy:** Goliath Hospital is like a **complete medical check-up**. It doesn't just take your temperature (Headless Validator) — it takes X-rays, blood tests, electrocardiogram, MRI, and calls 10 different specialists to examine every aspect of your health. If a department finds a problem, it flags it with the appropriate severity. The final result is a **complete report** with prioritized recommendations.

Ultimate ML Coach Debugger (`tools/Ultimate_ML_Coach_Debugger.py`) — neural belief falsification:

This tool performs a **3-phase audit** on the ML pipeline:

1. **Structural Verification** — Tensor sizes, model architecture, learnable parameters
2. **Behavioral Verification** — Forward pass with real data, gradient flow, loss convergence

3. **Belief Falsification** — Tests whether the model "believes" wrong things: overconfident predictions, systematic biases, degenerate patterns

Other tools:

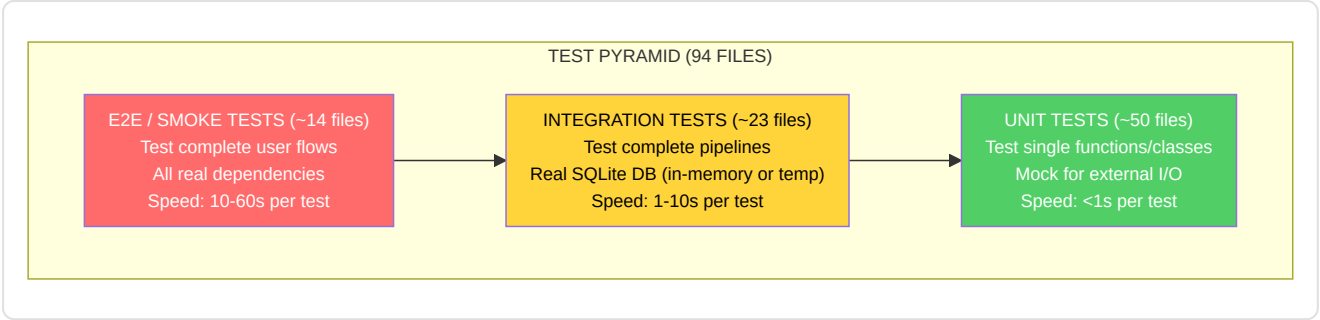
Tool	File	Purpose
`backend_validator.py`	`tools/`	Verifies backend import and interface consistency
`build_tools.py`	`tools/`	PyInstaller build automation
`user_tools.py`	`tools/`	Utilities for the end user (reset, export)
`dev_health.py`	`tools/`	Quick development health check
`context_gatherer.py`	`tools/`	Gathers context for debug reports
`dead_code_detector.py`	`tools/`	Identifies unreferenced dead code
`sync_integrity_manifest.py`	`tools/`	Updates `integrity_manifest.json` with SHA-256 hashes
`project_snapshot.py`	`tools/`	Complete snapshot of project state
`ui_diagnostic.py`	`tools/`	UI-specific diagnostics (Qt/Kivy)
`build_pipeline.py`	Root `tools/`	Complete build pipeline
`Feature_Audit.py`	Root	Audit of the 25-dim feature vector
`Sanitize_Project.py`	Root	Cleanup of temporary files, cache, artifacts
`audit_binaries.py`	Root	Verifies integrity of executables and .pt

12.18 Test Suite Architecture (`tests/`)

Directory: `Programma_CS2_RENAN/tests/` **Total files:** 94 test files + `conftest.py` + 10 forensics scripts + 15 verification scripts

The test suite is organized according to the **test pyramid principle**: many unit tests (fast, isolated), fewer integration tests (slower, with real dependencies), and few end-to-end tests (complete but costly).

> **Analogy:** The test suite is like the **exam system of a school**. **Unit tests** are daily quizzes: fast, focused on a single topic, easy to grade. **Integration tests** are mid-term exams: they verify that multiple topics work together. **E2E tests** are final exams: they simulate the entire student experience from enrollment to graduation. **Forensics** are special investigations: when something goes wrong, they dig deep to find the root cause.



Conftest (`tests/conftest.py`) — shared fixtures:

Fixture	Scope	Description
`test_db`	`function`	In-memory SQLite database with all tables created, auto-cleanup
`sample_match_stats`	`function`	`PlayerMatchStats` with realistic data derived from real matches
`mock_demo_data`	`function`	Mock `DemoData` with frames and events for parsing tests
`tmp_models_dir`	`function`	Temporary directory for `.pt` checkpoints
`coaching_service`	`function`	`CoachingService` configured with test DB

Automated Suite (`tests/automated_suite/`):

Category	File	Coverage
Smoke	`test_smoke_*.py`	Critical imports, model instantiation, DB connection
Unit	`test_unit_*.py`	Pure functions, calculations, transformations
Functional	`test_functional_*.py`	Complete pipelines, workflows
E2E	`test_e2e_*.py`	Demo upload → parsing → coaching → insight
Regression	`test_regression_*.py`	Specific bug fixes (G-01 label leakage, G-07 Bayesian, etc.)

Test per subsystem:

Subsystem	Test files	What they test
NN Core	`test_jepa_model.py`, `test_rap_model.py`, `test_advanced_nn.py`	Forward pass, tensor dimensions, gradient flow
Coaching	`test_coaching_service.py`, `test_hybrid_engine.py`, `test_correction_engine.py`	Coaching pipeline, insight prioritization, fallback chain
Processing	`test_vectorizer.py`, `test_heatmap.py`, `test_feature_eng.py`	Feature extraction, normalization, METADATA_DIM=25
Storage	`test_database.py`, `test_models.py`, `test_backup.py`	CRUD, schema, backup/restore, WAL mode
Data Sources	`test_demo_parser.py`, `test_hltv_scraper.py`, `test_steam_api.py`	Parsing, scraping, API timeout/retry
Analysis	`test_game_theory.py`, `test_belief.py`, `test_role_engine.py`	Analysis engines, calibration, heuristics
Knowledge	`test_rag.py`, `test_experience_bank.py`, `test_knowledge_graph.py`	Retrieval, COPER effectiveness, KG query
Ingestion	`test_ingest_pipeline.py`, `test_registry.py`	Complete pipeline, deduplication

Forensics (`tests/forensics/`, 10 scripts):

Forensics scripts are diagnostic tools for post-mortem investigations:

Script	Purpose
`diagnose_training_failure.py`	Analyzes training logs to identify divergence, NaN, gradient explosion
`inspect_model_weights.py`	Weight distribution, layer statistics, dead neurons
`replay_ingestion.py`	Re-runs ingestion of a specific demo with verbose logging
`trace_coaching_path.py`	Traces the path of an insight from demo to UI
`db_consistency_check.py`	Verifies consistency across the 3 databases

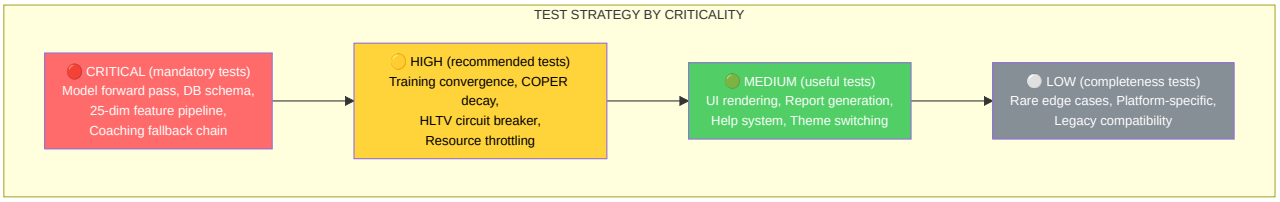
> **Forensics Analogy:** Forensics scripts are like an **airplane's black box**. After an incident (bug, crash, unexpected result), you use them to reconstruct exactly what happened, step by step. They are not executed routinely — only when investigating a specific problem is needed.

Verification Scripts (15 files in `tests/` root):

One-shot verification scripts to validate specific aspects of the system:

Script	Verification
`verify_feature_pipeline.py`	METADATA_DIM=25 respected in all paths
`verify_training_cycle.py`	4 training phases complete without error
`verify_db_schema.py`	21 tables present with correct schema
`verify_coaching_pipeline.py`	Demo → insight end-to-end path
`verify_imports.py`	All modules importable without circular errors
`verify_rag_index.py`	Knowledge base indexed with correct dimensions (384-dim)
`verify_pro_baseline.py`	Pro baseline loaded and valid
`verify_hltv_sync.py`	HLTV sync service configured correctly

Test strategy by criticality:



Identified coverage gaps:

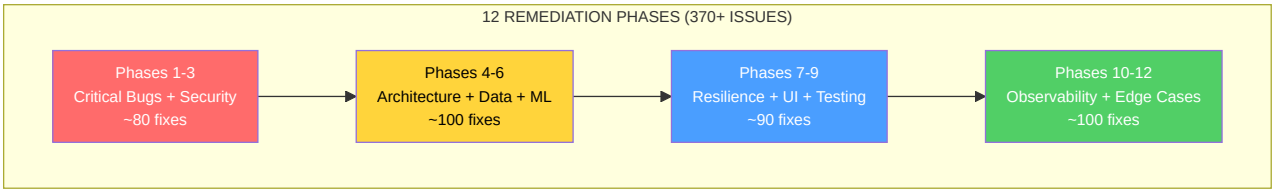
The following modules have high complexity (>500 LOC) but limited test coverage:

Module	LOC	Test coverage	Risk
`training_orchestrator.py`	733	Low	HIGH — multi-phase orchestration
`session_engine.py`	538	Medium	MEDIUM — testable only with subprocess
`coaching_service.py`	585	Medium	MEDIUM — 4 fallback levels
`experience_bank.py`	751	Low	HIGH — complex COPER logic
`tensor_factory.py`	686	Low	HIGH — DataLoader/Dataset
`coach_manager.py`	878	Low	HIGH — global training state

12.19 The 12 Phases of Systematic Remediation

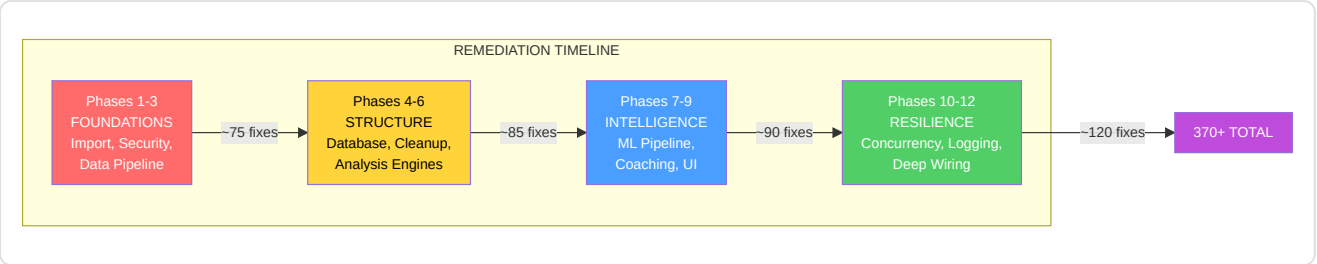
The project went through a **12-phase remediation** process that overall resolved **370+ issues** identified during progressive quality audits. Each phase focused on a specific category of problems, from critical bug fixes to architectural restructuring.

> **Analogy:** The 12 remediation phases are like the **progressive restoration of a historic building**. Phase 1 repaired the **foundations** (critical bugs preventing operation). The intermediate phases reinforced the **load-bearing walls** (architecture, security, data). The final phases completed the **finishes** (logging, edge cases, documentation). Each phase produced a detailed report in the `reports/` directory.



Detail per phase:

Phase	Focus	Issues resolved	Key example
1	Import and base structure	~20	Circular imports, path resolution, missing `__init__.py`
2	Security and secrets	~25	Hard-coded API keys → env vars/keyring, input validation
3	Data pipeline and features	~30	G-01 label leakage, G-02 normalization bounds, feature alignment
4	Database and schema	~30	WAL mode enforcement, missing indexes, schema migration safety
5	Dead code and cleanup	~25	G-06 elimination of `nn/advanced/`, unused imports, duplicate files
6	Analysis engines	~30	Graceful degradation for all 10 engines, edge case handling
7	ML pipeline	~35	G-07 Bayesian calibration, gradient clipping, checkpoint versioning
8	Coaching and COPER	~30	G-08 experience decay, RAG index validation, fallback chain
9	UI and UX	~25	F8-XX visual feedback, state consistency, error prevention
10	Resilience and concurrency	~40	F5-35 threading.Event, timeout enforcement, circuit breaker
11	Observability and logging	~35	F5-33 structured logging, correlation IDs, Sentry integration
12	Deep debug and wiring	~45	18 issues: wiring verification, integration testing, edge case audit



Directory `reports/`:

Each remediation phase produced a detailed report saved in the `reports/` directory. Each report includes:

- Numbered list of issues found with code (G-XX or F-XX)
- Problem description with affected files and lines
- Implemented solution with conceptual diff
- Verification: confirmation that the fix did not introduce regressions

Key fixes by code (G-XX):

Code	Phase	Description	Impact
G-01	3	Label leakage in `ConceptLabeler.label_tick()` — future features leaked into training labels	CRITICAL — invalidated VL-JEPA training
G-02	4	Danger zone in `vectorizer.py` — normalization without bounds check	HIGH — NaN propagation in training
G-06	5	Dead code in `backend/nn/advanced/` — 3 unreferenced files	MEDIUM — confusion and accidental imports
G-07	7	Bayesian death estimator not calibrated — static model without auto-calibration	HIGH — inaccurate predictions
G-08	8	COPER experience bank without decay — stale experiences never removed	MEDIUM — coaching based on outdated data

Fixes per feature (F-XX):

Code	Area	Description
F3-29	Processing	ResNet resize inconsistent — image dimensions not normalized
F5-19	Services	Matplotlib rendering without error handling — crash on empty stats
F5-21	Storage	Explicit `session.commit()` inside context manager — double commit
F5-22	Security	Hard-coded API keys in source — migrated to env vars + keyring
F5-33	Observability	`print()` debugging — migrated to structured logging
F5-35	Control	`time.sleep()` loop for stop — migrated to `threading.Event`
F6-XX	Analysis	Analysis engines without graceful degradation — crash on incomplete input
F7-XX	Knowledge	RAG without index validation — inconsistent embedding dimensions
F8-XX	UI	Kivy widget without visual feedback — silent actions confuse the user

---

12.20 Pre-commit Hooks and Quality Gates

File: `.pre-commit-config.yaml`

The project uses a **pre-commit hooks** system that activates automatically before every commit, preventing non-compliant code from reaching the repository.

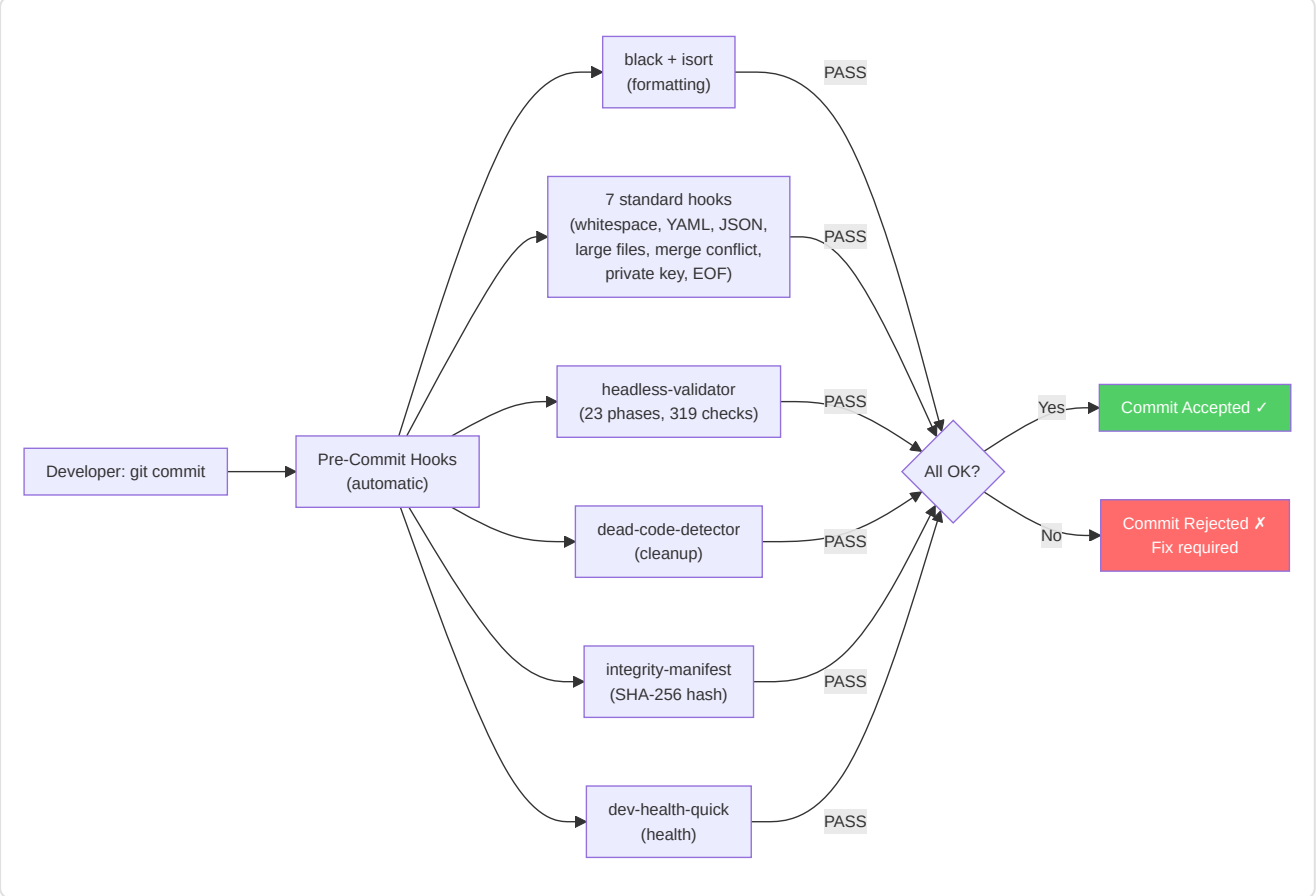
> **Analogy:** Pre-commit hooks are like the **metal detectors at an airport entrance**. Before any baggage (code) boards the plane (repository), it must pass through a series of automatic checks. If a check fails, the baggage is rejected and the passenger (developer) must fix the problem before trying again.

4 Local Hooks (project custom):

Hook	Script	Timeout	Description
`headless-validator`	`tools/headless_validator.py`	20s	23 phases, 319 checks regression gate — the most important
`dead-code-detector`	`tools/dead_code_detector.py`	15s	Identifies unreferenced imports and functions
`integrity-manifest-check`	`tools/sync_integrity_manifest.py`	10s	Verifies SHA-256 manifest hash consistency
`dev-health-quick`	`tools/dev_health.py`	10s	Quick project health check

Standard Hooks (7 from `pre-commit/pre-commit-hooks` + 2 Python):

Hook	Source	Description
`trailing-whitespace`	`pre-commit/pre-commit-hooks`	Removal of trailing whitespace
`end-of-file-fixer`	`pre-commit/pre-commit-hooks`	Required final newline
`check-yaml`	`pre-commit/pre-commit-hooks`	YAML syntax validation
`check-json`	`pre-commit/pre-commit-hooks`	JSON syntax validation
`check-added-large-files`	`pre-commit/pre-commit-hooks`	Blocks files >1MB (prevents accidental `.pt`)
`check-merge-conflict`	`pre-commit/pre-commit-hooks`	Detects unresolved merge conflict markers
`detect-private-key`	`pre-commit/pre-commit-hooks`	Blocks commits of private keys
`black`	`psf/black`	Automatic Python formatting (line length 100, target py3.12)
`isort`	`pycqa/isort`	Automatic import sorting (black profile, line length 100)



12.21 Build, Packaging, and Deployment

Key files: `packaging/windows_installer.iss`, `setup_new_pc.bat`, `export_env.bat`

The project includes a build and packaging system for distributing the desktop application on Windows.

> **Analogy:** The build system is like an **automotive assembly line**. The build process takes all the components (Python sources, ML models, UI assets) and assembles them into a single executable package. `windows_installer.iss` (Inno Setup) is the **bodywork**: it wraps the package in a professional installer with installation wizard, icons, and desktop shortcut. The 3 requirements files are the **bills of materials**: they specify exactly which parts are needed for each type of assembly.

3 Requirements Files:

File	Purpose	Dependencies
`requirements.txt`	**Base** — main dependencies for execution	28 packages (torch, kivy, sqlmodel, demoparser2, playwright, httpx, etc.)
`requirements-ci.txt`	**CI/CD** — adds test and analysis tools	pytest, coverage, mypy, black, isort, pre-commit
`requirements-lock.txt`	**Lock** — exact versions for reproducible builds	Exact pin of each dependency and sub-dependency

Windows Packaging (`packaging/windows_installer.iss`):

- Inno Setup installer with graphical wizard
- Custom CS2 Analyzer icon
- Desktop and Start menu shortcut
- Estimated size: ~500MB (includes PyTorch and models)
- Target: Windows 10/11 64-bit

Setup Scripts:

Script	Purpose
`setup_new_pc.bat`	Environment setup from scratch: Python, pip, venv, dependencies
`export_env.bat`	Export of required environment variables
`run_full_training_cycle.py`	Full ML training cycle orchestration

12.22 Alembic Migration System

Directory: `alembic/` (root) + `Programma_CS2_RENAN/backend/storage/migrations/`

The project uses **two separate Alembic setups** to manage database schema migrations in an organized way.

> **Analogy:** The two Alembic setups are like two **workbooks** for an architect. The main workbook (`alembic/` root) contains the **complete construction plan**: 13 revisions documenting every structural change since the project's beginning. The secondary workbook (`backend/storage/migrations/`) contains the **site notes**: 2 more specific revisions for changes to the storage layer. Each revision is numbered, dated, and reversible — like the versions of a legal document.

Alembic Root (`alembic/`, 13 versions):

Contains the main schema migrations, from initial table creation to the most recent changes. Each migration:

- Has a unique hash as identifier
- Contains `upgrade()` (applies change) and `downgrade()` (reverts change)
- Is idempotent — can be reapplied without errors
- Is tested on a production-like schema before deploy

Backend Migrations (`backend/storage/migrations/`, 2 versions):

Migrations specific to the storage layer, kept separate for modularity. They handle changes to tables that are the exclusive responsibility of the `backend/storage/` module.

Automatic execution: Migrations are executed automatically during phase 3 of the startup sequence (`main.py`), ensuring the schema is always up to date before any DB operation.

12.23 Main Ingestion Orchestrator (`run_ingestion.py`)

File: `Programma_CS2_RENAN/run_ingestion.py`

`run_ingestion.py` is the **orchestrating heart** of the entire ingestion pipeline. It is the largest file dedicated to ingestion and coordinates all phases from demo discovery to persisting results in the database.

> **Analogy:** If the ingestion pipeline (section 12.6) is the path of a letter through the post office, `run_ingestion.py` is the **post office director** in person. He doesn't sort the letters himself — he delegates to the postmen (DemoLoader), the archivists (FeatureExtractor), and the doctors (CoachingService) — but he coordinates the overall flow, decides priorities, handles errors, and keeps the record of everything.

12+ main functions:

Function	Role
`discover_demos()`	Scans configured directories, filters valid `.dem` files
`validate_demo_file()`	Checks magic bytes, size, pre-parsing integrity
`process_single_demo()`	Complete pipeline for a demo: parse → extract → enrich → persist
`batch_process()`	Processes multiple demos in sequence with progress tracking
`enrich_round_stats()`	Post-processing: computes trade kills, blind kills, flash assists
`compute_hltv_rating()`	Computes HLTV 2.0 rating for each player
`assign_dataset_split()`	Assigns 70% train / 15% val / 15% test with temporal split
`generate_coaching_insights()`	Invokes CoachingService to generate 5-20 insights
`update_ingestion_task()`	Updates task status in DB (queued → processing → completed/failed)
`cleanup_failed_tasks()`	Cleans up failed tasks, resets to queued if recoverable
`report_progress()`	Structured logging of overall progress
`handle_duplicate_detection()`	Verifies via Registry whether the demo has already been processed

ResourceManager (`ingestion/resource_manager.py`):

The ResourceManager manages **hardware resources** during ingestion to avoid system overload:

Parameter	Default Value	Purpose
CPU threshold	80%	Pauses ingestion if CPU exceeds this threshold
RAM threshold	85%	Pauses ingestion if RAM exceeds this threshold
Disk check	Yes	Verifies sufficient disk space before processing
Throttle delay	2s	Delay between consecutive tasks for cooldown

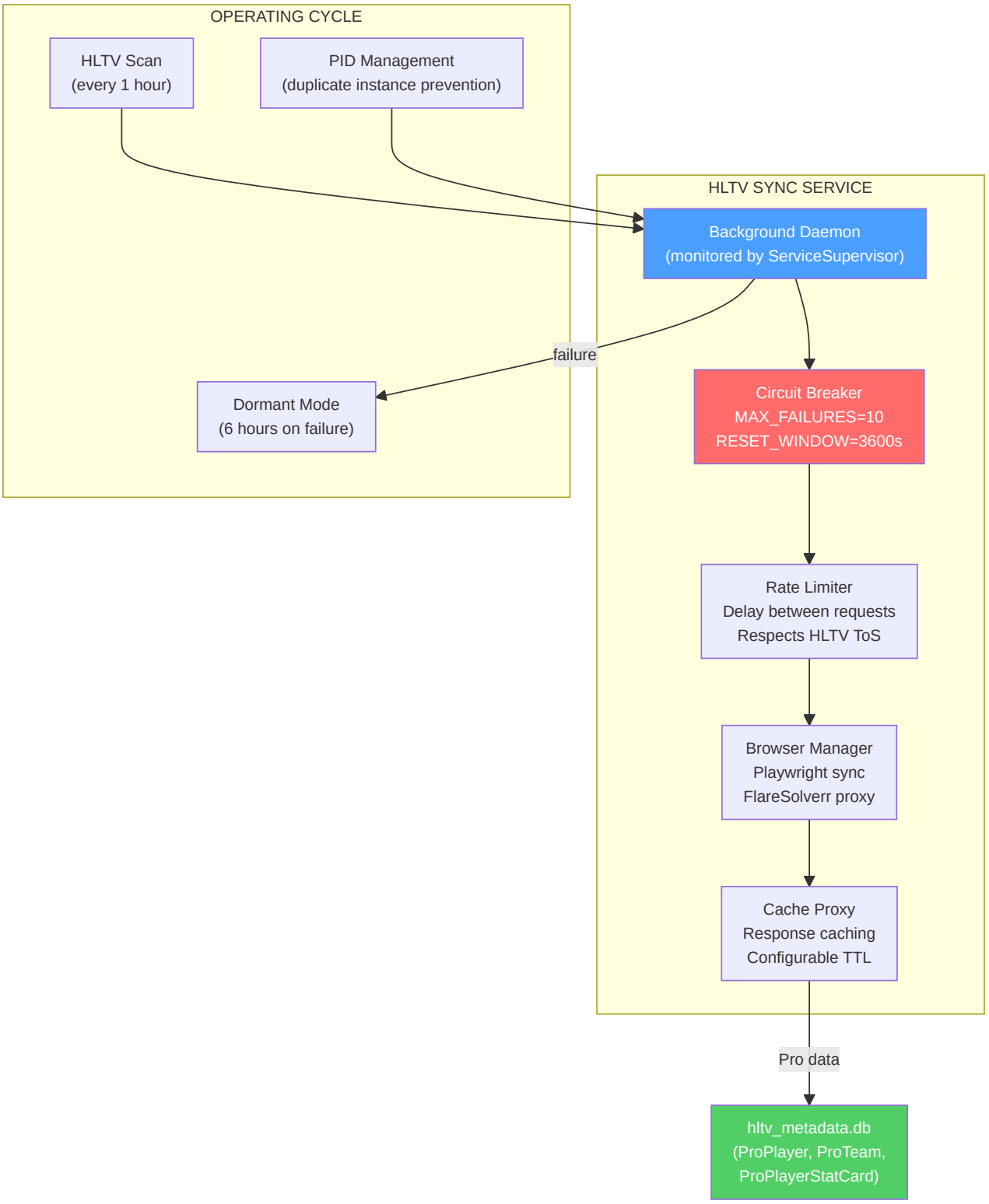
> **Analogy:** The ResourceManager is like the **thermostat of an industrial oven**. If the temperature (CPU/RAM) rises too much, it automatically slows down production (ingestion) to avoid overheating. When the temperature drops, it resumes full throttle. It doesn't ask the operator for permission — it acts autonomously to protect the system.

12.24 HLTV Sync Service and Background Daemon

File: `Programma_CS2_RENAN/hltv_sync_service.py` **Related files:** `backend/data_sources/hltv/` , `backend/services/telemetry_client.py`

The HLTV Sync Service is a **background daemon** that automatically synchronizes pro player data from HLTV.org. It operates as a service monitored by the Console's `ServiceSupervisor` .

> **Analogy:** The HLTV Sync Service is like a **sports correspondent** who lives near the stadium (HLTV.org). Every hour he checks match results, downloads player statistics, and sends them to the newsroom (the database). If the stadium closes (HLTV down), the correspondent enters "dormant mode" for 6 hours before trying again. If he is kicked out of the stadium too many times (rate limit), the Circuit Breaker stops him for an hour to avoid a permanent ban.



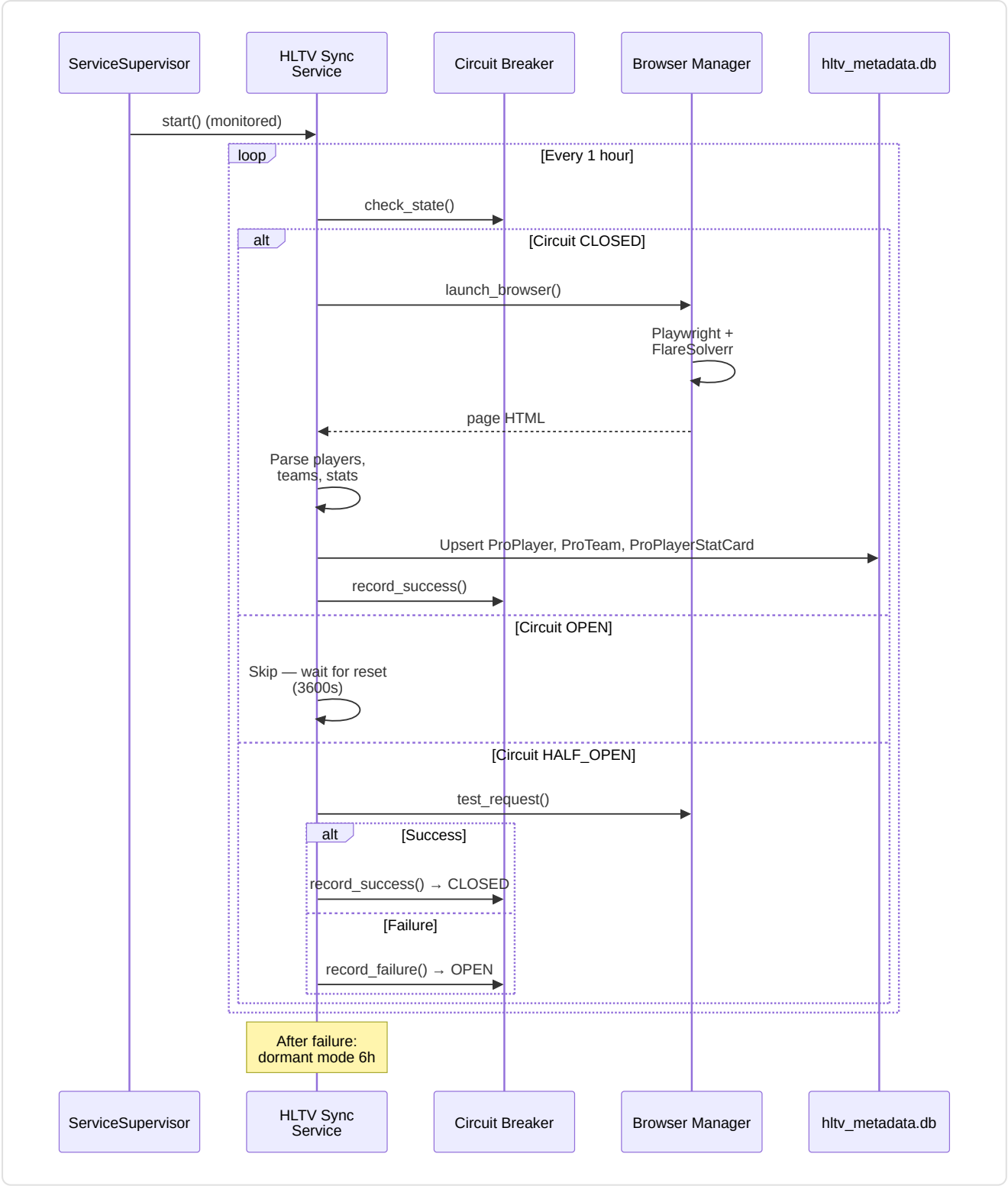
Component	File	Role
CircuitBreaker	`circuit_breaker.py`	Protects against cascade failure: CLOSED→OPEN (after 10 fails)→HALF_OPEN (test)→CLOSED
BrowserManager	`browser_manager.py`	Headless Playwright browser management with FlareSolverr for Cloudflare bypass
CacheProxy	`cache_proxy.py`	Local cache of HLTV responses to reduce requests
RateLimiter	`rate_limiter.py`	Configurable delay between requests to respect ToS

HLTV Internal Architecture — Collectors and Selectors:

The `ingestion/hltv/` tree also contains specialized modules for data collection:

Module	Type	Description
`player_collector.py`	Collector	Collects pro player profiles from HLTV pages
`team_collector.py`	Collector	Collects rosters and team statistics
`match_collector.py`	Collector	Collects match results and demo links
`stat_selector.py`	Selector	Extracts specific statistics from parsed HTML pages
`demo_selector.py`	Selector	Identifies and downloads demo links from match pages

Detailed operating cycle:



HLTV Circuit Breaker Analogy: The Circuit Breaker is like a **smart electrical fuse**. When the circuit is CLOSED (normal), current flows freely (HLTV requests work). If there are too many short circuits (10 consecutive failures), the fuse trips → OPEN (all requests blocked for 1 hour). After the cooldown period, the fuse switches to HALF_OPEN: it lets a single test request through. If the test succeeds, the fuse closes (CLOSED). If it fails, it reopens (OPEN) for another hour. This mechanism prevents "hammering" of a downed server, which would lead to IP bans or blacklisting.

12.25 RASP Guard — Runtime Code Integrity

File: `Programma_CS2_RENAN/observability/rasp.py` **Data file:** `data/integrity_manifest.json`

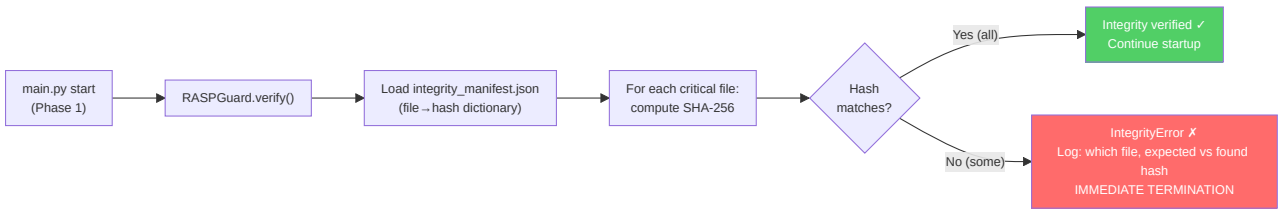
The RASP (Runtime Application Self-Protection) Guard is the **first check** executed on application startup (Phase 1 of the boot sequence). It verifies that no source file has been modified compared to the integrity manifest.

> **Analogy:** The RASP Guard is like a **burglar alarm system with sensors on every door and window**. When the alarm is armed (application start), the system checks that every door (Python file) is in the same position it was left (SHA-256 hash). If someone has forced a window (modified a file), the alarm goes off immediately and the building goes into lockdown (`IntegrityError` → termination). This protects against accidental tampering (corrupt updates) and intentional ones (supply chain attacks).

RASP Guard Components:

Component	Description
`RASPGuard`	Main class — loads the manifest, verifies hashes, raises exception
`integrity_manifest.json`	JSON file with SHA-256 hashes of each critical source file
`IntegrityError`	Custom exception — immediate termination with detailed log
`sync_integrity_manifest.py`	Tool to update the manifest after legitimate changes

Verification flow:



12.26 MatchVisualizer — Advanced Rendering

File: `Programma_CS2_RENAN/reporting/visualizer.py`

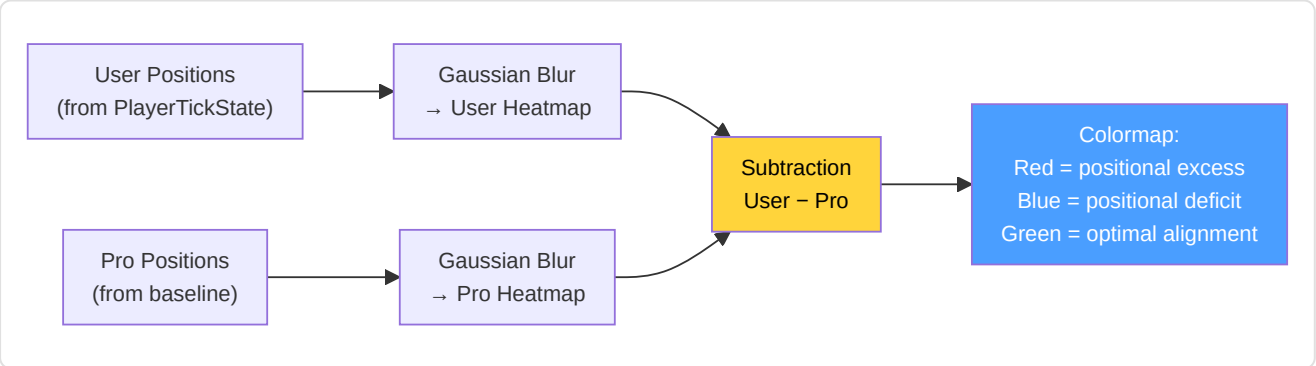
The `MatchVisualizer` extends the reporting system (section 12.13) with **6 specialized rendering methods** for generating charts and visualizations.

> **Analogy:** The MatchVisualizer is like a **professional illustrator** who transforms dry numerical data into at-a-glance understandable illustrations. Each method is a different technique: the radar chart is a **spider portrait** of skills, the differential heatmap is a **heat map** showing where you stand too much and where too little, critical moments are **comic panels** highlighting the key scenes of the match.

6 Rendering methods:

Method	Output	Description
`render_skill_radar()`	PNG radar chart	5-axis pentagon (Mechanics, Positioning, Utility, Timing, Decision) with pro overlay
`render_trend_chart()`	PNG line chart	Historical trend of key metrics (KPR, ADR, KAST) over N matches
`render_heatmap()`	PNG heatmap	2D Gaussian occupancy on tactical map
`render_differential_heatmap()`	PNG diff heatmap	user−pro subtraction: red=too much, blue=too little, green=optimal
`render_critical_moments()`	PNG annotated timeline	Timeline with color markers for key moments (kills, deaths, plants)
`render_comparison_table()`	PNG/HTML table	user vs pro comparison table with percentage delta for each metric

Differential heatmap — algorithm:



12.27 Runtime Data and Configuration Files

The project includes several data files that are read or written during execution. These files are not Python code but are essential for the system to function.

`data/map_config.json` — CS2 map configuration:

Contains the coordinate transformation parameters for each supported map (Dust2, Mirage, Inferno, Nuke, Vertigo, Overpass, Ancient, Anubis):

```
{
  "de_dust2": {
    "pos_x": -2476, "pos_y": 3239, "scale": 4.4,
    "z_cutoff": null, "level": "single"
  },
  "de_nuke": {
    "pos_x": -3453, "pos_y": 2887, "scale": 7.0,
    "z_cutoff": -495, "level": "multi"
  }
}
```

> **Analogy:** `map_config.json` is like an **atlas with cartographic projections**. Each map has its unique projection: origin point, scale, and, for multi-level maps, the altitude that separates the floors. Without this file, the Tactical Viewer wouldn't know where to position players on the 2D map.

`data/integrity_manifest.json` — RASP integrity manifest:

Dictionary `{file_path: sha256_hash}` for all critical Python files. Generated/updated by `tools/sync_integrity_manifest.py`. Verified on startup by `RASPGuard` (section 12.25).

`data/training_progress.json` — Training progress:

Persists training state between restarts: current epoch, best loss, learning rate, last processed demo. Consulted by the Teacher daemon to resume training from the exact point where it stopped.

`user_settings.json` — User settings:

Level 2 configuration file (cf. section 12.3). Saved in the project directory, contains customizable preferences: demo paths, UI theme, language, feature flags.

12.28 Root-Level Entry Points

Besides `main.py`, the project includes several **executable scripts** at the root level that serve as alternative entry points for specific operations:

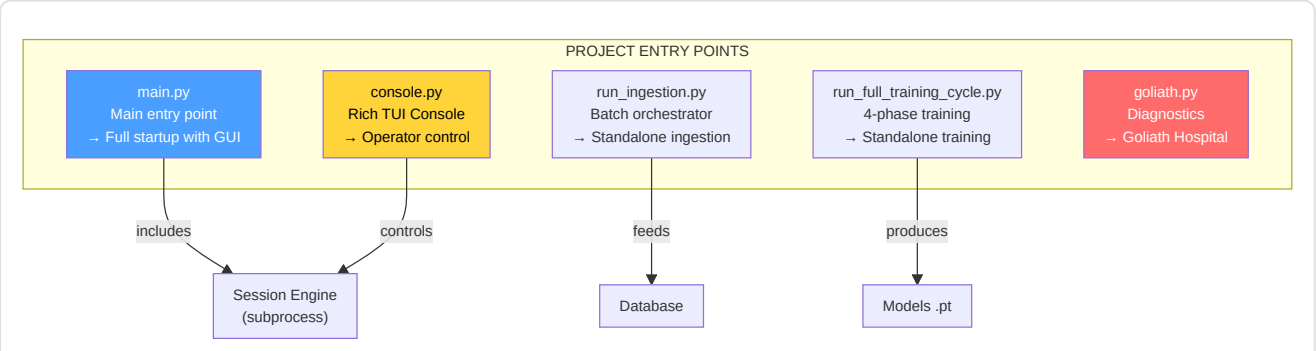
Script	Lines	Purpose	Invocation
`console.py`	~61KB	Complete interactive TUI console with Rich — command registry, CLI and interactive modes, complete system management	`python console.py`
`run_ingestion.py`	1,057	Standalone ingestion orchestrator (cf. 12.23) — batch demo processing	`python run_ingestion.py [path]`
`goliath.py`	~200	Launcher for Goliath Hospital diagnostics (cf. 12.17)	`python goliath.py`
`schema.py`	~100	Generation and visualization of the current DB schema	`python schema.py`
`run_full_training_cycle.py`	~150	Standalone 4-phase complete training (JEPA → Pro → User → RAP)	`python run_full_training_cycle.py`
`hflayers.py`	~50	Utility for Hopfield layers (NCPs integration)	Import only

Interactive console (`console.py` , ~61KB):

The console is the **most powerful entry point** for advanced operators. It offers a Rich-based TUI (Terminal User Interface) with:

- **Command Registry**: System of registered commands with autocompletion
- **CLI Mode**: Single-shot command execution (`console.py status` , `console.py train`)
- **Interactive Mode**: REPL shell with custom prompt and command history
- **Rich Panels**: Dashboard with color tables, progress bars, status tree view

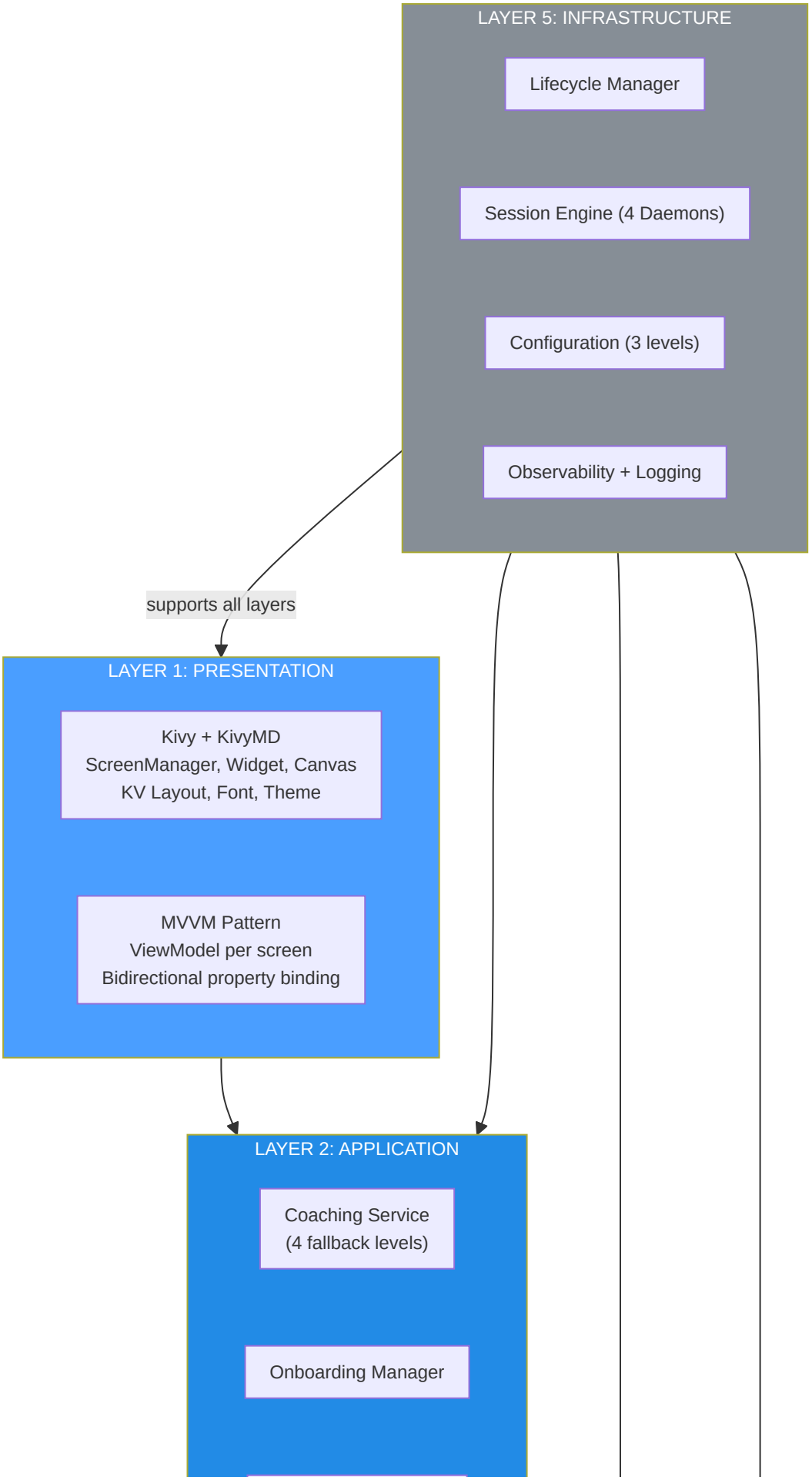
> **Analogy:** The console is like the **terminal of a Linux system administrator**. While the graphical interface (Kivy) is the user-friendly desktop for the end user, the console is the SSH terminal for the advanced operator who wants to control everything manually: start/stop services, inspect the database, launch training, run diagnostics.

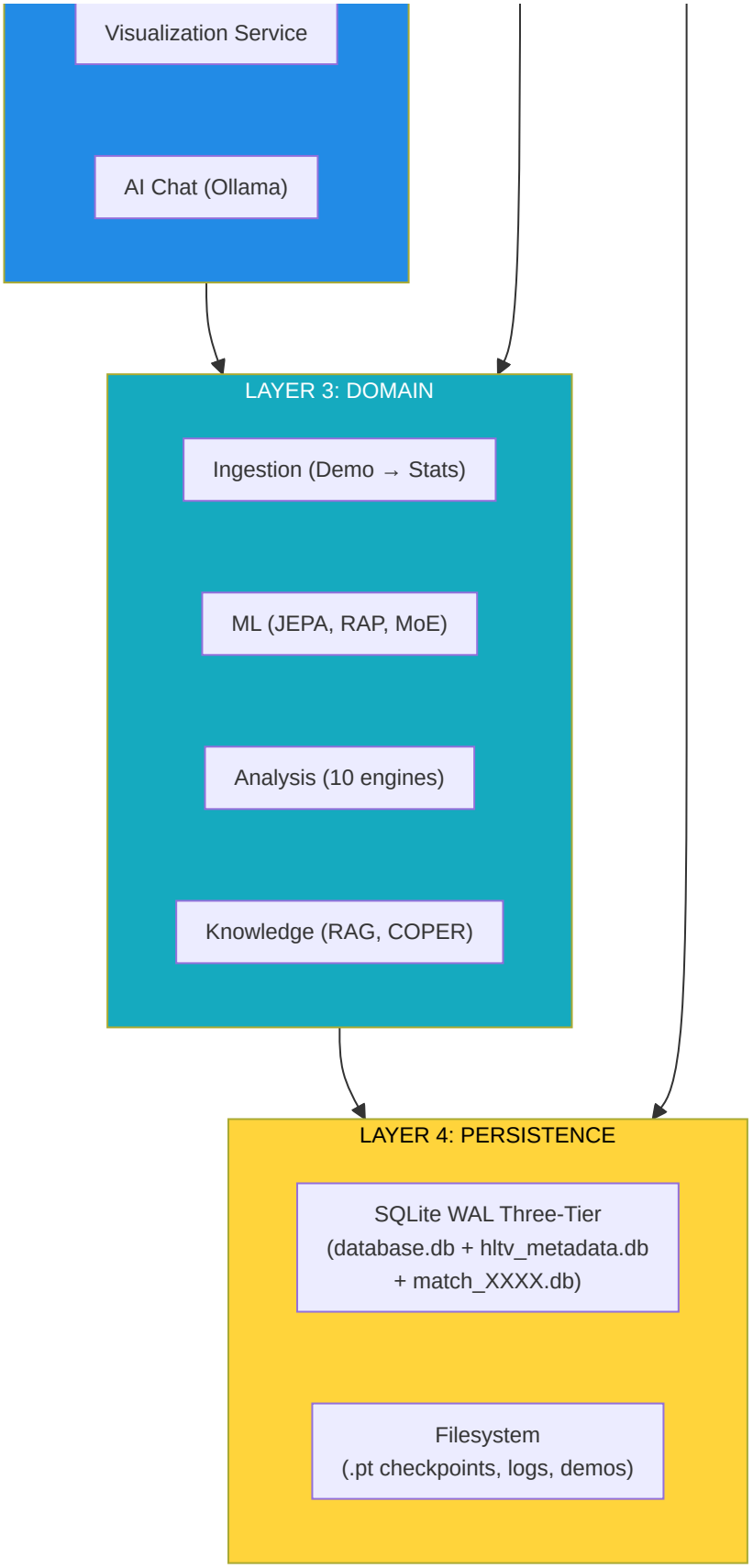


Architectural Summary

Macena CS2 Analyzer is a **layered and modular** application organized across 5 architectural layers with 3 separate processes.

> **Analogy:** The architecture is like a **5-layer cake**, where each layer has a precise role and communicates only with the layers immediately adjacent to it. The top layer (Presentation) is the frosting: what the user sees and touches. The next layer (Application) is the filling: it combines ingredients from various sources. The central layer (Domain) is the sponge cake: the real substance of the system. The lower layer (Persistence) is the plate: it supports everything and never changes shape. The base layer (Infrastructure) is the table: the silent foundation on which everything rests.


LAYER 5: INFRASTRUCTURE
Lifecycle Manager
Session Engine (4 Daemons)
Configuration (3 levels)
Observability + Logging
supports all layers
LAYER 1: PRESENTATION
Kivy + KivyMD
ScreenManager, Widget, Canvas
KV Layout, Font, Theme
MVVM Pattern
ViewModel per screen
Bidirectional property binding
LAYER 2: APPLICATION
Coaching Service
(4 fallback levels)
Onboarding Manager



Complete technology stack (28 dependencies):

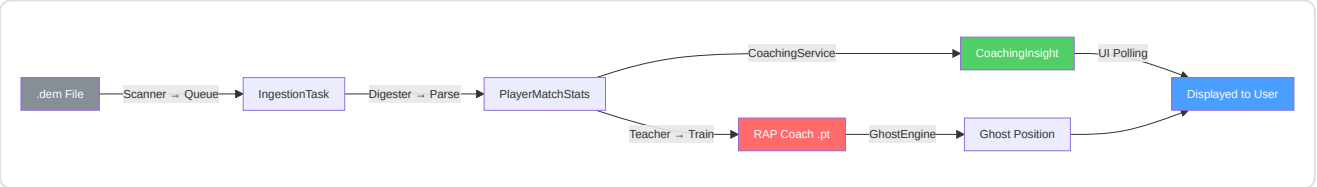
Category	Library	Version	Role in the project
ML Core	PyTorch	2.x	Neural networks (JEPA, RAP, MoE, NeuralRoleHead, WinProb)
ML Ext	ncps	latest	Liquid Time-Constant Networks (LTC) for RAP memory
ML Ext	hflayers	latest	Hopfield layers for associative attention
UI	Kivy	2.x	Cross-platform UI framework
UI	KivyMD	1.x	Material Design widgets
DB	SQLModel	latest	ORM (Pydantic + SQLAlchemy)
DB	SQLAlchemy	2.x	Underlying database engine
DB	Alembic	1.x	Schema migrations
HTTP	requests	2.x	Steam/FACEIT API (synchronous)
HTTP	httpx	latest	Telemetry (asynchronous)
Scraping	playwright	latest	Headless browser for HLTV
Parsing	demoparser2	latest	CS2 demo parser (Rust-based, fast)
Data	pandas	2.x	Tabular data manipulation
Data	numpy	1.x	Numerical computations
Viz	matplotlib	3.x	Charts and visualizations
NLP	sentence-transformers	latest	384-dim embeddings for RAG
TUI	rich	latest	Color console, tables, progress bars
Logging	logging (stdlib)	—	Structured logging
Security	keyring	latest	Secure management of API keys
Monitoring	sentry-sdk	latest	Remote error tracking (optional)
Testing	pytest	latest	Test framework
Formatting	black	latest	Code formatting
Import	isort	latest	Import sorting
QA	pre-commit	latest	Pre-commit hooks
Crypto	hashlib (stdlib)	—	SHA-256 for RASP manifest
Concurrency	threading (stdlib)	—	Daemon threads, MLControlContext
IPC	subprocess (stdlib)	—	Session Engine as subprocess
Config	json (stdlib)	—	user_settings.json, map_config.json

> **Stack Analogy:** The technology stack is like the **toolbox of an expert craftsman**. PyTorch is the **CNC lathe** — the most expensive and powerful precision tool. Kivy is the **workbench** — the surface on which everything is assembled. SQLModel is the **organized filing cabinet** — where all projects are kept. Playwright is the **reconnaissance drone** — it goes to gather data in the field (HLTV). Rich is the **finishing paint** — it makes everything more beautiful and readable. Each tool has a precise role, and none is redundant.

The 3 application processes:

Process	Type	Responsibility	Communication
Main	Kivy GUI	User interface, rendering, interaction	DB polling every 10-15s
Daemon	Subprocess (Session Engine)	Scanner, Digester, Teacher, Pulse	stdin pipe (IPC) + shared DB
Optional Services	External processes	HLTV sync, local Ollama LLM	HTTP/API + Console supervision

Main data flows:



Remediation Note — Eliminated Code (G-06)

During the 12-phase remediation process (370+ issues resolved overall — cf. section 12.19), the `backend/nn/advanced/` directory was **completely eliminated** because it contained unreferenced dead code:

- `superposition_net.py` — An experimental superposition network never integrated into training or inference flow. Active superposition functionality is implemented in `layers/superposition.py` (used by the RAP Strategy layer).
- `brain_bridge.py` — An experimental model bridge never called by any module.
- `feature_engineering.py` — Duplicate feature engineering; the canonical version resides in `backend/processing/feature_engineering/`.

> **Rationale (G-06):** Keeping dead code creates risks of confusion (which `superposition` is the real one?), increases maintenance surface, and can introduce accidental imports. The elimination was verified through static dependency analysis: no file in the project imported from `backend/nn/advanced/`.

Shared Utility — `round_utils.py`

File: `Programma_CS2_RENAN/backend/knowledge/round_utils.py`

The function `infer_round_phase(equipment_value)` is a **shared utility** used by both the coaching service and the knowledge system to classify the economic phase of a round. It resides in `backend/knowledge/` and is imported by modules in `services/` and `processing/`.

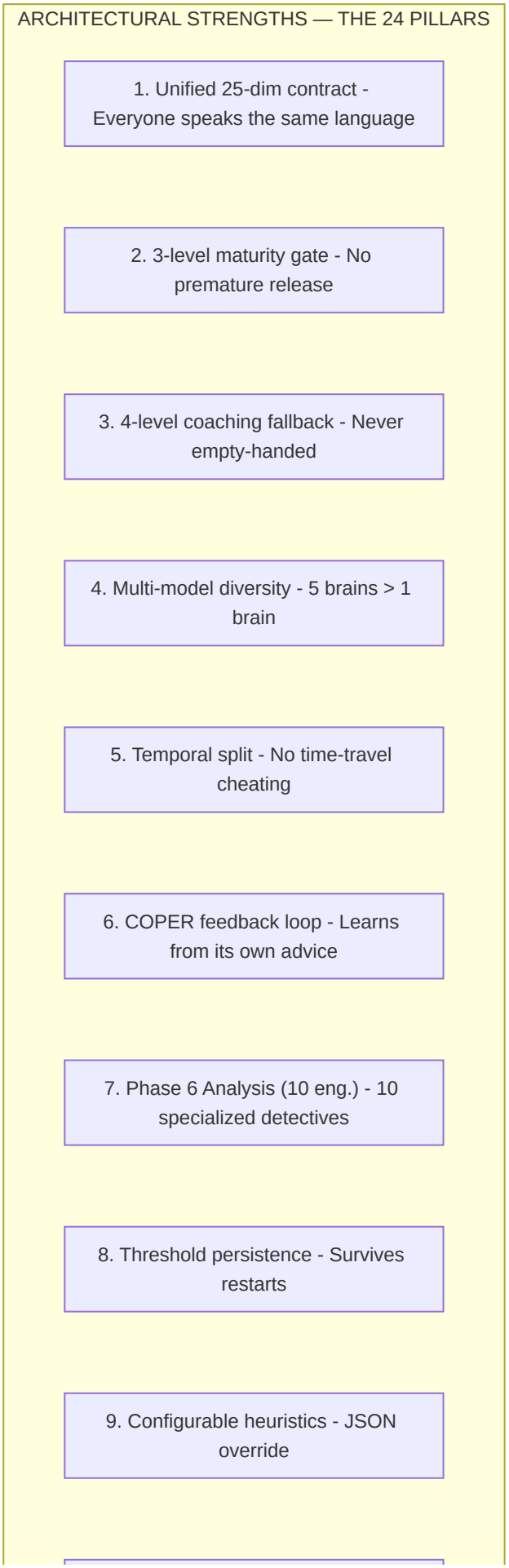
Equipment value	Returned phase
< $1,500	`"pistol"`
$1,500 – $2,999	`"eco"`
$3,000 – $3,999	`"force"`
≥ $4,000	`"full_buy"`

> **Analogy:** `round_utils.py` is like a **shared dictionary** consulted by multiple departments. Instead of having each office define its own rules for classifying "eco" vs "force buy", they all consult the same dictionary, ensuring consistency across coaching, analysis, and knowledge base.

Architectural Strengths

1. **25-dimensional unified feature contract** — `METADATA_DIM = 25` enforces system-wide training/inference parity.
2. **3-level maturity gating** — Prevents premature model deployment with insufficient data.
3. **4-level coaching fallback** — COPER → Hybrid → RAG → Base guarantees insights are always delivered.
4. **Multi-model diversity** — JEPA, VL-JEPA, LSTM+MoE, RAP, and NeuralRoleHead contribute complementary inductive biases.
5. **Temporal split** — Prevents data leakage by ensuring chronological ordering.
6. **COPER feedback loop** — EMA-based effectiveness monitoring with decay of stale experience.
7. **Phase 6 analysis suite** — 10 analysis engines (role, win probability, game tree, belief, deception, momentum, entropy, blind spots, utility and economy, engagement distance).
8. **Threshold persistence** — Role thresholds survive restarts via the DB table `RoleThresholdRecord`.
9. **Configurable heuristics** — `HeuristicConfig` externalizes normalization bounds to JSON.

10. **LLM polishing** — Optional Ollama integration for natural language coaching narratives.
11. **Training Observatory** — 4-level introspection (Callback, TensorBoard, Maturity State Machine, Embedding Projector) with zero impact when disabled and callbacks isolated from errors.
12. **Neural Role Consensus** — Dual heuristic + NeuralRoleHead MLP classification with cold-start protection, guaranteeing reliable role assignment even with partial data.
13. **Per-Round Statistical Isolation** — The `RoundStats` model prevents contamination across rounds, enabling granular round-level coaching and per-round HLTV 2.0 evaluations.
14. **Quad-Daemon Architecture** — Complete separation between GUI and heavy work, with coordinated shutdown and automatic zombie task cleanup.
15. **Pervasive graceful degradation** — Every component has a fallback plan: the system never crashes, it always degrades in a controlled way.
16. **Three-Tier Storage Architecture** — Separation of `database.db` (core + knowledge, 18 tables), `hltv_metadata.db` (pro data, 3 tables), and `match_data/{id}.db` (per-match telemetry) to eliminate WAL contention and prevent uncontrolled monolith growth.
17. **Live Bayesian Calibration (G-07)** — The death estimator self-calibrates with `extract_death_events_from_db()` → `auto_calibrate()`, transforming from a static model into an adaptive system.
18. **Live Training Control (MLControlContext)** — Real-time pause/resume/stop/throttle of training via `threading.Event`, with custom exception `TrainingStopRequested` instead of `StopIteration`.
19. **Resilient Circuit Breaker** — `_CircuitBreaker` for external APIs (HLTV) with MAX_FAILURES=10, RESET_WINDOW_S=3600, prevents cascade failure with CLOSED→OPEN→HALF_OPEN pattern.
20. **5-Level Validation Pyramid** — Headless Validator → pytest → Backend Validator → Goliath Hospital → Brain Verify: each level progressively deeper, from quick smoke test (10s) to complete audit of 118 rules.
21. **RASP Guard** — Runtime Application Self-Protection with SHA-256 manifest: verifies source code integrity on startup, prevents accidental and intentional tampering.
22. **13-Hook Pre-commit Gate** — 4 custom hooks + 7 standard + 2 Python prevent non-compliant code from reaching the repository. Formatting, validation, dead code, integrity, merge conflict, and private key verified automatically.
23. **Hardware-Aware ResourceManager** — Automatic CPU/RAM throttling during ingestion: the system autonomously slows down if hardware resources are under pressure, without human intervention.
24. **Test Forensics** — 10 post-mortem investigation scripts to diagnose training failure, weight anomalies, coaching path traces, and DB consistency — the system's "black box".


ARCHITECTURAL STRENGTHS — THE 24 PILLARS
1. Unified 25-dim contract - Everyone speaks the same language
2. 3-level maturity gate - No premature release
3. 4-level coaching fallback - Never empty-handed
4. Multi-model diversity - 5 brains > 1 brain
5. Temporal split - No time-travel cheating
6. COPER feedback loop - Learns from its own advice
7. Phase 6 Analysis (10 eng.) - 10 specialized detectives
8. Threshold persistence - Survives restarts
9. Configurable heuristics - JSON override

10. LLM polish (Ollama) - Advice sounds natural

11. Training Observatory - Report card for the brain

12. Neural Role Consensus - Two teachers compare notes

13. Per-Round isolation - Evaluates every question, not just the test

14. Quad-Daemon architecture - Responsive GUI, heavy work in background

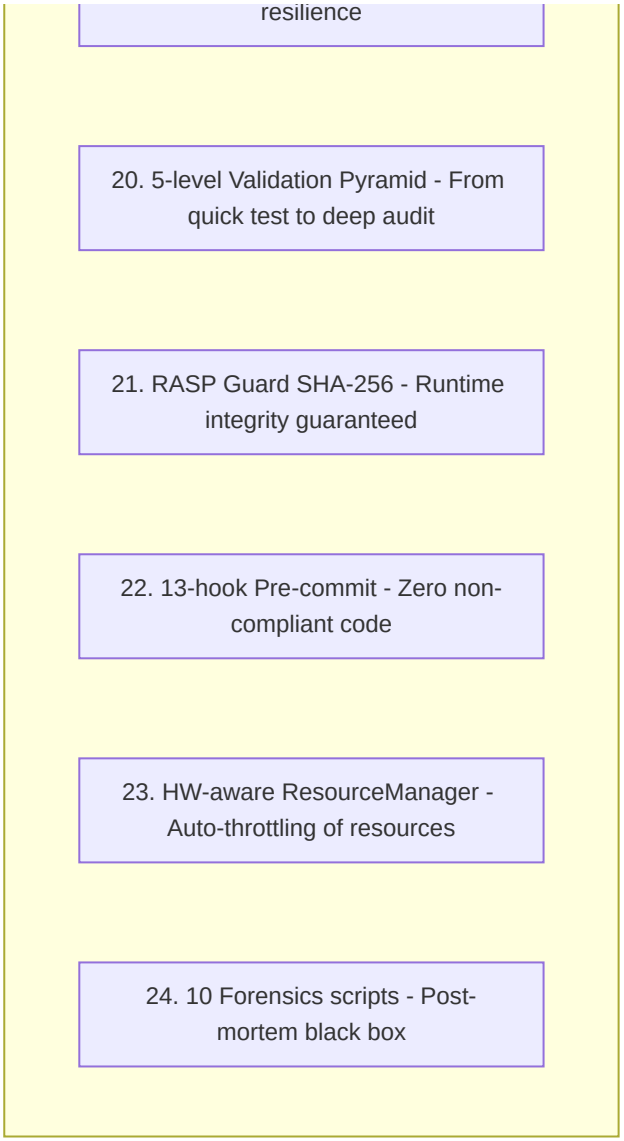
15. Graceful degradation - The system never crashes

16. Three-Tier Storage - No contention between write and read

17. Live Bayesian calibration - Self-calibrates with the data

18. Live Training control - Pause/Stop without loss

19. Circuit Breaker - External API



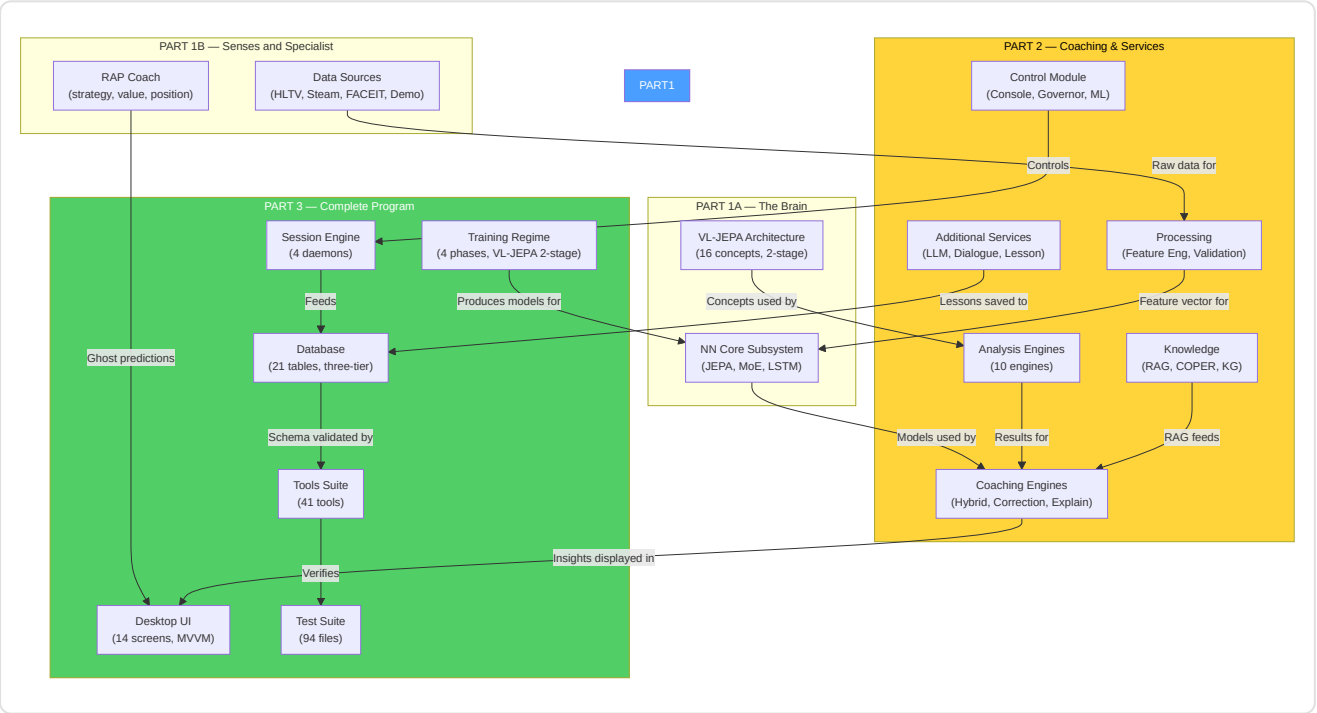
End of document — Complete guide to Macena CS2 Analyzer

Total `.py` files in the project: **~406** (in `Programma_CS2_RENAN/`, ~480 project-wide) Total verified Python lines of code: **≈ 102,000+** AI subsystems covered: **8** (NN Core, VL-JEPA, RAP Coach, Coaching Services, Coaching Engines, Knowledge, Analysis, Processing + Training Observatory) Program subsystems covered: **18** (Startup, Lifecycle, Configuration, Session Engine, Desktop UI, Ingestion, Storage, Observability, Control Console, RASP Guard, HLTV Sync, Ingestion Orchestrator, ResourceManager, Tools Suite, Test Suite, Pre-commit, Build/Packaging, Alembic Migrations) Documented models: **6** (AdvancedCoachNN/TeacherRefinementNN, JEPA, VL-JEPA, RAPCoachModel, NeuralRoleHead, WinProbabilityNN) Documented analysis engines: **10** (Role, WinProb, GameTree, Belief, Deception, Momentum, Entropy, Blind Spots, Utility and Economy, Engagement Distance) Documented coaching engines: **7** (HybridEngine, CorrectionEngine, ExplainabilityGenerator, NNRefinement, ProBridge, TokenResolver, LongitudinalEngine) Additional services documented: **7** (CoachingDialogue, LessonGenerator, LLMService, VisualizationService, ProfileService, AnalysisService, TelemetryClient) Database tables documented: **21** (distributed across three-tier storage architecture: `database.db`, `hltv_metadata.db`, `match_data/{id}.db`) UI screens documented: **14** (Wizard, Home, Coach, Tactical

Viewer, Settings, Help, Match History, Match Detail, Performance, User Profile, Profile, Pro Comparison, Steam Config, FACEIT Config) — Qt/PySide6 (primary) + Kivy/KivyMD (legacy) Documented daemons: **4** (Scanner, Digester, Teacher, Pulse) Documented validation tools: **41** (Headless Validator, Brain Verify, Goliath Hospital, DB Inspector, Demo Inspector, ML Coach Debugger, Backend Validator, Dead Code Detector, Dev Health, Feature Audit, etc.) Documented test files: **94** (+ conftest.py, 10 forensics, 15 verification scripts) Headless validator phases: **23** (319 automated checks) Documented pre-commit hooks: **13** (4 custom local + 7 standard + 2 Python) Architectural pillars: **24** (including Three-Tier Storage, Live Bayesian Calibration, Live Training Control, Circuit Breaker, Validation Pyramid, RASP Guard, Pre-commit Gate, HW-aware ResourceManager, Forensics) Issues resolved through remediation: **368** (in 12 systematic phases) Documented remediation phases: **12** (with G-XX and F-XX codes)

Cross-Part Interconnection Map

The three parts of the documentation form an interconnected system. This map shows the main dependencies between sections:



Key cross-references:

From (Part)	To (Part)	Connection
Part 1 §VL-JEPA	Part 3 §10 (Training)	The two-stage protocol described in Part 1 is implemented by the training regime in Part 3
Part 1 §Data Sources	Part 3 §12.6 (Ingestion)	Data sources feed the ingestion pipeline
Part 2 §Coaching Engines	Part 3 §12.16 Flow 1	Coaching engines generate the insights displayed in the UI
Part 2 §Control Module	Part 3 §12.4 (Session Engine)	The control module manages the 4 daemons
Part 2 §Processing	Part 1 §NN Core	The vectorizer produces the METADATA_DIM=25 consumed by the models
Part 3 §9 (Database)	Part 2 §Knowledge	The RAG/COPER tables are used by the knowledge system
Part 3 §11 (Loss)	Part 1 §VL-JEPA	The documented loss functions implement VL-JEPA training
Part 3 §12.17 (Tools)	Part 3 §12.18 (Tests)	The validation pyramid includes both tools and test suite

Technical Glossary

Term	Definition
ADR	Average Damage per Round — average damage dealt per round
BCE	Binary Cross-Entropy — loss for binary classification
COPER	Coaching through Past Experiences and References — coaching based on past experiences
EMA	Exponential Moving Average — exponential moving average used for JEPA target encoder
HLTV 2.0	HLTV.org rating system based on KPR, survival, impact, damage
InfoNCE	Noise Contrastive Estimation — contrastive loss to align representations
JEPA	Joint-Embedding Predictive Architecture — architecture that predicts target embeddings from context
KAST	Kill/Assist/Survive/Trade — percentage of rounds with positive contribution
KG	Knowledge Graph — knowledge graph with entities and relationships
KPR	Kills Per Round — average kills per round
LTC	Liquid Time-Constant — neural networks with learned time constants
METADATA_DIM	Dimension of the unified feature vector (25 in the project)
MoE	Mixture of Experts — network with specialized experts and gating
MVVM	Model-View-ViewModel — architectural pattern for UI/logic separation
RAG	Retrieval-Augmented Generation — generation enriched by retrieval
RAP	Reasoning Action Planning — model for strategic coaching
RASP	Runtime Application Self-Protection — runtime application protection
TUI	Terminal User Interface — user interface in the terminal
VICReg	Variance-Invariance-Covariance Regularization — regularization for embedding diversity
VL-JEPA	Vision-Language JEPA — extension with linguistic concepts
WAL	Write-Ahead Logging — SQLite mode for concurrent reads/writes
Z-score	Number of standard deviations from the mean — measures how far a value is from the norm

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